RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)
Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS LABORATORY RECORD

III Year / V Semester

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Academic Year : 2025 – 2026

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EXPERIMENT 01

Study of Computer Forensics and Different Tools Used for Forensic Investigation

Aim

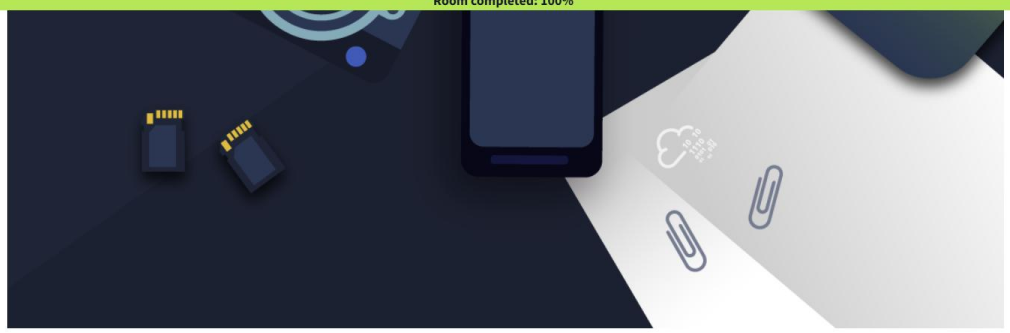
To study the fundamentals of computer forensics and explore various commercial, open-source, and field tools used in digital crime investigations.

Overview of Digital Forensics

- **Concept:** Digital forensics is the field of identifying, collecting, examining, and analyzing digital evidence to determine responsibility for cyber intrusions or computer crimes.

- **Core Principle:** Perpetrators leave digital traces (log entries, registry modifications, deleted files, malware remnants) that serve as investigative evidence.





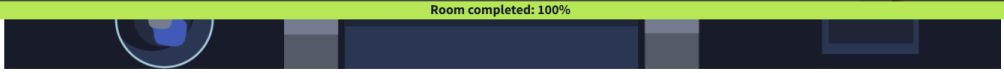
Answer the questions below

Consider the desk in the photo above. In addition to the smartphone, camera, and SD cards, what would be interesting for digital forensics?

laptop ✓ Correct Answer

Task 2 Digital Forensics Process

Practical Example of Digital Forensics



Assuming this employee is suspected in the figure above, we can quickly see the digital devices that might be of interest to an investigation. We notice a tablet, a smartphone, a digital camera, and a USB flash memory in addition to a desktop computer. Any of these devices might contain a trove of information that can help with an investigation. Processing these as evidence would require digital forensics.

More formally, digital forensics is the application of computer science to investigate digital evidence for a legal purpose. Digital forensics is used in two types of investigations:

1. **Public-sector investigations** refer to the investigations carried out by government and law enforcement agencies. They would be part of a crime or civil investigation.
2. **Private-sector investigations** refer to the investigations carried out by corporate bodies by assigning a private investigator, whether in-house or outsourced. They are triggered by corporate policy violations.

Whether investigating a crime or a corporate policy violation, part of the evidence is related to digital devices and digital media. This is where digital forensics comes into play and tries to establish what has happened. Without trained digital forensics investigators, it won't be possible to process any digital evidence properly.

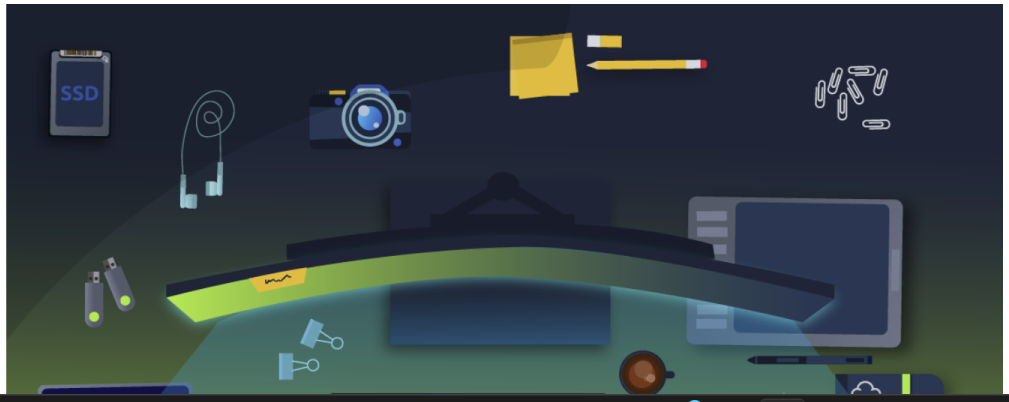


dgsivakumar/drsivakumardh | CR23532 material - Classroo | CR23532 -LAB MANUAL.pdf | TryHackMe | Intro to Digital | Ask Gemini
tryhackme.com/room/introdigitalforensics
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Room completed: 100%

Forensics is the application of science to investigate crimes and establish facts. With the use and spread of digital systems, such as computers and smartphones, a new branch of forensics was born to investigate related crimes: computer forensics, which later evolved into, *digital forensics*.

Think about the following scenario. The law enforcement agents arrive at a crime scene; however, part of this crime scene includes digital devices and media. Digital devices include desktop computers, laptops, digital cameras, music players, and smartphones, to name a few. Digital media includes CDs, DVDs, USB flash memory drives, and external storage. A few questions arise:

- How should the police collect digital evidence, such as smartphones and laptops? What are the procedures to follow if the computer and smartphone are running?
- How to transfer the digital evidence? Are there certain best practices to follow when moving computers, for instance?
- How to analyze the collected digital evidence? Personal device storage ranges between tens of gigabytes to several terabytes; how can this be analyzed?



Search
ENG
IN
41%
17:19
15-10-2026

Classifications & Forensic Tools Studied

Category	Tools Studied	Key Purpose / Feature
Commercial Suites	EnCase Forensic, FTK, ProDiscover	Comprehensive forensic suites with advanced reporting used extensively in law enforcement.
Open-Source Suites	The Sleuth Kit (TSK), Helix, Knoppix	Free, accessible forensic suites providing core investigation capabilities without licensing costs.
Disk & Partition Tools	FTK Imager, Active@ Partition Recovery	Bit-by-bit disk imaging, file recovery, and restoring deleted partitions.
Log & Analysis Utilities	Log Parser Lizard GUI, Noxcivis Field Toolkit (NFT)	Log file parsing and evidence collection from host machines.
Forensic Platforms	Autopsy	Open-source digital forensics platform and graphical interface for The Sleuth Kit.

Result

The study of computer forensics concepts and various forensic investigation tools was successfully completed.

EXPERIMENT 02

Live Forensics Case Investigation using Autopsy

Aim

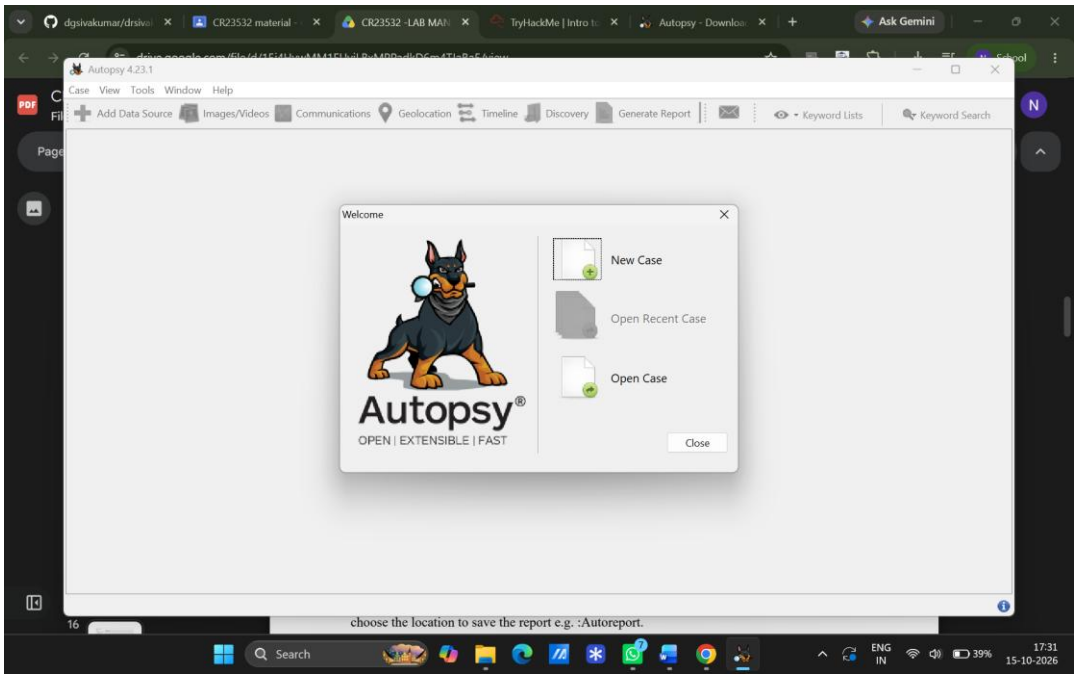
To perform a live digital forensics case investigation and analyze system evidence using the Autopsy forensic tool.

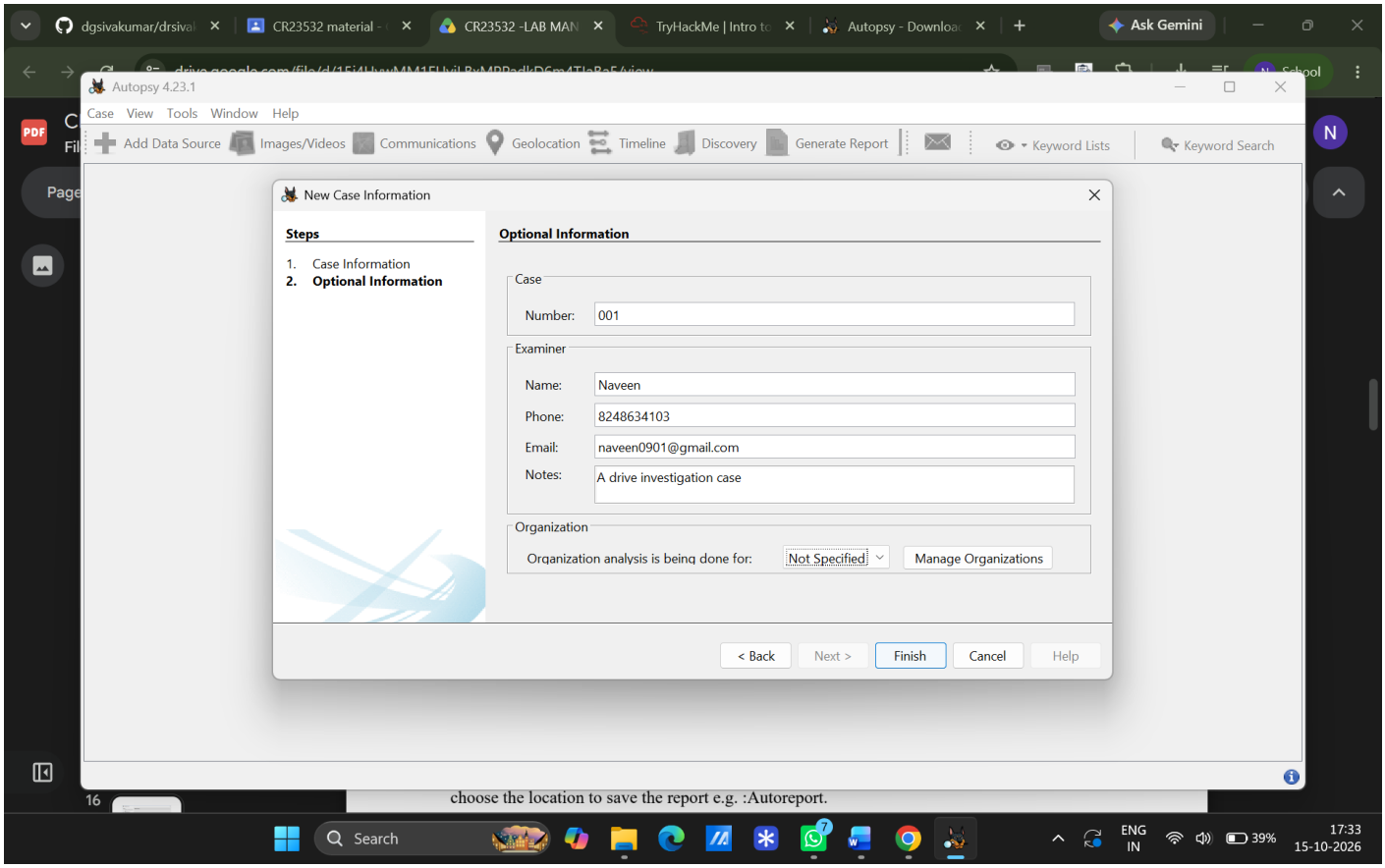
Software / Tools Required

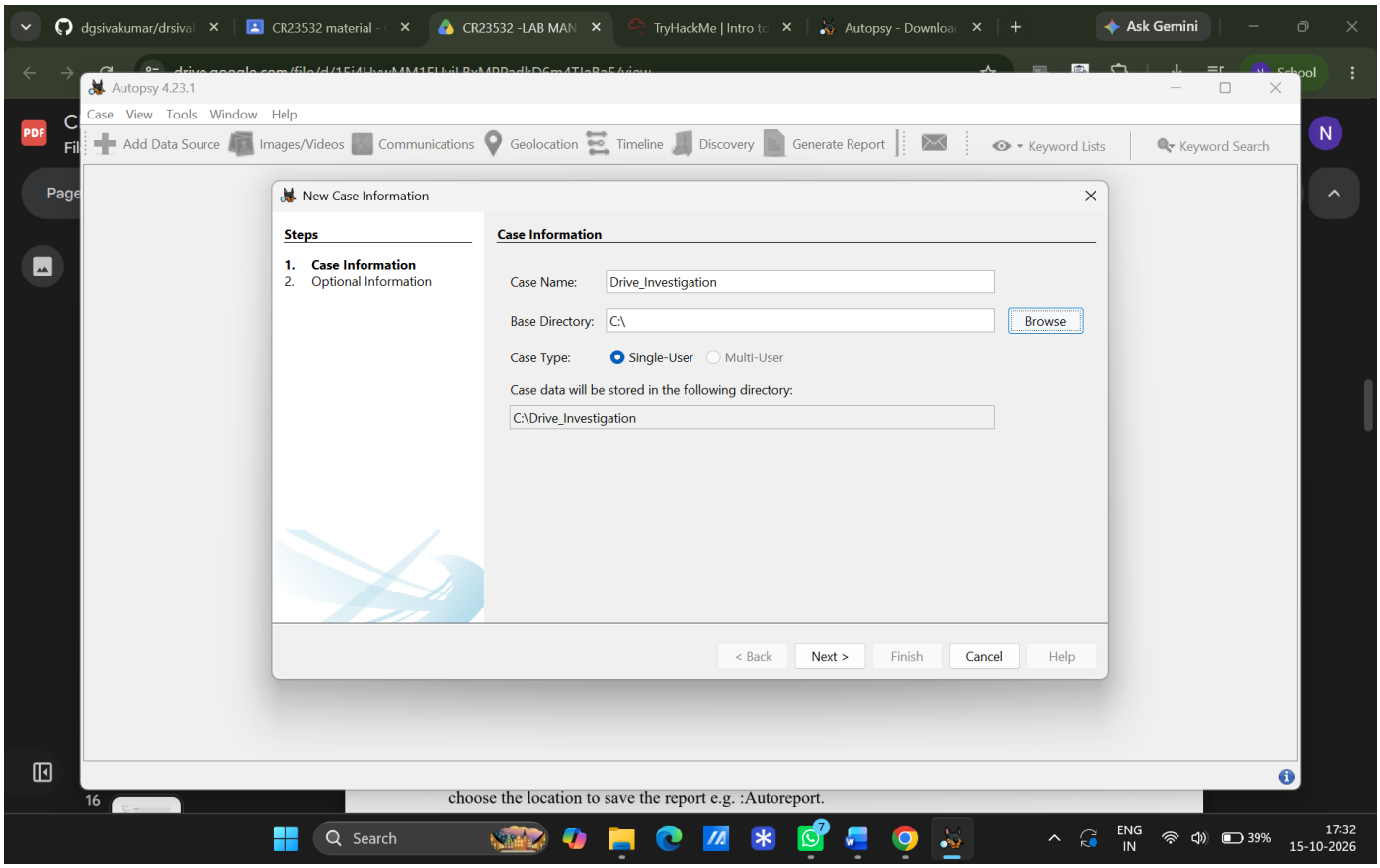
- Autopsy Digital Forensics Platform
- Target System / Evidence Image / Data Source

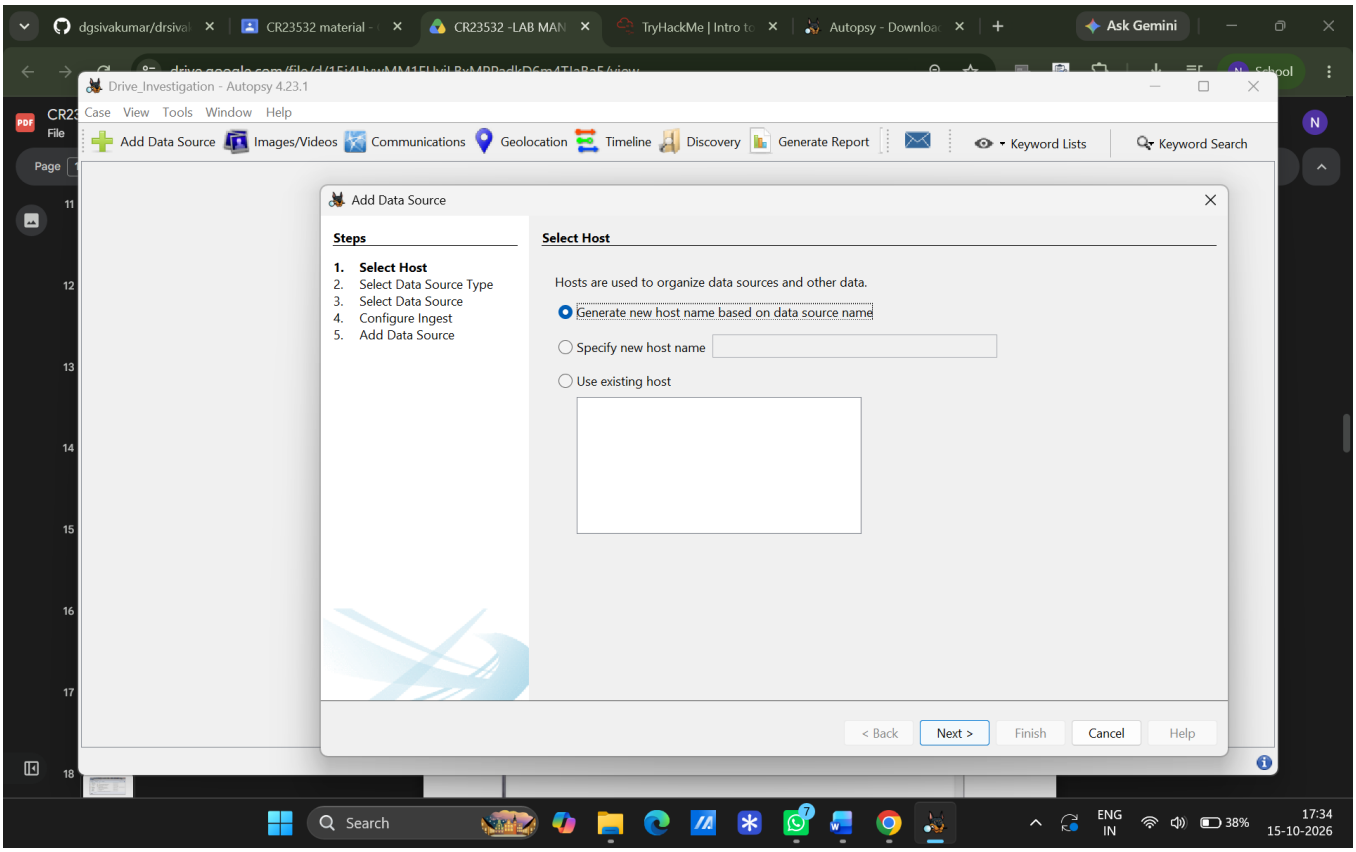
Procedure

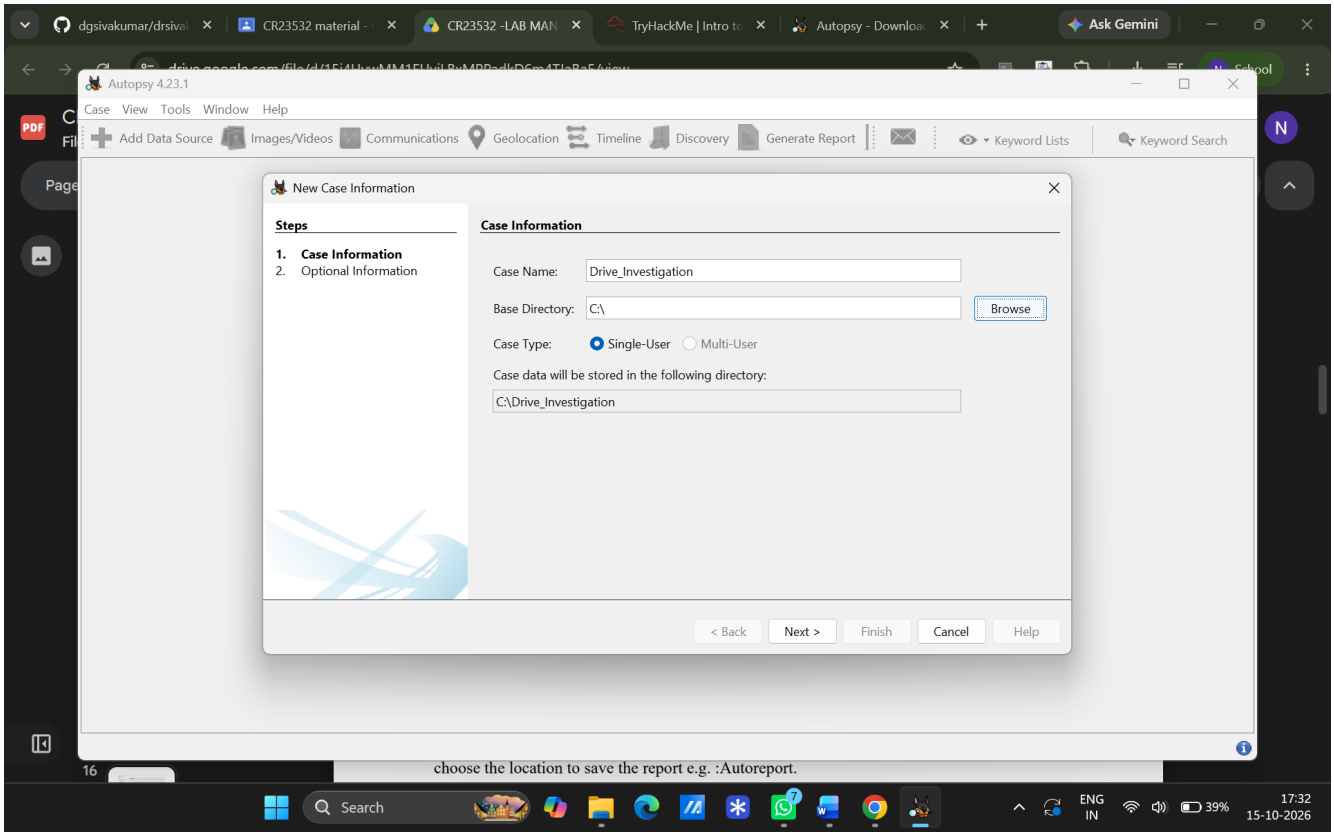
1. **Case Creation:** Launched Autopsy, selected **New Case**, and configured the *Case Name*, *Base Directory*, *Case Number*, and *Examiner Details*.
2. **Data Source Addition:** Added the target drive/disk path as the data source to the local database.
3. **Ingest Modules Configuration:** Enabled necessary ingest modules to scan and extract evidence.
4. **Evidence Analysis:** Processed and navigated through the generated investigation results.



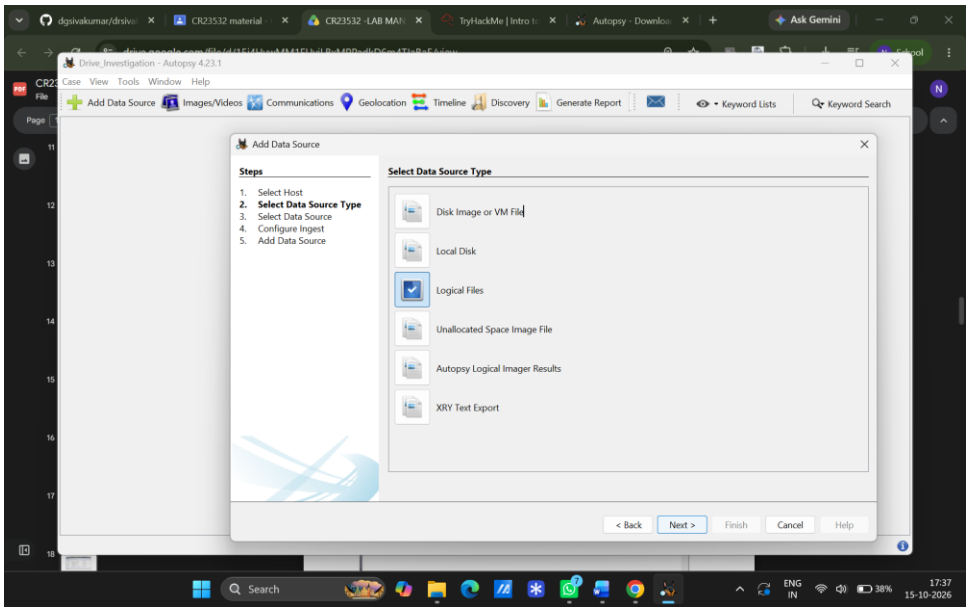

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number: 001
Examiner
Name: Naveen
Phone: 8248634103
Email: naveen0901@gmail.com
Notes: A drive investigation case
Organization
Organization analysis is being done for: Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help
choose the location to save the report e.g. :Autoreport.

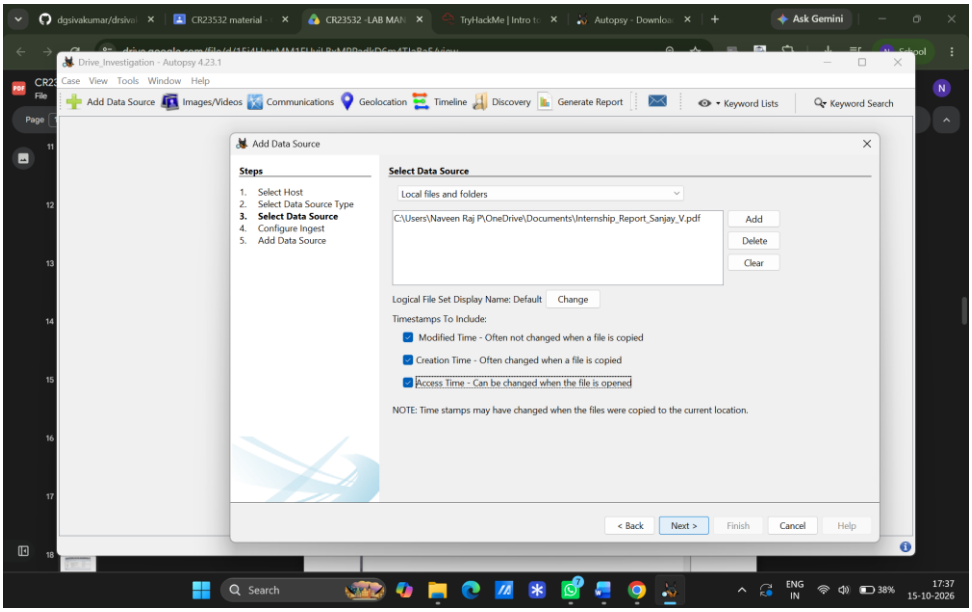

Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Keyword Lists
Keyword Search
New Case Information
Steps
1. Case Information
2. Optional Information
Case Information
Case Name: Drive_Investigation
Base Directory: C:\
Browse
Case Type: Single-User Multi-User
Case data will be stored in the following directory:
C:\Drive_Investigation
< Back
Next >
Finish
Cancel
Help
choose the location to save the report e.g. :Autoreport.

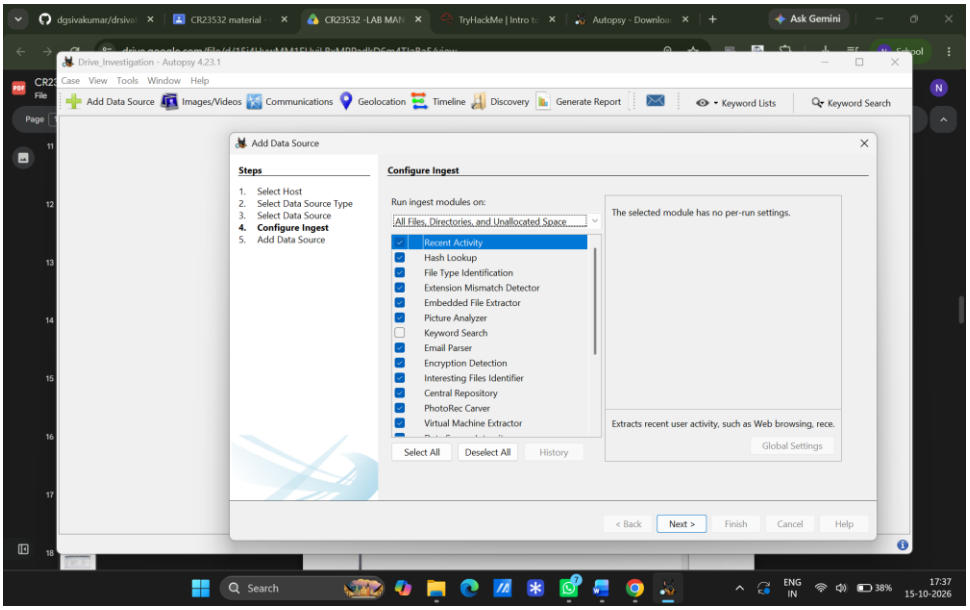


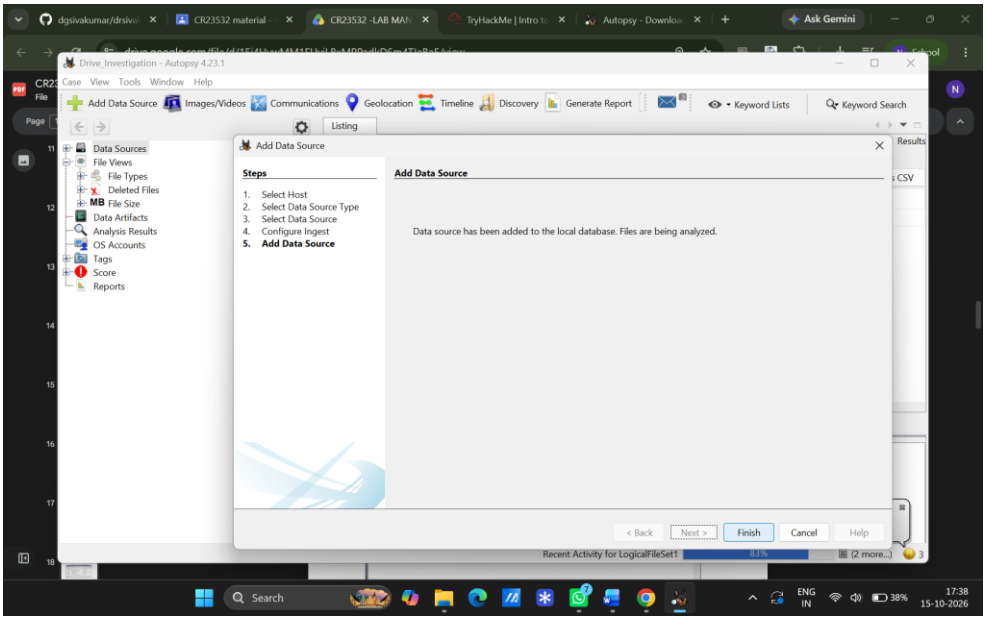


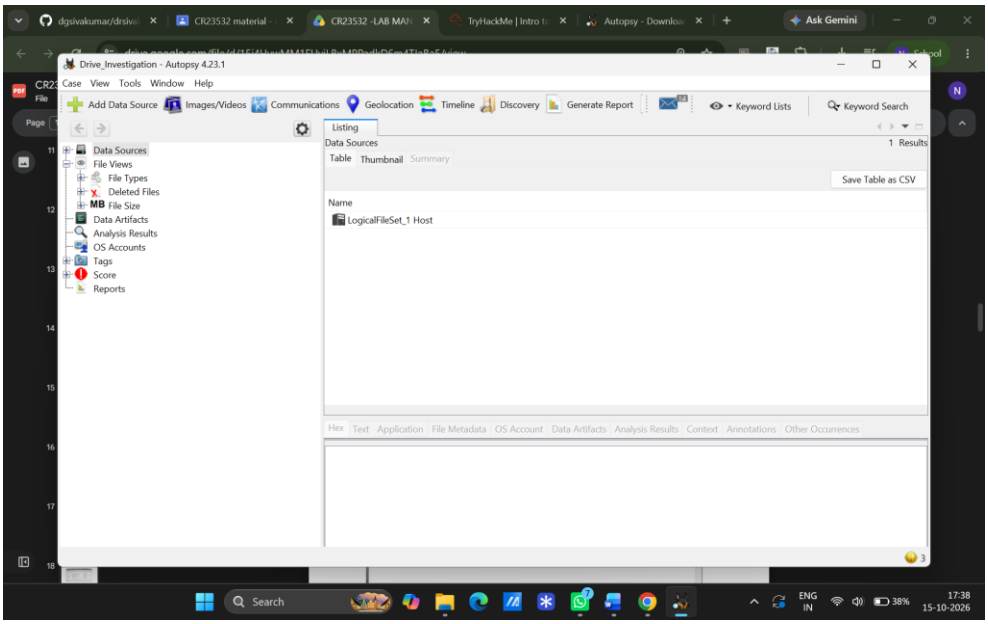
choose the location to save the report e.g. :Autoreport.

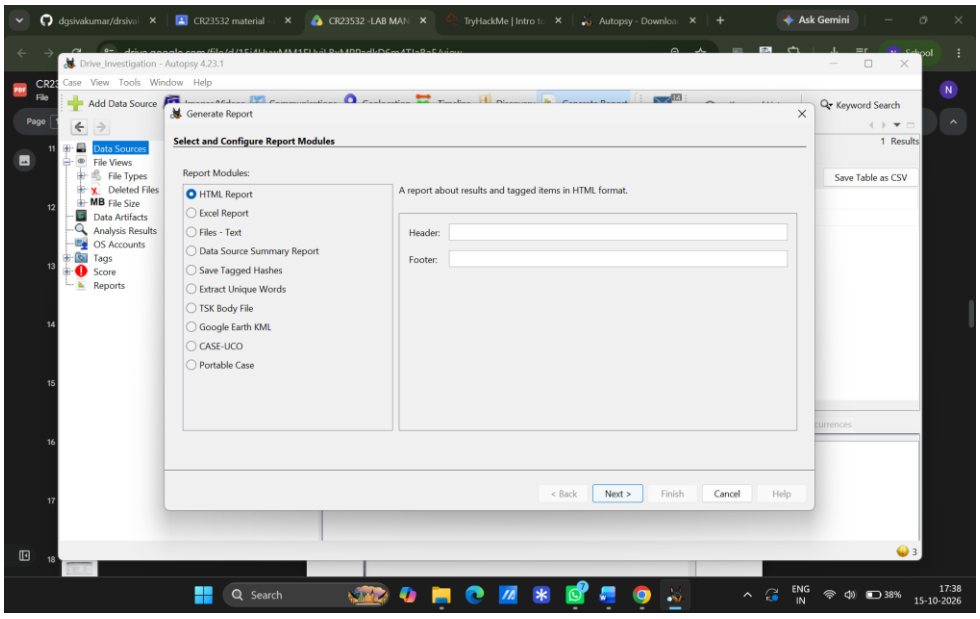

Drive_Investigation - Autopsy 4.23.1
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

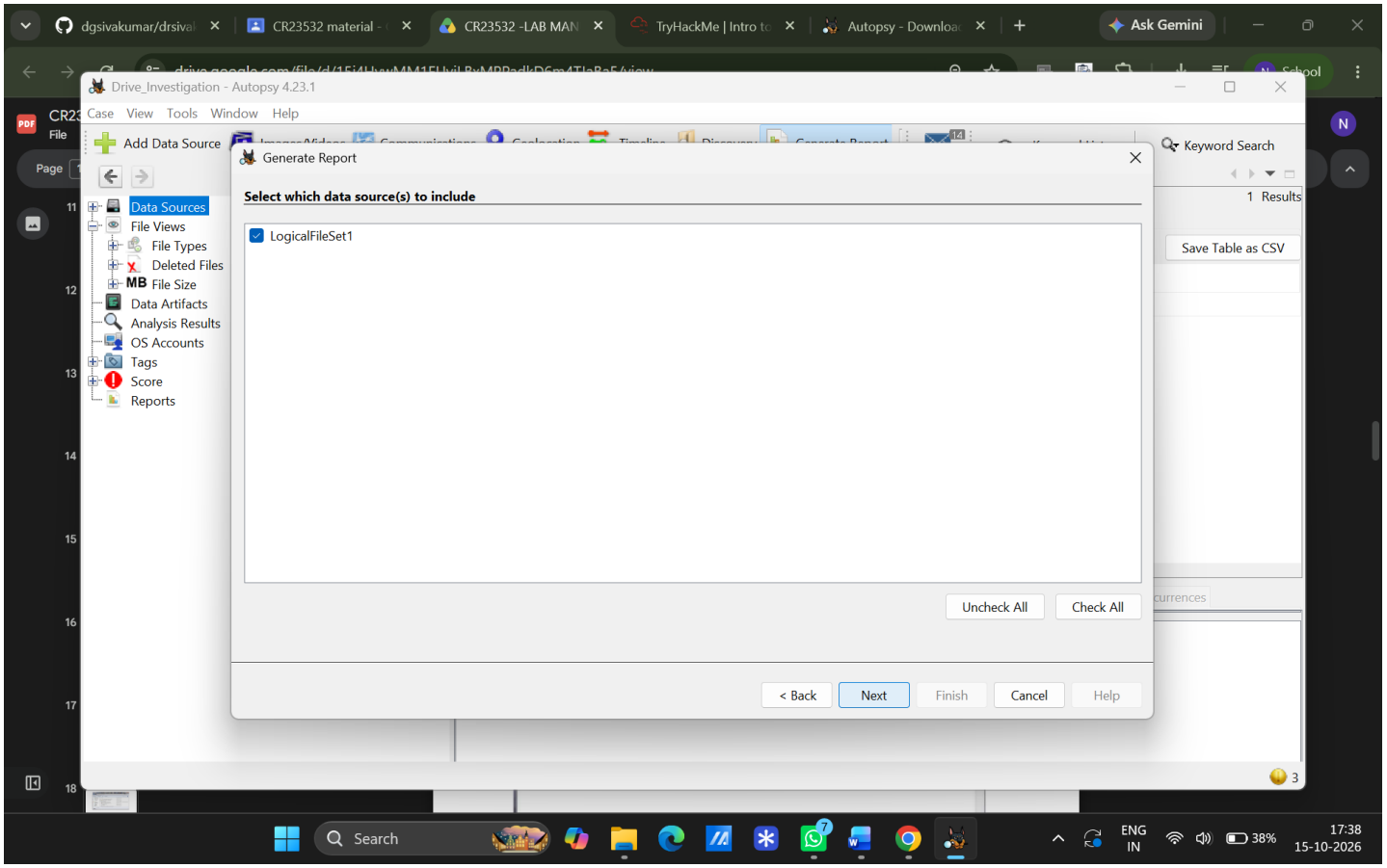

Drive_Investigation - Autopsy 4.23.1
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Local files and folders
C:\Users\Naveen Raj P\OneDrive\Documents\Internship_Report_Sanjay_V.pdf
Add
Delete
Clear
Logical File Set Display Name: Default
Change
Timestamps To Include:
Modified Time - Often not changed when a file is copied
Creation Time - Often changed when a file is copied
Access Time - Can be changed when the file is opened
NOTE: Time stamps may have changed when the files were copied to the current location.
< Back
Next >
Finish
Cancel
Help

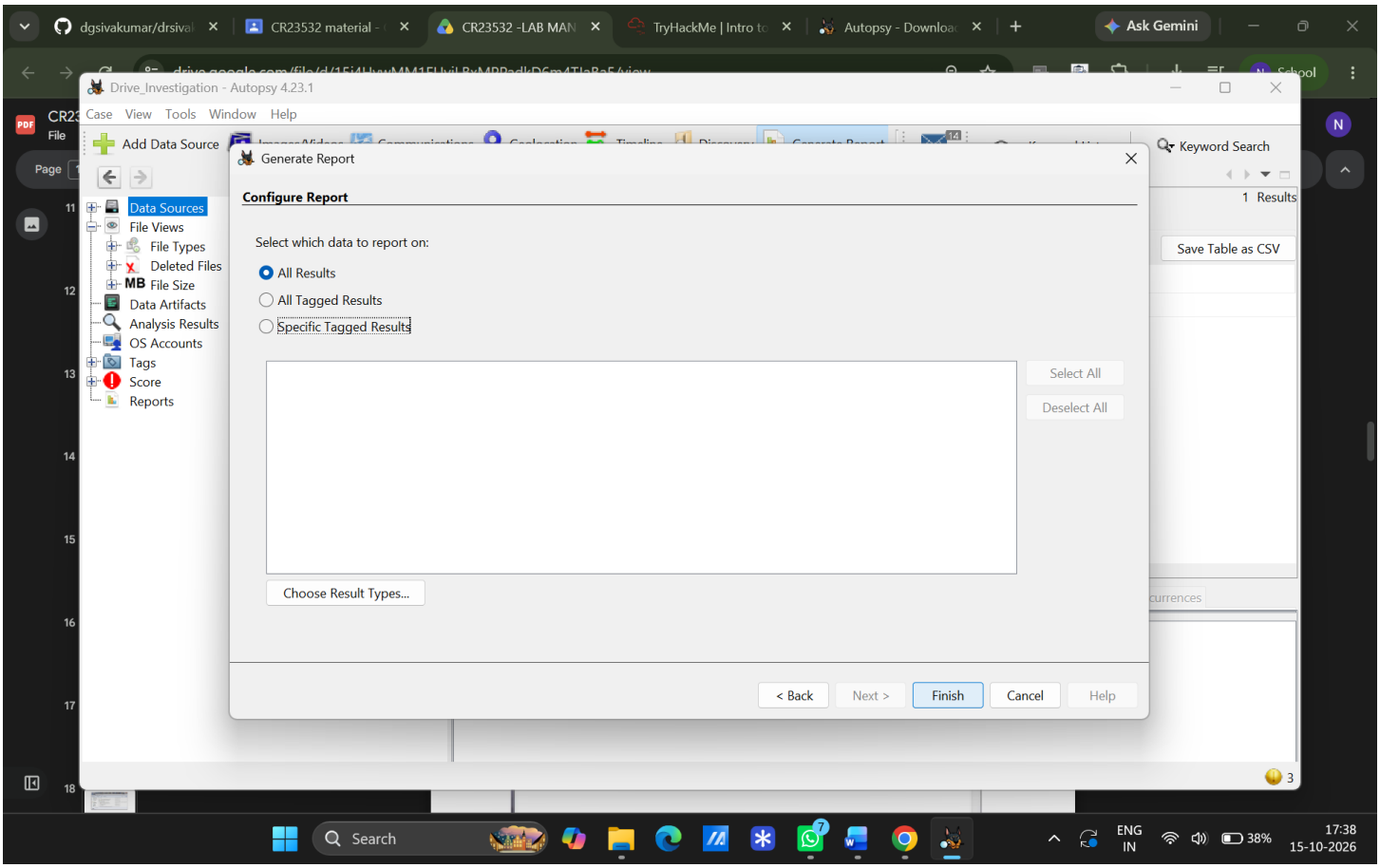

Drive_Investigation - Autopsy 4.23.1
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rece.
Global Settings
Select All
Deselect All
History
< Back
Next >
Finish
Cancel
Help

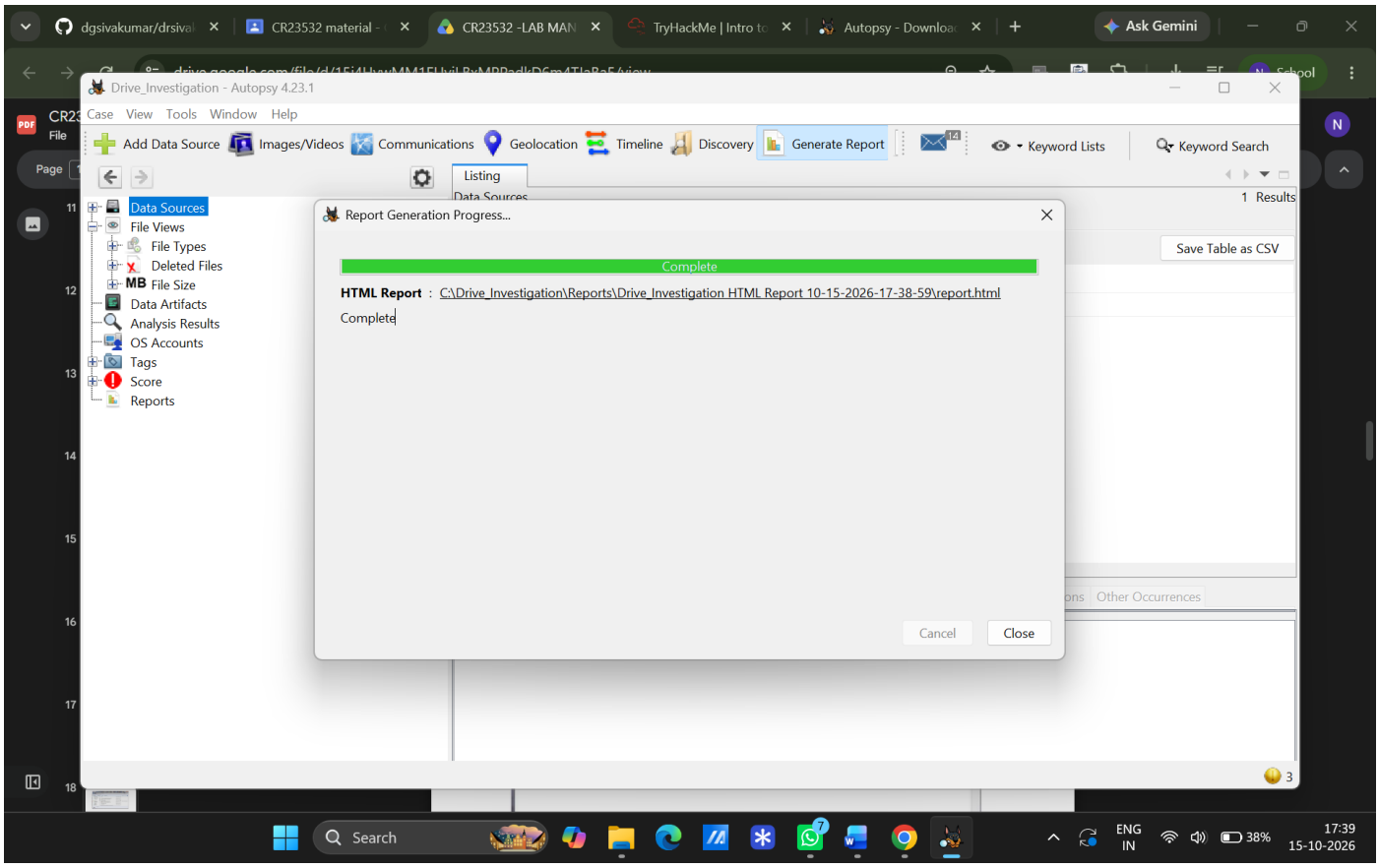

Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Keyword Lists
Keyword Search
Listing
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Data source has been added to the local database. Files are being analyzed.
< Back
Next >
Finish
Cancel
Help
Recent Activity for LogicalFileSet1
Search
ENG
IN
38%
17:38
15-10-2026

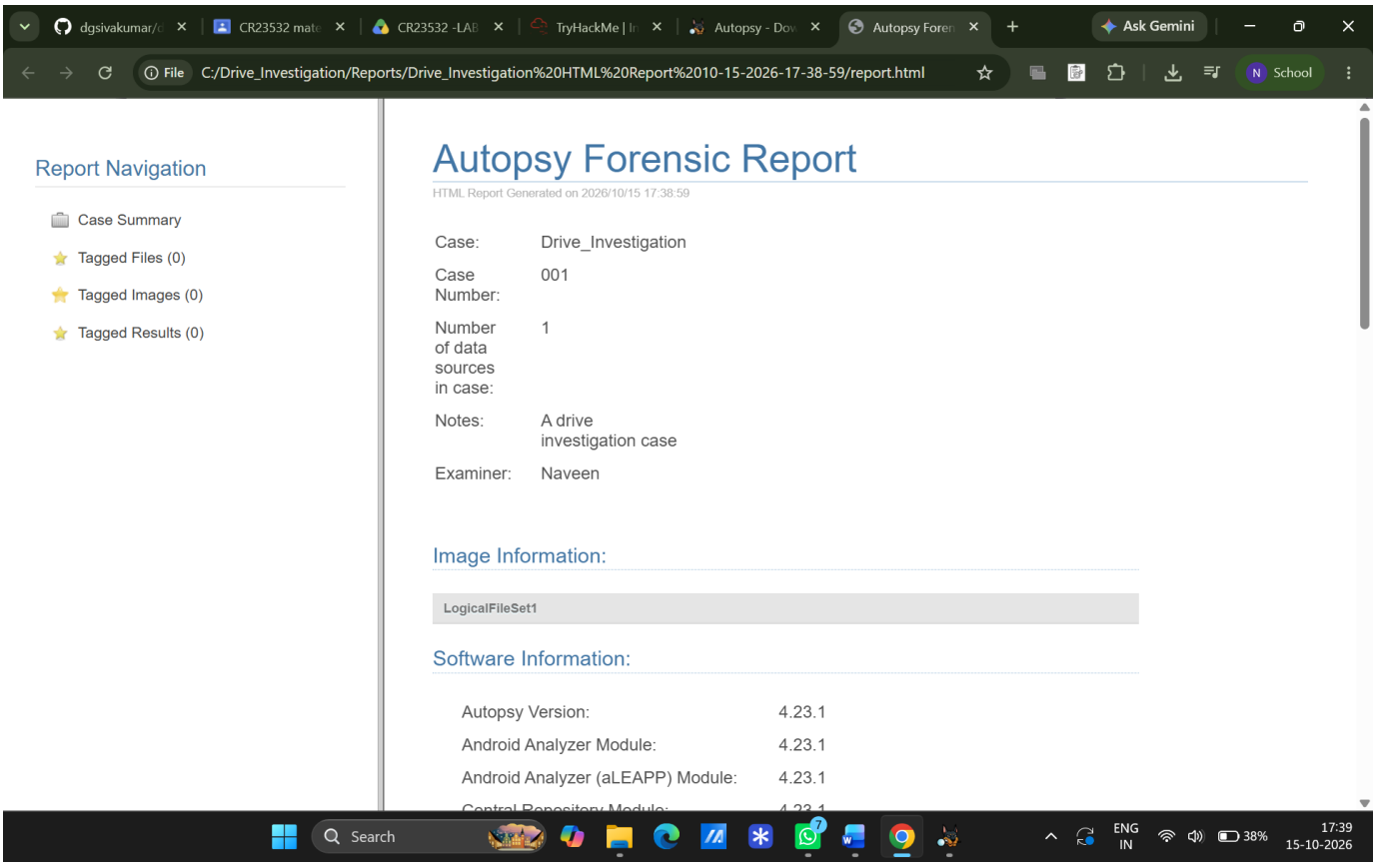

Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Keyword Lists
Keyword Search
Listing
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Data Sources
1 Results
Table
Thumbnail
Summary
Save Table as CSV
Name
LogicalFileSet_1 Host
Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences
Search
ENG
IN
38%
17:38
15-10-2026

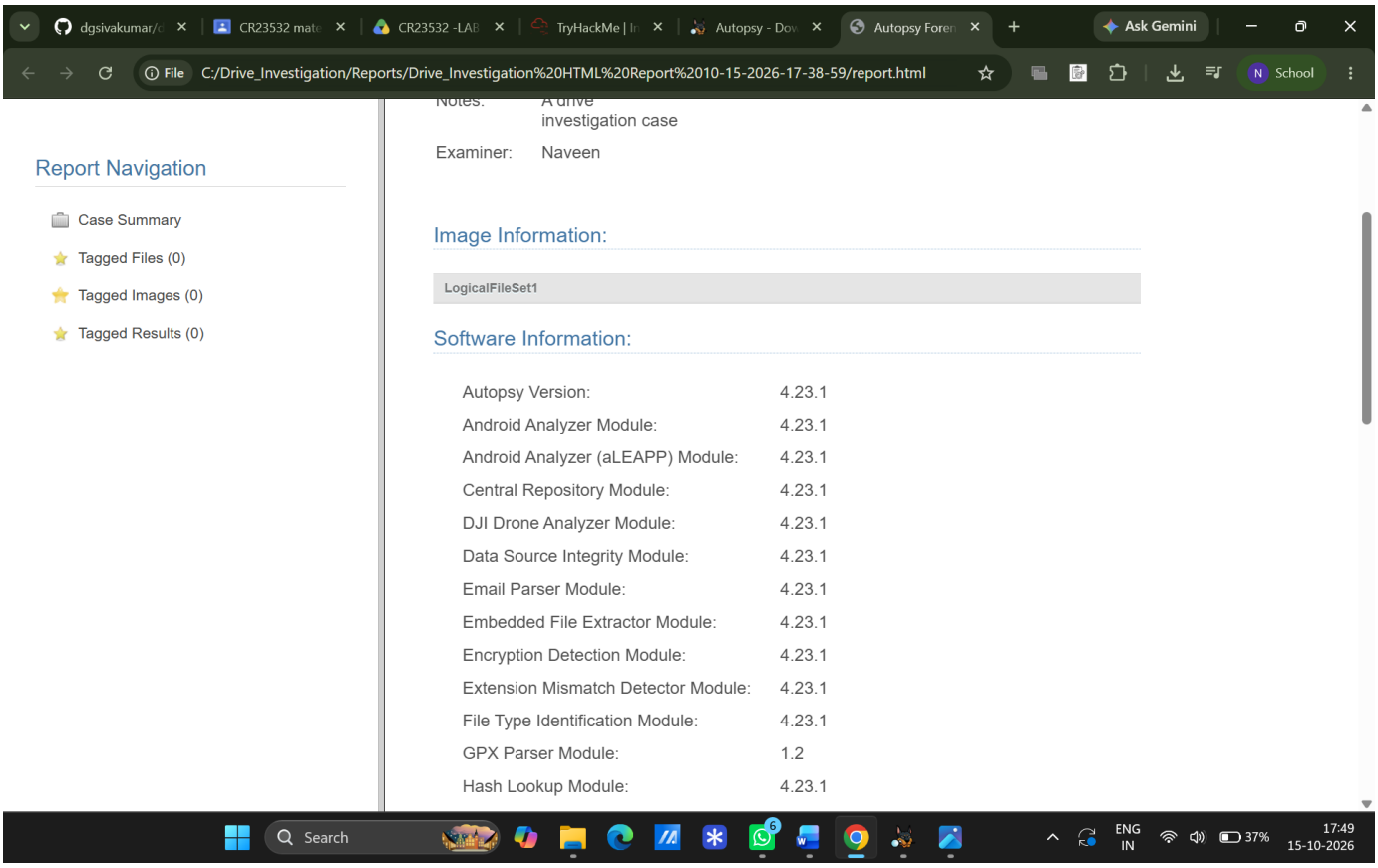

Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Keyword Search
Generate Report
Select and Configure Report Modules
Report Modules:
HTML Report
Excel Report
Files - Text
Data Source Summary Report
Save Tagged Hashes
Extract Unique Words
TSK Body File
Google Earth KML
CASE-UCO
Portable Case
A report about results and tagged items in HTML format.
Header:
Footer:
< Back
Next >
Finish
Cancel
Help
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
1 Results
Save Table as CSV
Search
ENG
IN
38%
17:38
15-10-2026

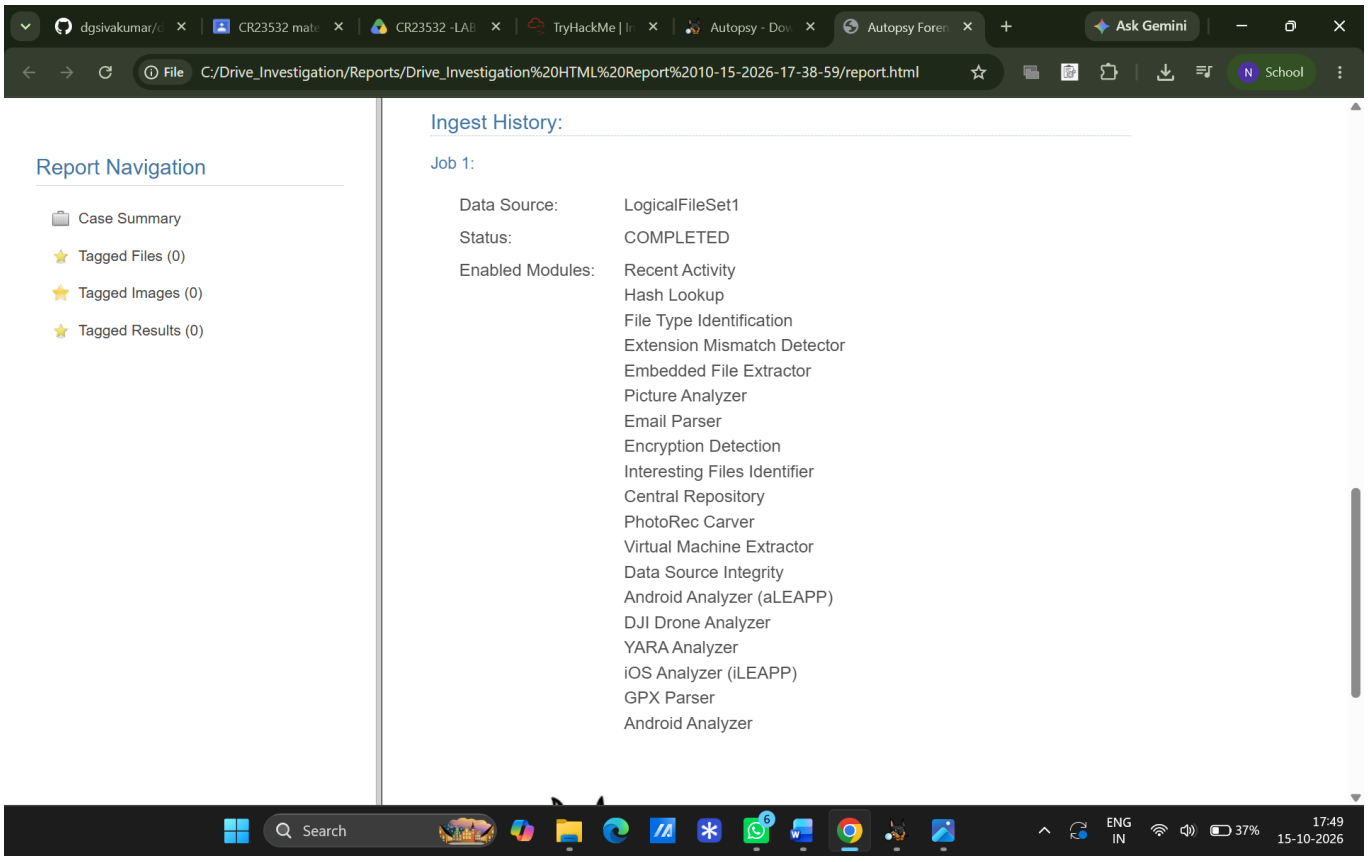

Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Keyword Search
Generate Report
Select which data source(s) to include
LogicalFileSet1
Uncheck All
Check All
< Back
Next
Finish
Cancel
Help
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
1 Results
Save Table as CSV


Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Keyword Search
Generate Report
Configure Report
Select which data to report on:
All Results
All Tagged Results
Specific Tagged Results
Select All
Deselect All
Choose Result Types...
< Back
Next >
Finish
Cancel
Help
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
1 Results
Save Table as CSV


Drive_Investigation - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source
Images/Videos
Communications
Geolocation
Timeline
Discovery
Generate Report
Keyword Lists
Keyword Search
Listing
Data Sources
1 Results
Save Table as CSV
Data Sources
File Views
File Types
Deleted Files
File Size
Data Artifacts
Analysis Results
OS Accounts
Tags
Score
Reports
Report Generation Progress...
Complete
HTML Report : C:\Drive_Investigation\Reports\Drive_Investigation HTML Report 10-15-2026-17-38-59\report.html
Complete
Cancel
Close
Other Occurrences
Search
ENG
IN
38%
17:39
15-10-2026


Ask Gemini
School
File C:/Drive_Investigation/Reports/Drive_Investigation%20HTML%20Report%2010-15-2026-17-38-59/report.html
Report Navigation
Case Summary
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Autopsy Forensic Report
HTML Report Generated on 2026/10/15 17:38:59
Case: Drive_Investigation
Case Number: 001
Number of data sources in case: 1
Notes: A drive investigation case
Examiner: Naveen
Image Information:
LogicalFileSet1
Software Information:
Autopsy Version: 4.23.1
Android Analyzer Module: 4.23.1
Android Analyzer (aLEAPP) Module: 4.23.1
Search
ENG
IN
38%
17:39
15-10-2026


Ask Gemini
File C:/Drive_Investigation/Reports/Drive_Investigation%20HTML%20Report%2010-15-2026-17-38-59/report.html
School
Notes: A drive investigation case
Examiner: Naveen
Report Navigation
Case Summary
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Image Information:
LogicalFileSet1
Software Information:
Autopsy Version: 4.23.1
Android Analyzer Module: 4.23.1
Android Analyzer (aLEAPP) Module: 4.23.1
Central Repository Module: 4.23.1
DJI Drone Analyzer Module: 4.23.1
Data Source Integrity Module: 4.23.1
Email Parser Module: 4.23.1
Embedded File Extractor Module: 4.23.1
Encryption Detection Module: 4.23.1
Extension Mismatch Detector Module: 4.23.1
File Type Identification Module: 4.23.1
GPX Parser Module: 1.2
Hash Lookup Module: 4.23.1
Search
ENG IN
37%
17:49
15-10-2026


Ask Gemini
File C:/Drive_Investigation/Reports/Drive_Investigation%20HTML%20Report%2010-15-2026-17-38-59/report.html
School
Report Navigation
Case Summary
Tagged Files (0)
Tagged Images (0)
Tagged Results (0)
Ingest History:
Job 1:
Data Source: LogicalFileSet1
Status: COMPLETED
Enabled Modules: Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
DJI Drone Analyzer
YARA Analyzer
iOS Analyzer (iLEAPP)
GPX Parser
Android Analyzer
Search
ENG IN
37%
17:49
15-10-2026

Observations & Artifacts Analyzed

Artifact Category	Information / Data Observed
System & Devices	Hardware and USB devices connected to the target system.
OS & User Accounts	Operating system details and list of active user accounts.
File Activity	Recent documents opened or created on the system.
Web & Browser Artifacts	Web bookmarks, cookies, web search queries, and browsing history.
Media Metadata	EXIF data from images and audio files.
Communication	Installed applications and discovered email addresses.

Result

The Autopsy tool was successfully executed, and digital evidence was captured, categorized, and analyzed.

EXPERIMENT 03

Recover Deleted Files using Forensics Tools

Aim

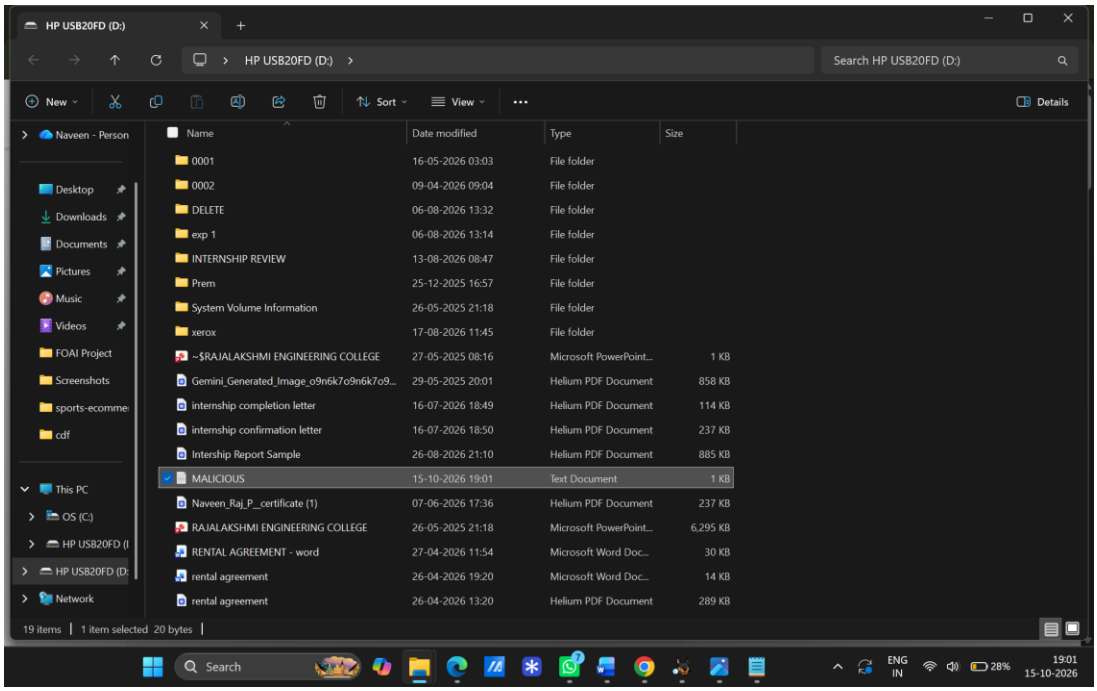
To practically demonstrate the process of recovering deleted files from a drive using digital forensic imaging and recovery tools.

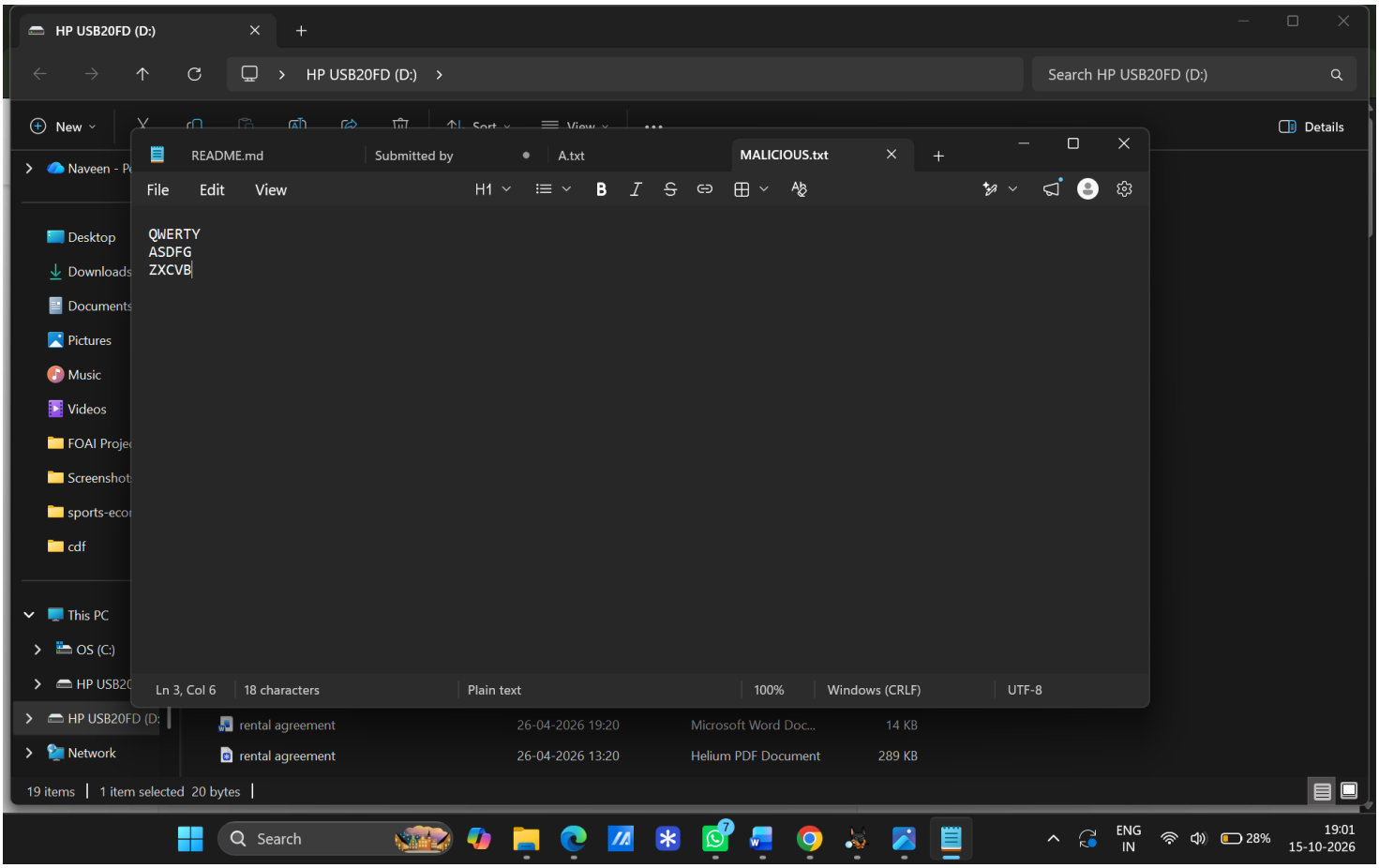
Software / Tools Required

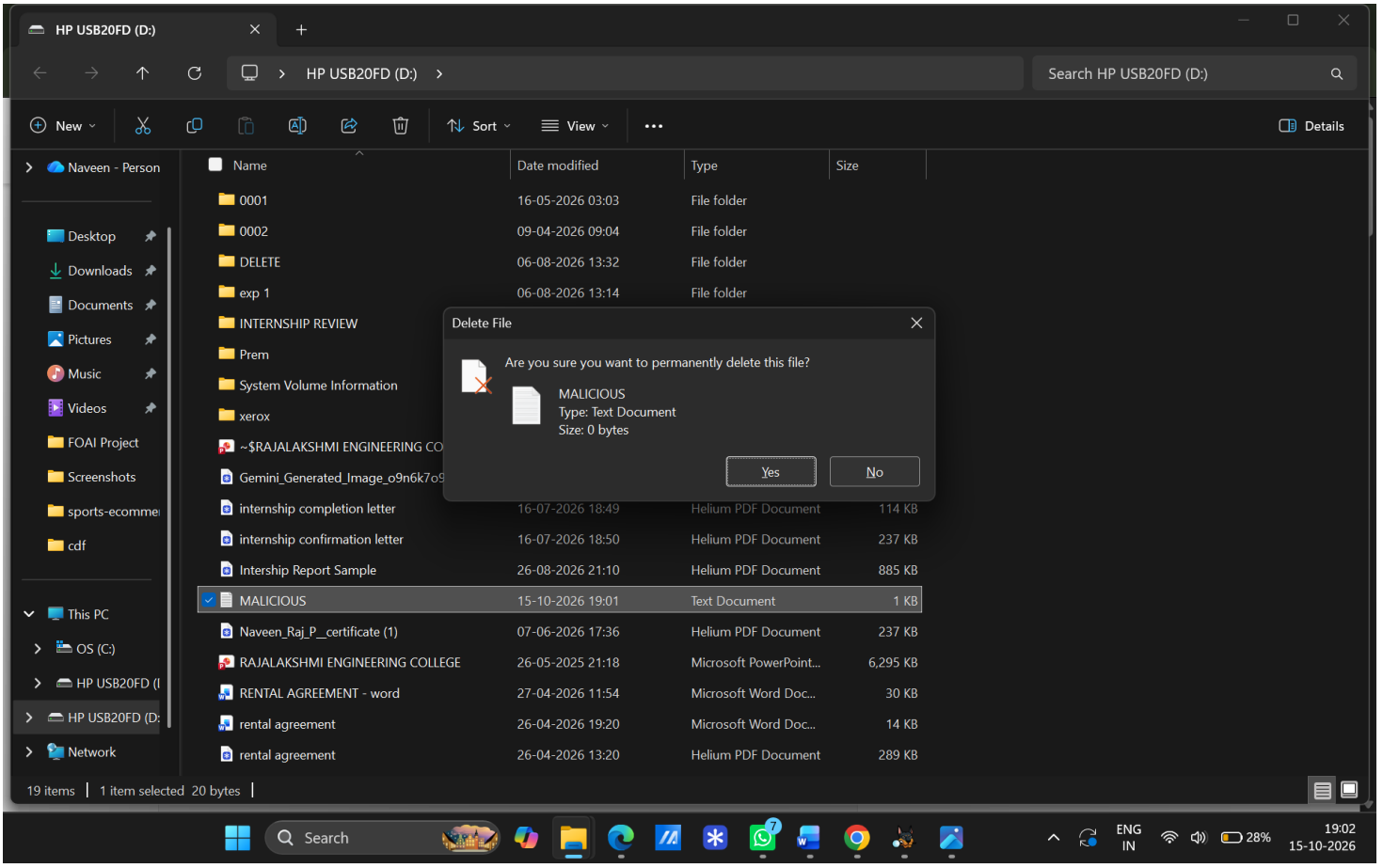
- **FTK Imager** (for bit-by-bit imaging)
- **RecoverMyFiles** (for scanning and recovery)
- Target Storage Device (e.g., USB Flash Drive)

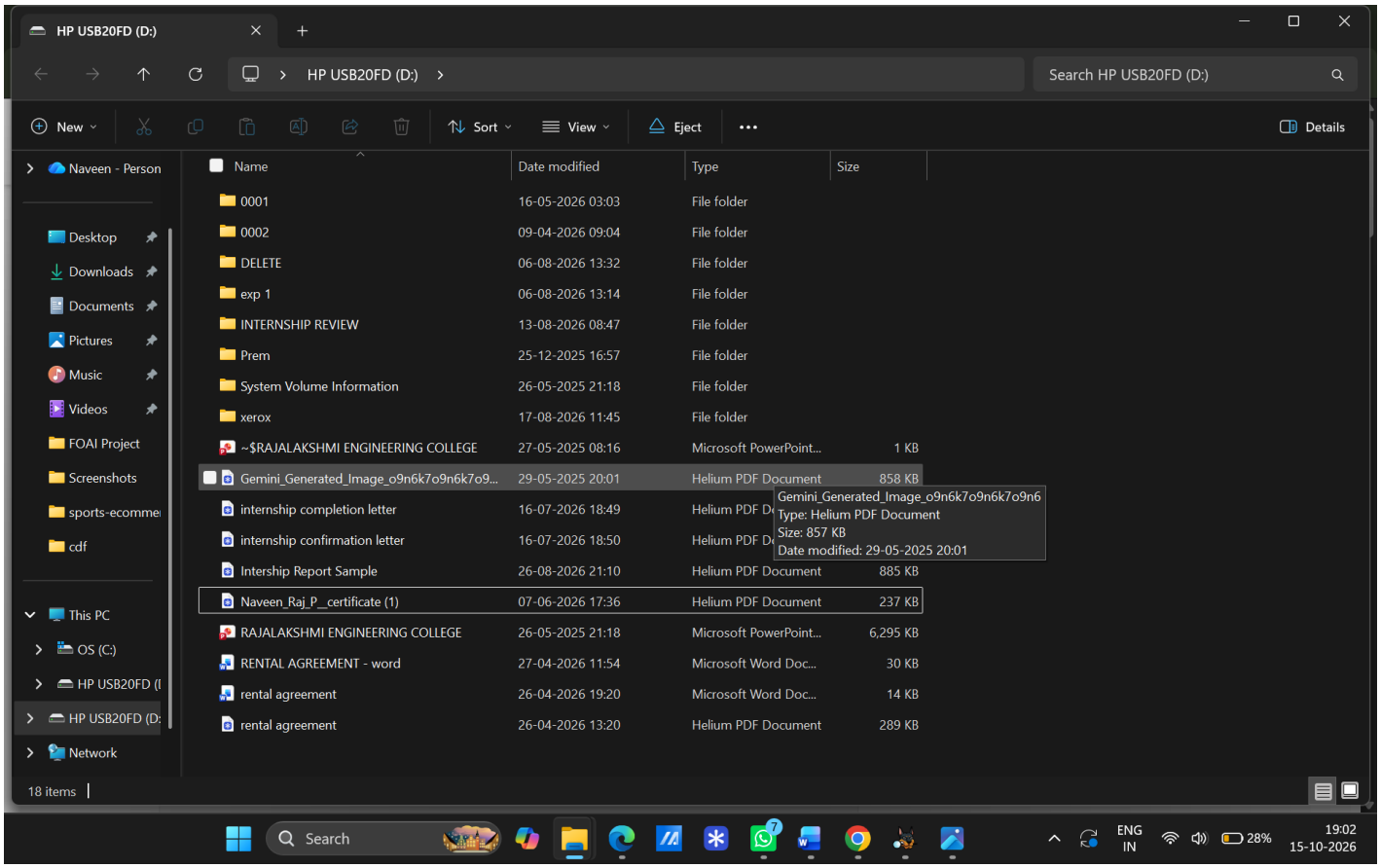
Procedure

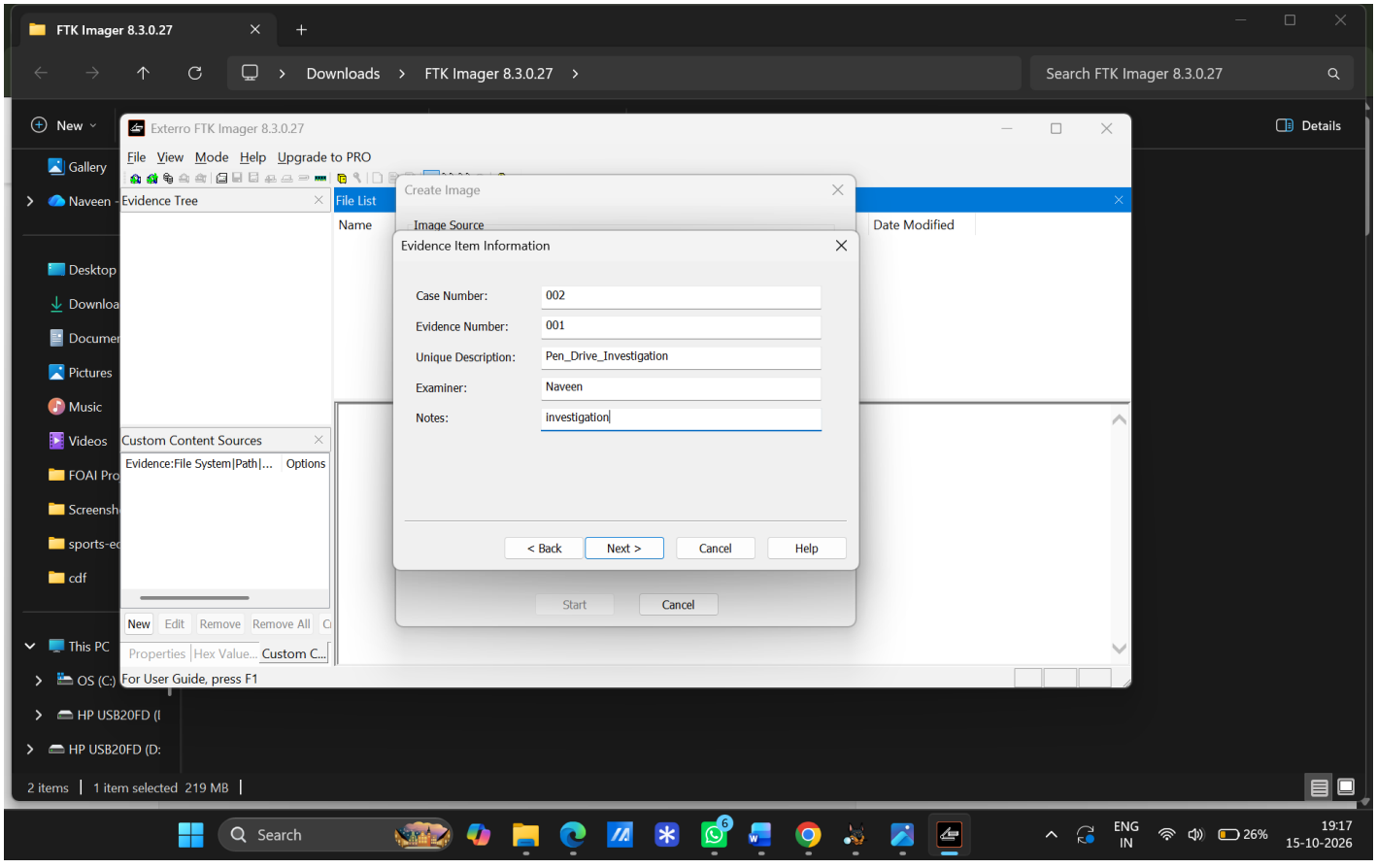
1. **Evidence Simulation:** Created a test text file (e.g., malicious.txt) on a flash drive and intentionally deleted it.
2. **Drive Imaging:** Used **FTK Imager** to select the physical drive and create a bit-by-bit forensic image.
3. **Data Recovery:** Opened **RecoverMyFiles**, selected "Recover a Drive", and loaded the created forensic image.
4. **File Extraction:** Executed the file recovery scan, filtered results by file type, and located the deleted file.

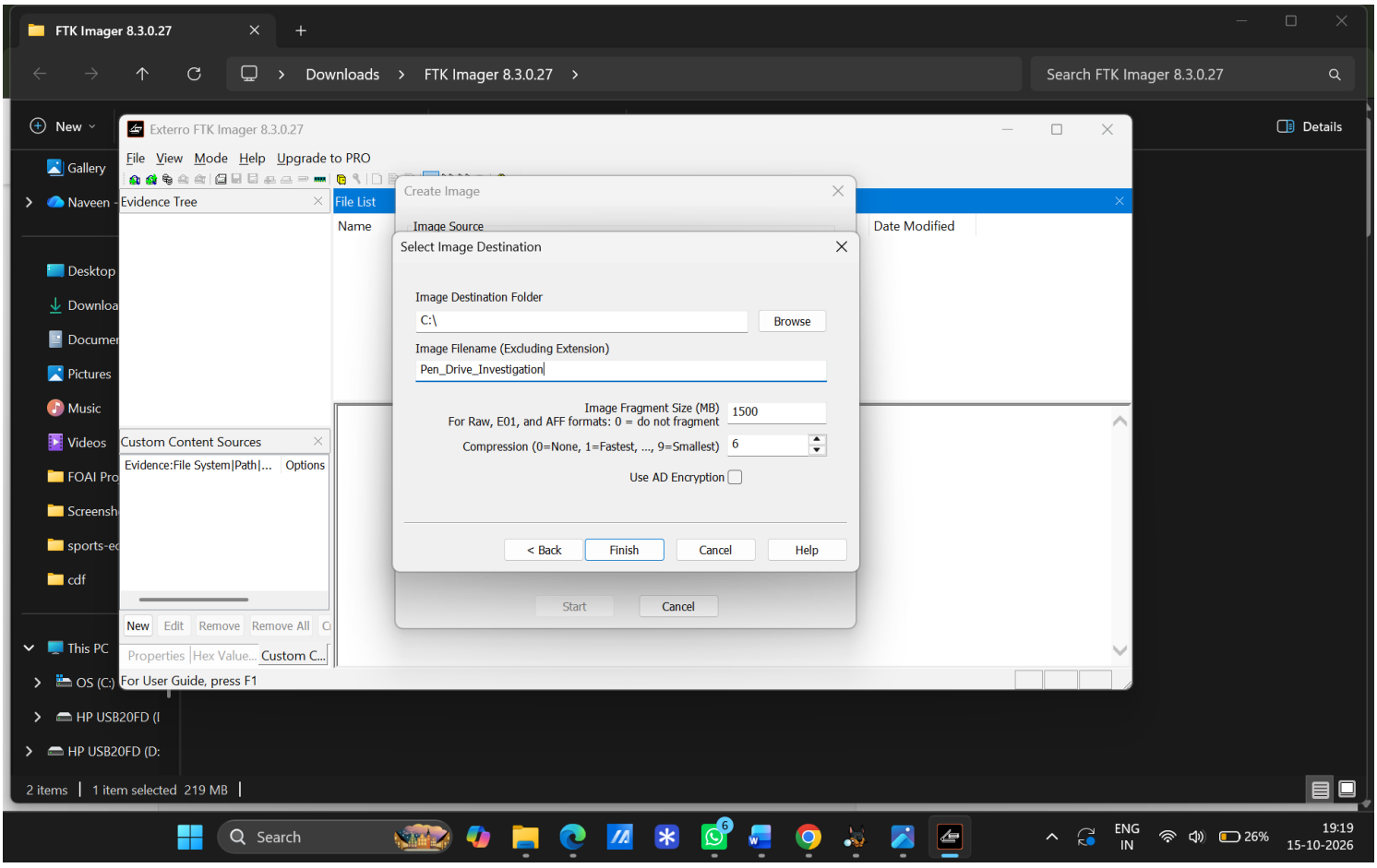


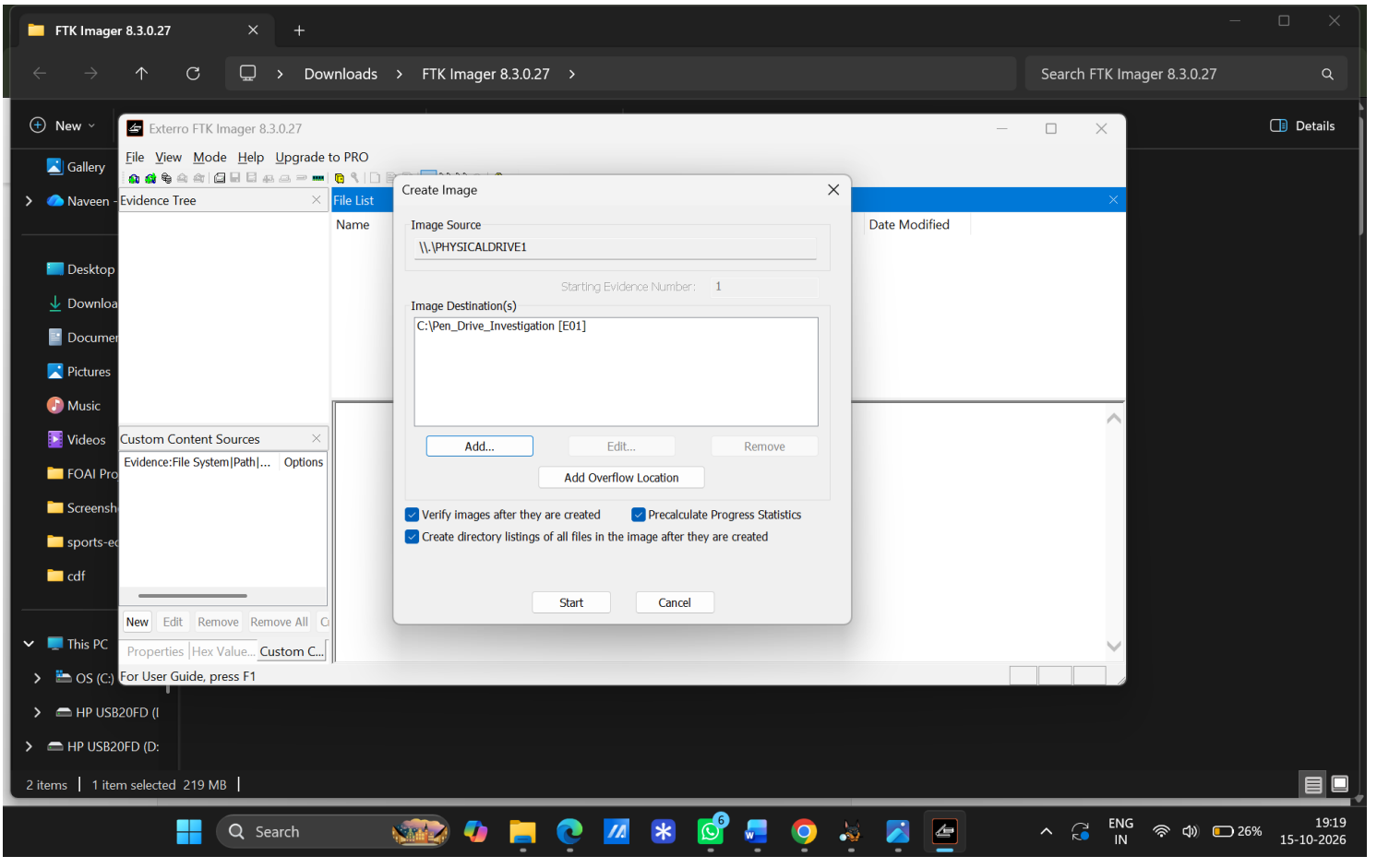

HP USB20FD (D:)
Search HP USB20FD (D:)
README.md
Submitted by
A.txt
MALICIOUS.txt
File Edit View
QWERTY
ASDFG
ZXCVB
Ln 3, Col 6
18 characters
Plain text
100%
Windows (CRLF)
UTF-8
rental agreement 26-04-2026 19:20 Microsoft Word Doc... 14 KB
rental agreement 26-04-2026 13:20 Helium PDF Document 289 KB
19 items | 1 item selected 20 bytes

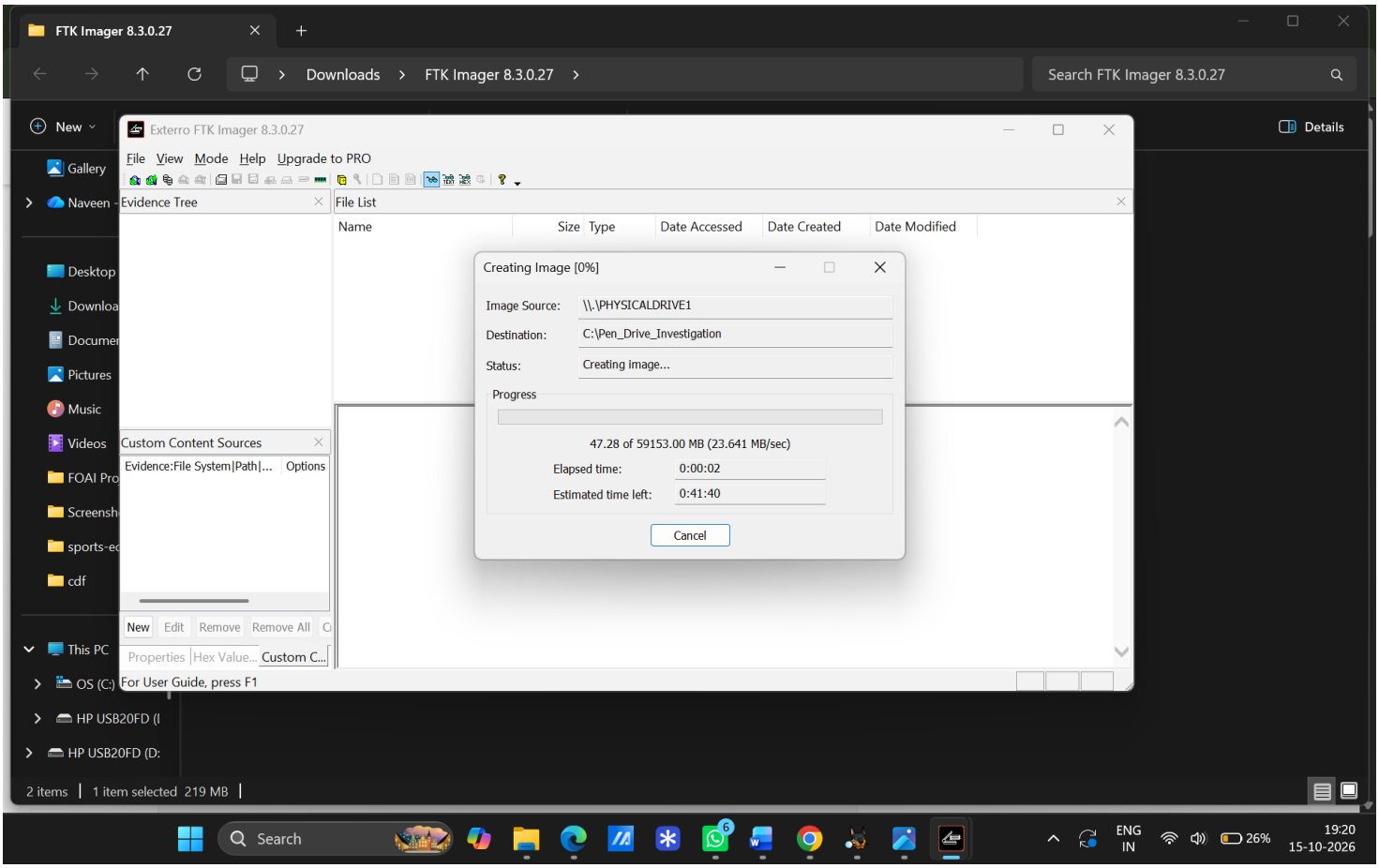

HP USB20FD (D:)
Search HP USB20FD (D:)
Name
Date modified
Type
Size
0001 16-05-2026 03:03 File folder
0002 09-04-2026 09:04 File folder
DELETE 06-08-2026 13:32 File folder
exp 1 06-08-2026 13:14 File folder
INTERNSHIP REVIEW
Prem
System Volume Information
xerox
~$RAJALAKSHMI ENGINEERING CO
Gemini_Generated_Image_o9n6k7o9
internship completion letter 16-07-2026 18:49 Helium PDF Document 114 KB
internship confirmation letter 16-07-2026 18:50 Helium PDF Document 237 KB
Intership Report Sample 26-08-2026 21:10 Helium PDF Document 885 KB
MALICIOUS 15-10-2026 19:01 Text Document 1 KB
Naveen_Raj_P_certificate (1) 07-06-2026 17:36 Helium PDF Document 237 KB
RAJALAKSHMI ENGINEERING COLLEGE 26-05-2025 21:18 Microsoft PowerPoint... 6,295 KB
RENTAL AGREEMENT - word 27-04-2026 11:54 Microsoft Word Doc... 30 KB
rental agreement 26-04-2026 19:20 Microsoft Word Doc... 14 KB
rental agreement 26-04-2026 13:20 Helium PDF Document 289 KB
Delete File
Are you sure you want to permanently delete this file?
MALICIOUS
Type: Text Document
Size: 0 bytes
Yes
No
19 items | 1 item selected 20 bytes

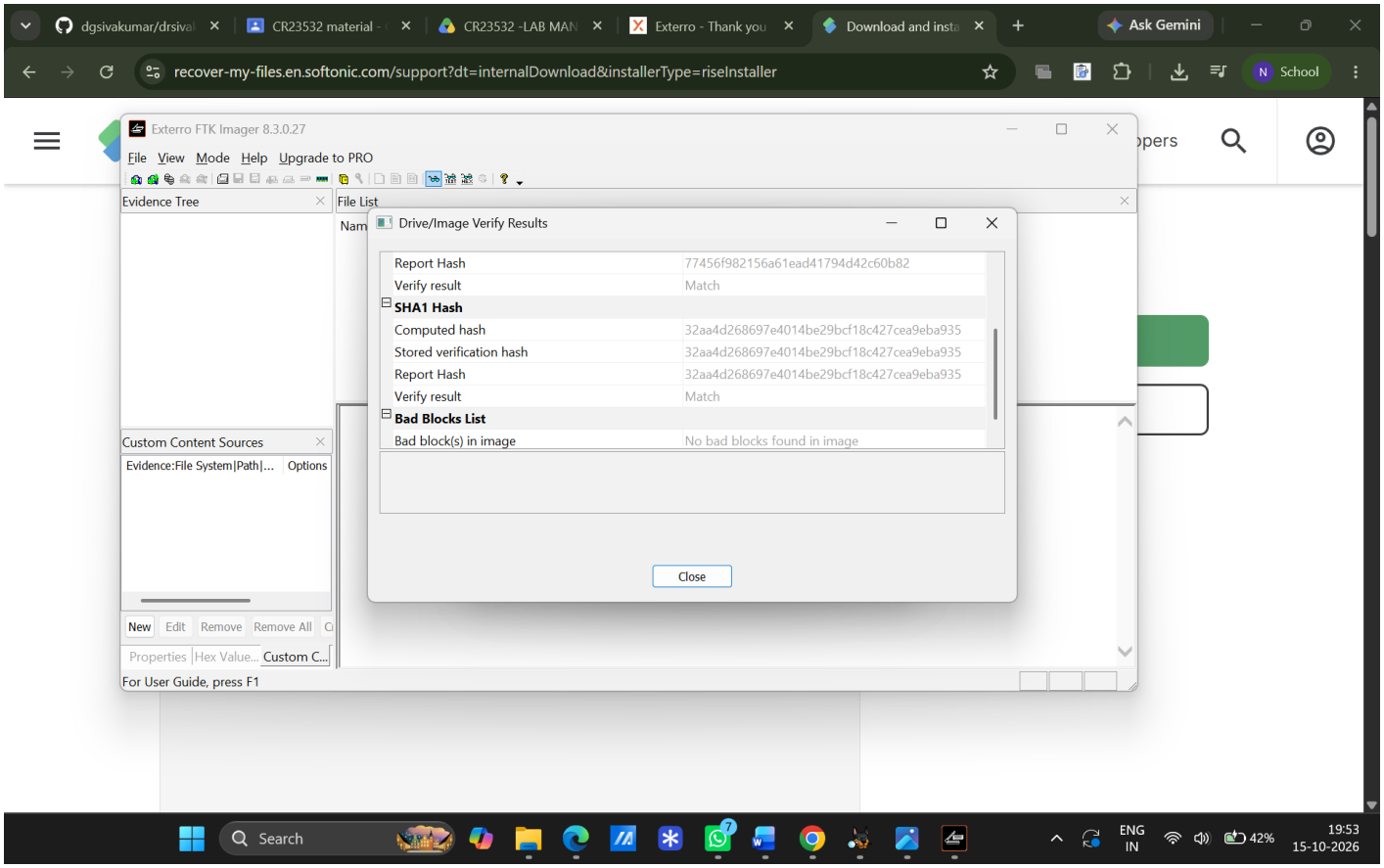

HP USB20FD (D:)
Search HP USB20FD (D:)
New
Sort
View
Eject
Details
Name
Date modified
Type
Size
0001 16-05-2026 03:03 File folder
0002 09-04-2026 09:04 File folder
DELETE 06-08-2026 13:32 File folder
exp 1 06-08-2026 13:14 File folder
INTERNSHIP REVIEW 13-08-2026 08:47 File folder
Prem 25-12-2025 16:57 File folder
System Volume Information 26-05-2025 21:18 File folder
xerox 17-08-2026 11:45 File folder
~$RAJALAKSHMI ENGINEERING COLLEGE 27-05-2025 08:16 Microsoft PowerPoint... 1 KB
Gemini_Generated_Image_o9n6k7o9n6k7o9... 29-05-2025 20:01 Helium PDF Document 858 KB
internship completion letter 16-07-2026 18:49 Helium PDF D
internship confirmation letter 16-07-2026 18:50 Helium PDF D
Intership Report Sample 26-08-2026 21:10 Helium PDF Document 885 KB
Naveen_Raj_P__certificate (1) 07-06-2026 17:36 Helium PDF Document 237 KB
RAJALAKSHMI ENGINEERING COLLEGE 26-05-2025 21:18 Microsoft PowerPoint... 6,295 KB
RENTAL AGREEMENT - word 27-04-2026 11:54 Microsoft Word Doc... 30 KB
rental agreement 26-04-2026 19:20 Microsoft Word Doc... 14 KB
rental agreement 26-04-2026 13:20 Helium PDF Document 289 KB
Gemini_Generated_Image_o9n6k7o9n6k7o9n6
Type: Helium PDF Document
Size: 857 KB
Date modified: 29-05-2025 20:01
18 items
Search
ENG IN
28%
19:02
15-10-2026

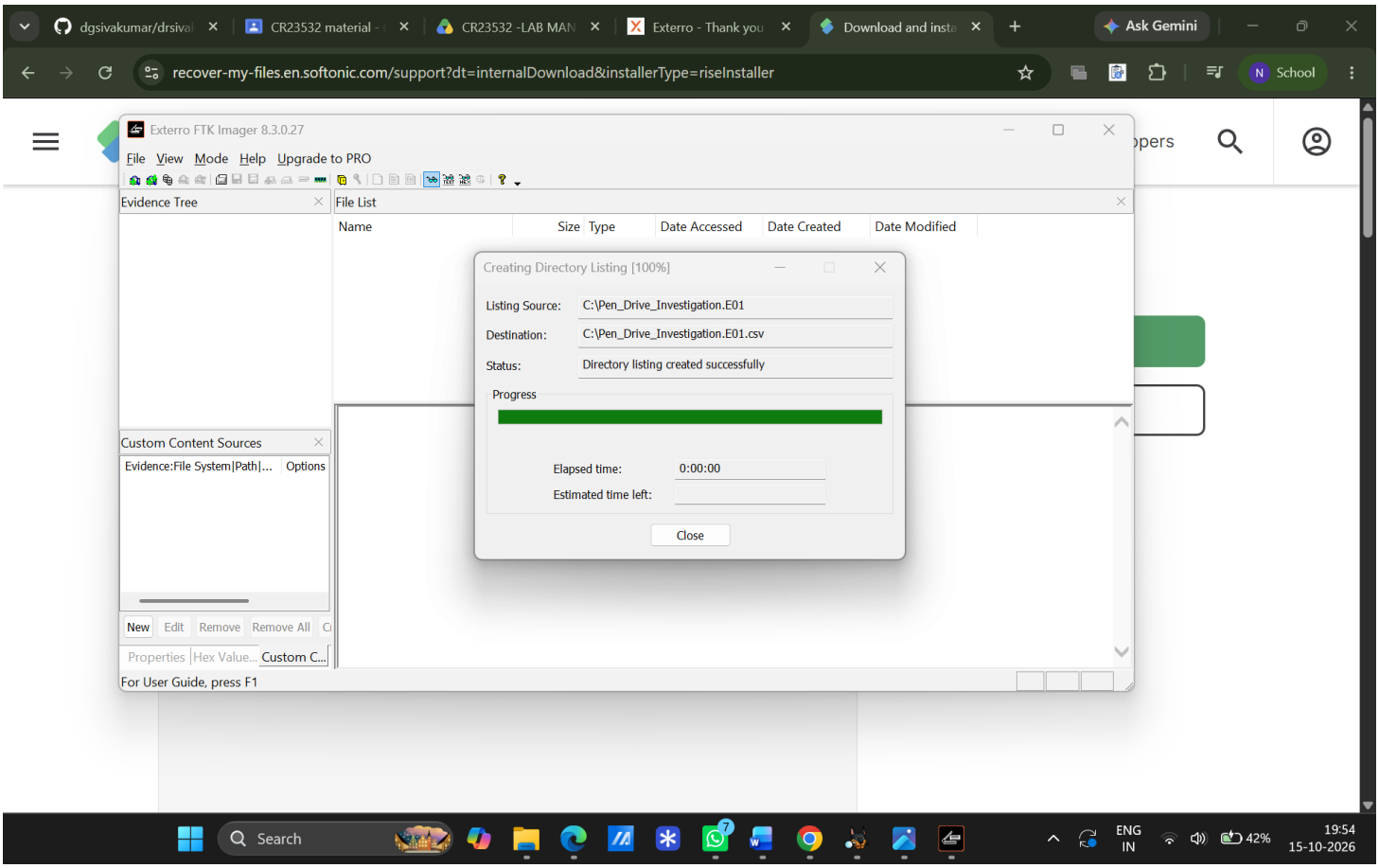

FTK Imager 8.3.0.27
Downloads
FTK Imager 8.3.0.27
Search FTK Imager 8.3.0.27
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name
Date Modified
Create Image
Image Source
Evidence Item Information
Case Number: 002
Evidence Number: 001
Unique Description: Pen_Drive_Investigation
Examiner: Naveen
Notes: investigation
< Back
Next >
Cancel
Help
Start
Cancel
Custom Content Sources
Evidence:File System|Path|... Options
New Edit Remove Remove All
Properties Hex Value... Custom C...
For User Guide, press F1
2 items | 1 item selected 219 MB
Search
ENG IN
26%
19:17
15-10-2026

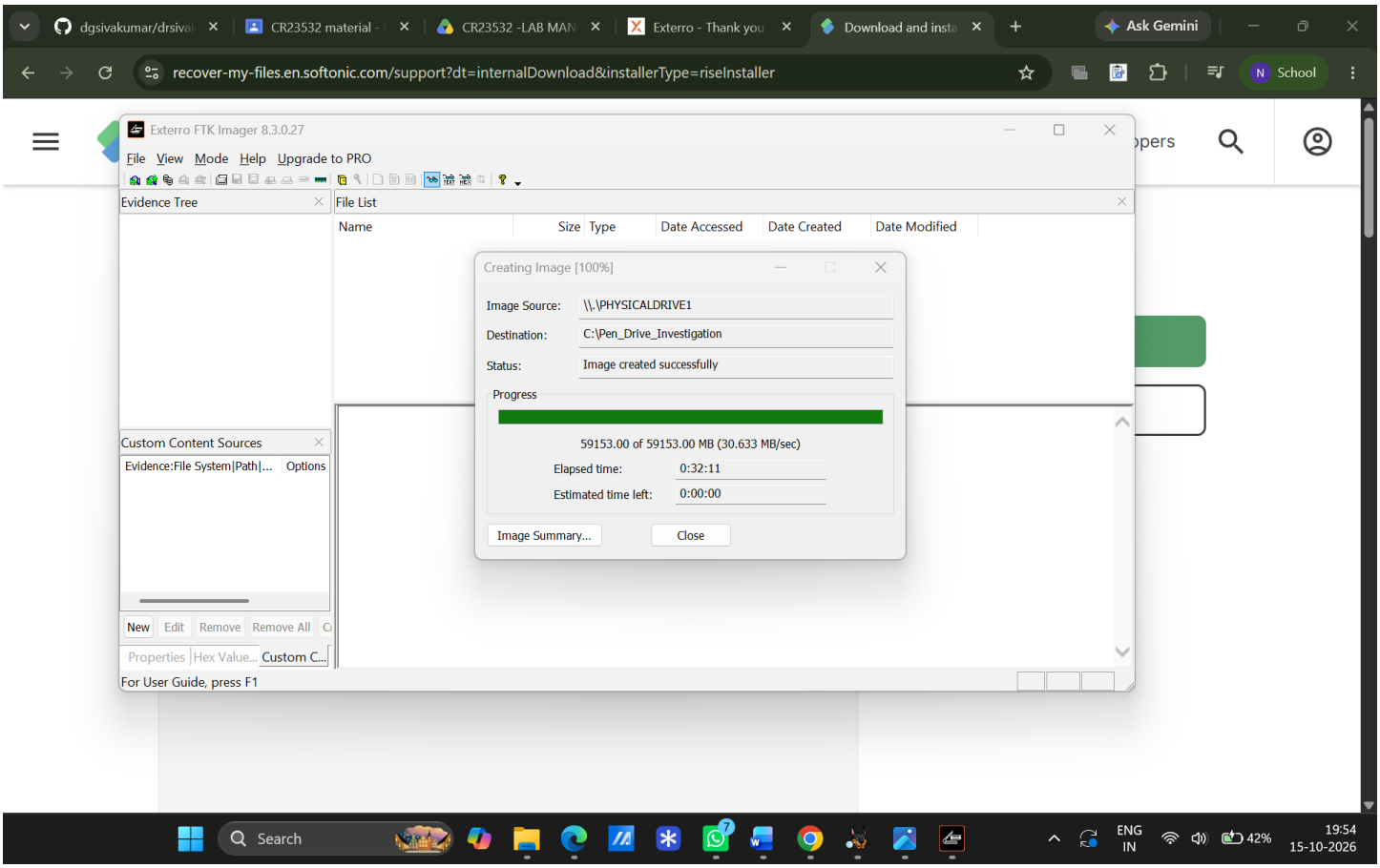

FTK Imager 8.3.0.27
Downloads
FTK Imager 8.3.0.27
Search FTK Imager 8.3.0.27
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name
Date Modified
Create Image
Image Source
Select Image Destination
Image Destination Folder
C:\
Browse
Image Filename (Excluding Extension)
Pen_Drive_Investigation
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
1500
Compression (0=None, 1=Fastest, ..., 9=Smallest)
6
Use AD Encryption
< Back
Finish
Cancel
Help
Start
Cancel
Custom Content Sources
Evidence:File System|Path|...
Options
New
Edit
Remove
Remove All
Properties
Hex Value...
Custom C...
For User Guide, press F1
2 items | 1 item selected 219 MB
Search
ENG
IN
26%
19:19
15-10-2026

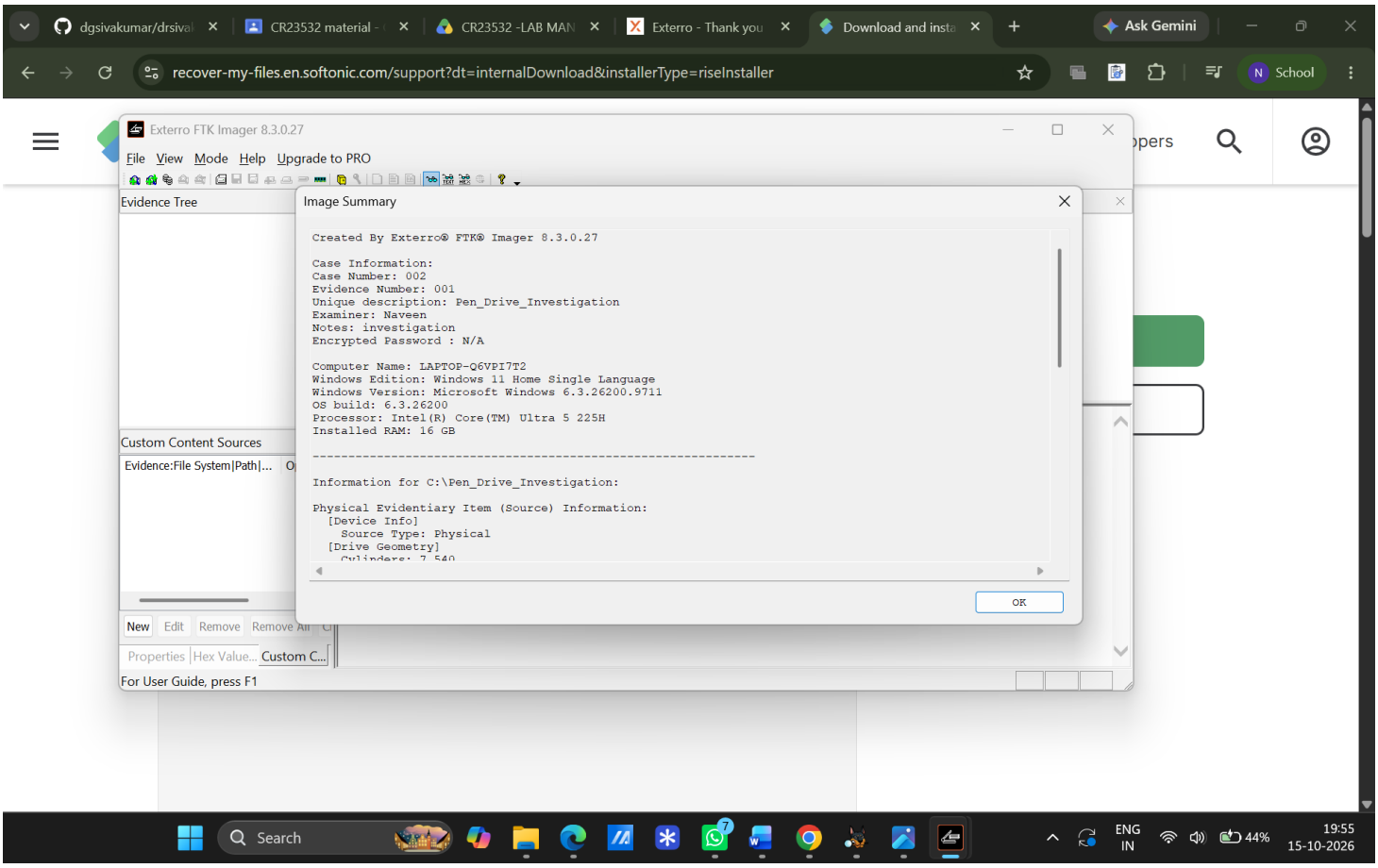

FTK Imager 8.3.0.27
Downloads
FTK Imager 8.3.0.27
Search FTK Imager 8.3.0.27
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name
Date Modified
Create Image
Image Source
\\.\PHYSICALDRIVE1
Starting Evidence Number:
1
Image Destination(s)
C:\Pen_Drive_Investigation [E01]
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel
Custom Content Sources
Evidence:File System|Path|...
Options
New
Edit
Remove
Remove All
Properties
Hex Value...
Custom C...
For User Guide, press F1
2 items | 1 item selected 219 MB
Search
ENG
IN
26%
19:19
15-10-2026

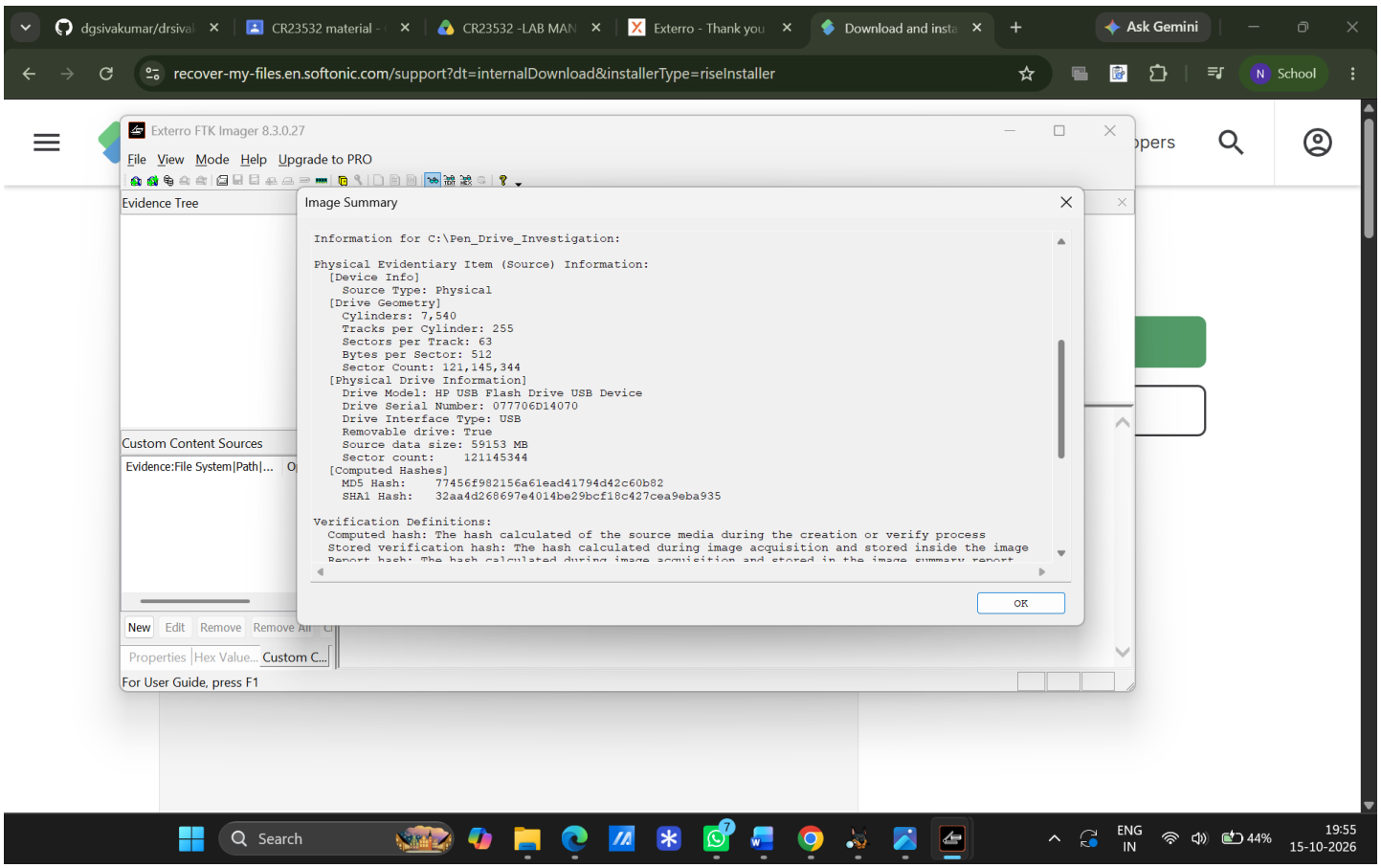

FTK Imager 8.3.0.27
Downloads
FTK Imager 8.3.0.27
Search FTK Imager 8.3.0.27
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name Size Type Date Accessed Date Created Date Modified
Creating Image [0%]
Image Source: \\.\PHYSICALDRIVE1
Destination: C:\Pen_Drive_Investigation
Status: Creating image...
Progress
47.28 of 59153.00 MB (23.641 MB/sec)
Elapsed time: 0:00:02
Estimated time left: 0:41:40
Cancel
Custom Content Sources
Evidence:File System|Path|... Options
New Edit Remove Remove All
Properties Hex Value... Custom C...
For User Guide, press F1
2 items | 1 item selected 219 MB
Search
ENG IN
26%
19:20
15-10-2026

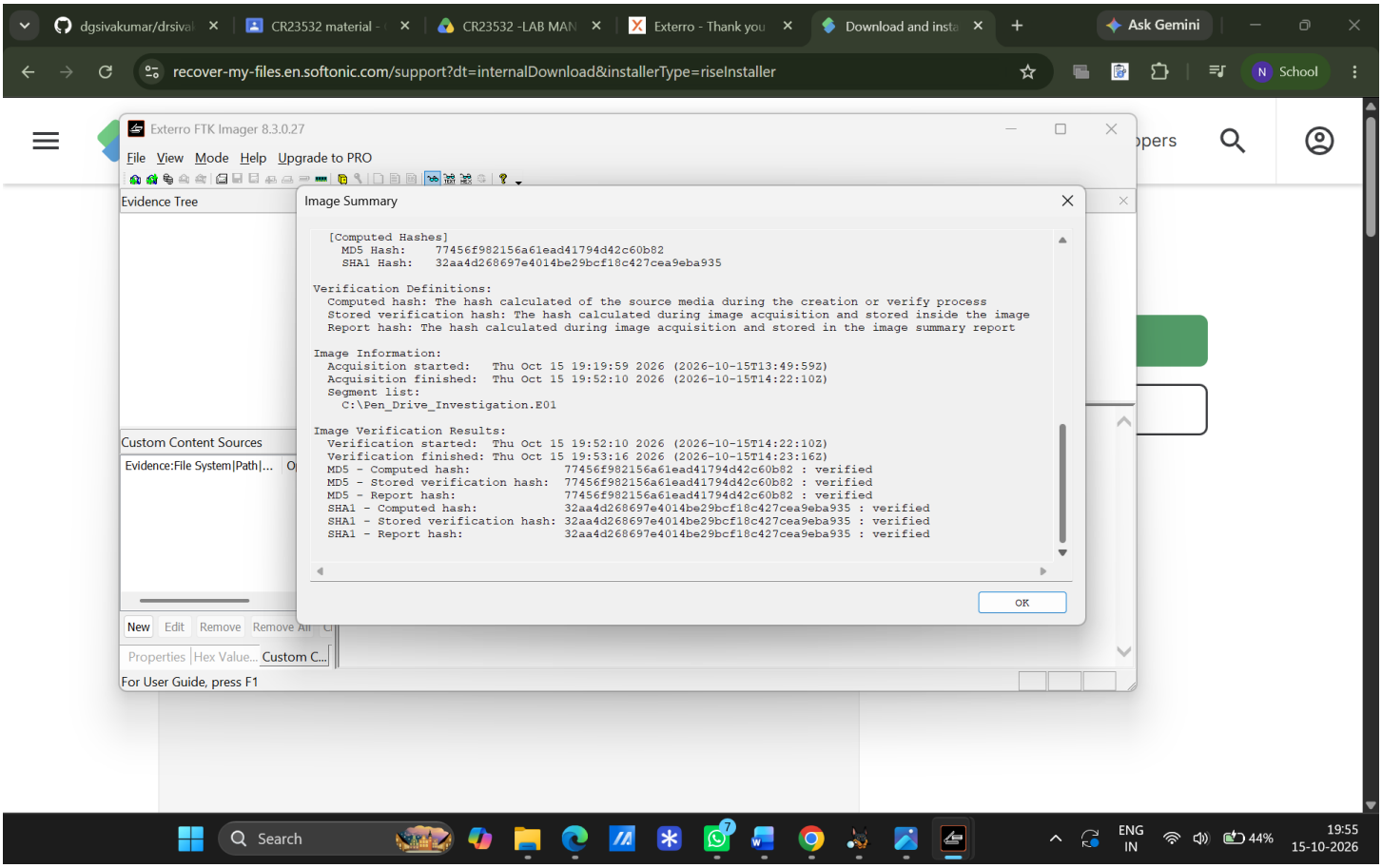

recover-my-files.en.softonic.com/support?dt=internalDownload&installerType=riseInstaller
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Drive/Image Verify Results
Report Hash 77456f982156a61ead41794d42c60b82
Verify result Match
SHA1 Hash
Computed hash 32aa4d268697e4014be29bcf18c427cea9eba935
Stored verification hash 32aa4d268697e4014be29bcf18c427cea9eba935
Report Hash 32aa4d268697e4014be29bcf18c427cea9eba935
Verify result Match
Bad Blocks List
Bad block(s) in image No bad blocks found in image
Close
Search
ENG IN
42%
19:53
15-10-2026

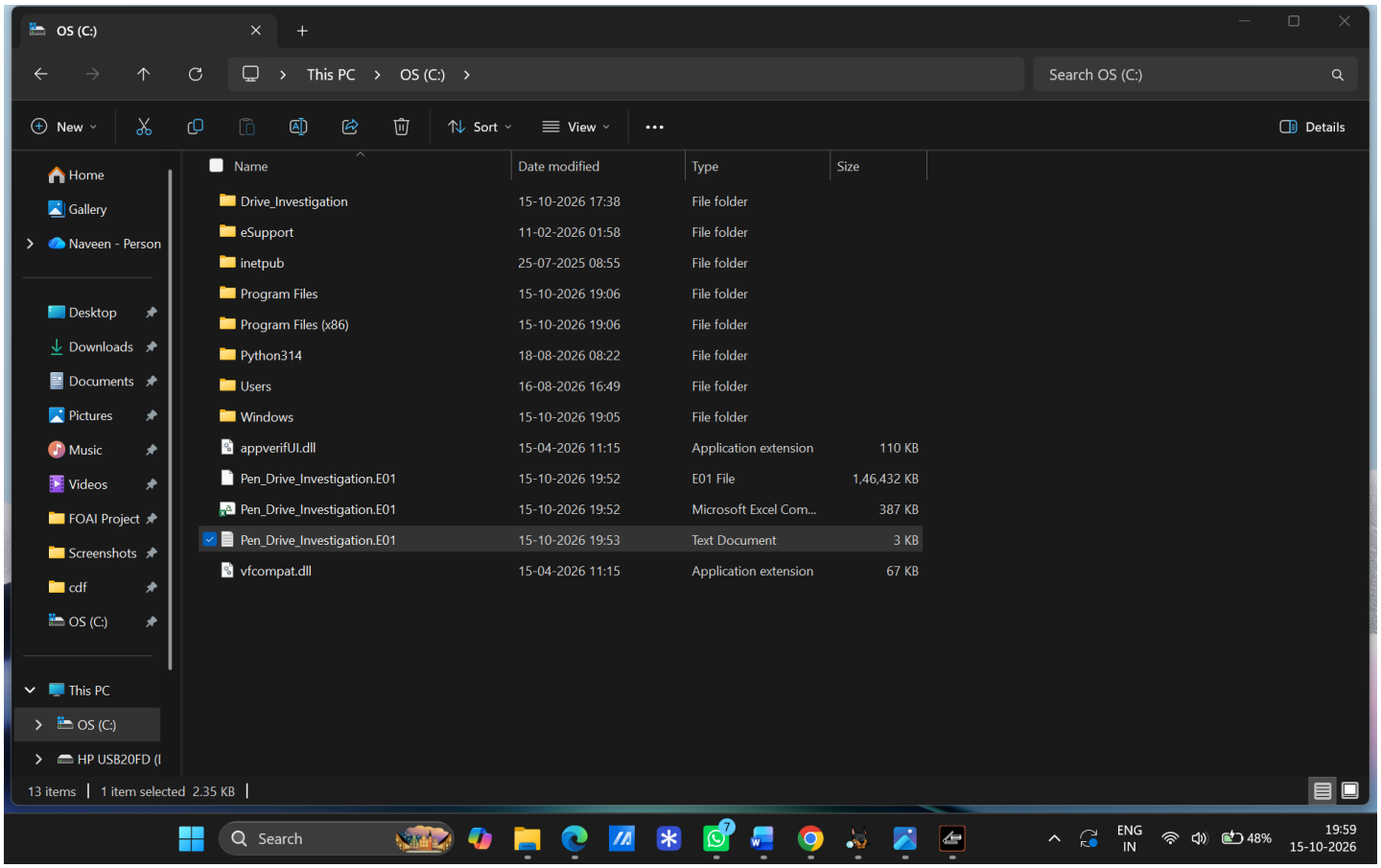

Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name Size Type Date Accessed Date Created Date Modified
Creating Directory Listing [100%]
Listing Source: C:\Pen_Drive_Investigation.E01
Destination: C:\Pen_Drive_Investigation.E01.csv
Status: Directory listing created successfully
Progress
Elapsed time: 0:00:00
Estimated time left:
Close
Custom Content Sources
Evidence:File System|Path|... Options
New Edit Remove Remove All
Properties Hex Value... Custom C...
For User Guide, press F1

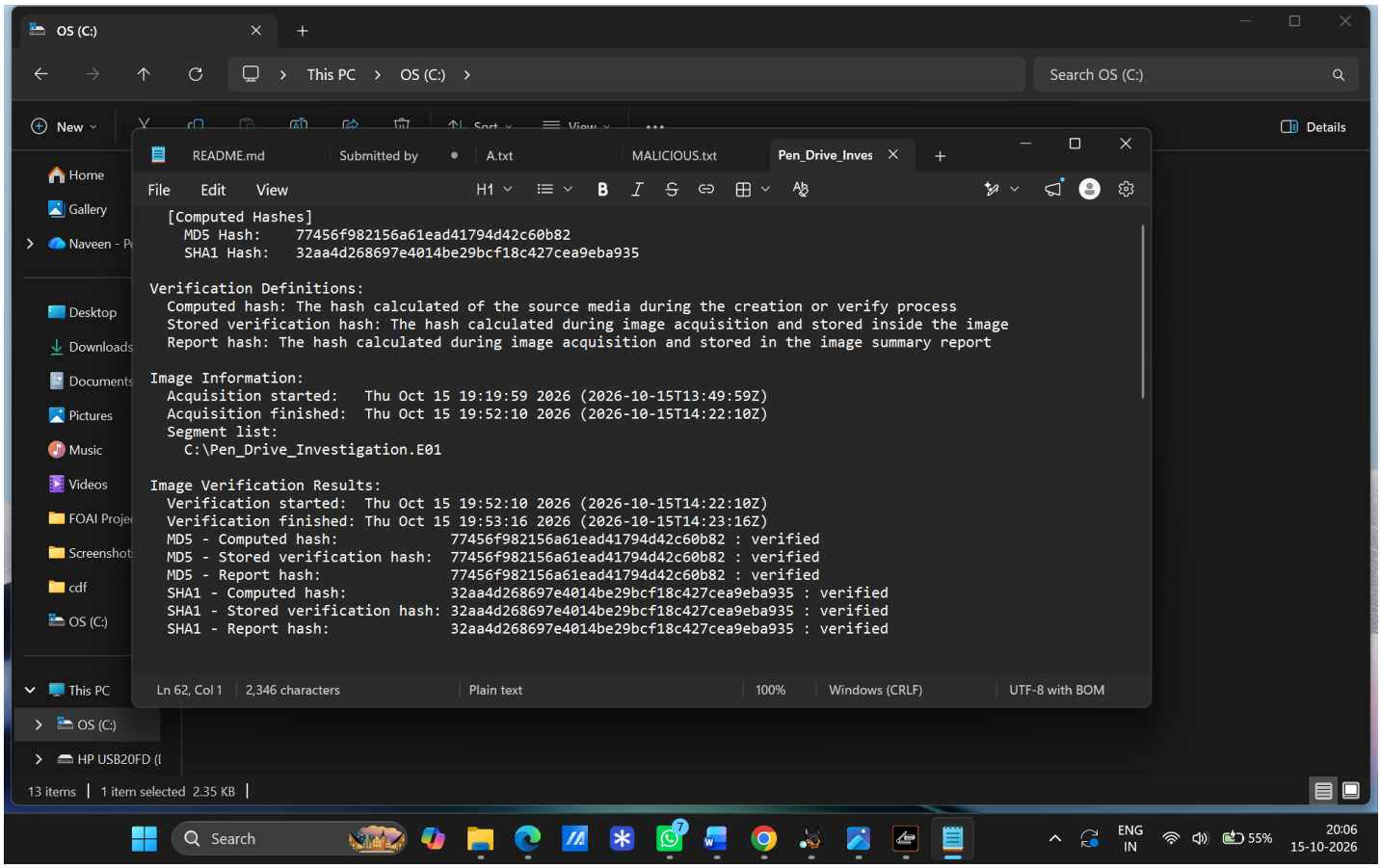

Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
File List
Name Size Type Date Accessed Date Created Date Modified
Creating Image [100%]
Image Source: \\.\PHYSICALDRIVE1
Destination: C:\Pen_Drive_Investigation
Status: Image created successfully
Progress
59153.00 of 59153.00 MB (30.633 MB/sec)
Elapsed time: 0:32:11
Estimated time left: 0:00:00
Image Summary...
Close
Custom Content Sources
Evidence:File System|Path|... Options
New Edit Remove Remove All
Properties Hex Value... Custom C...
For User Guide, press F1

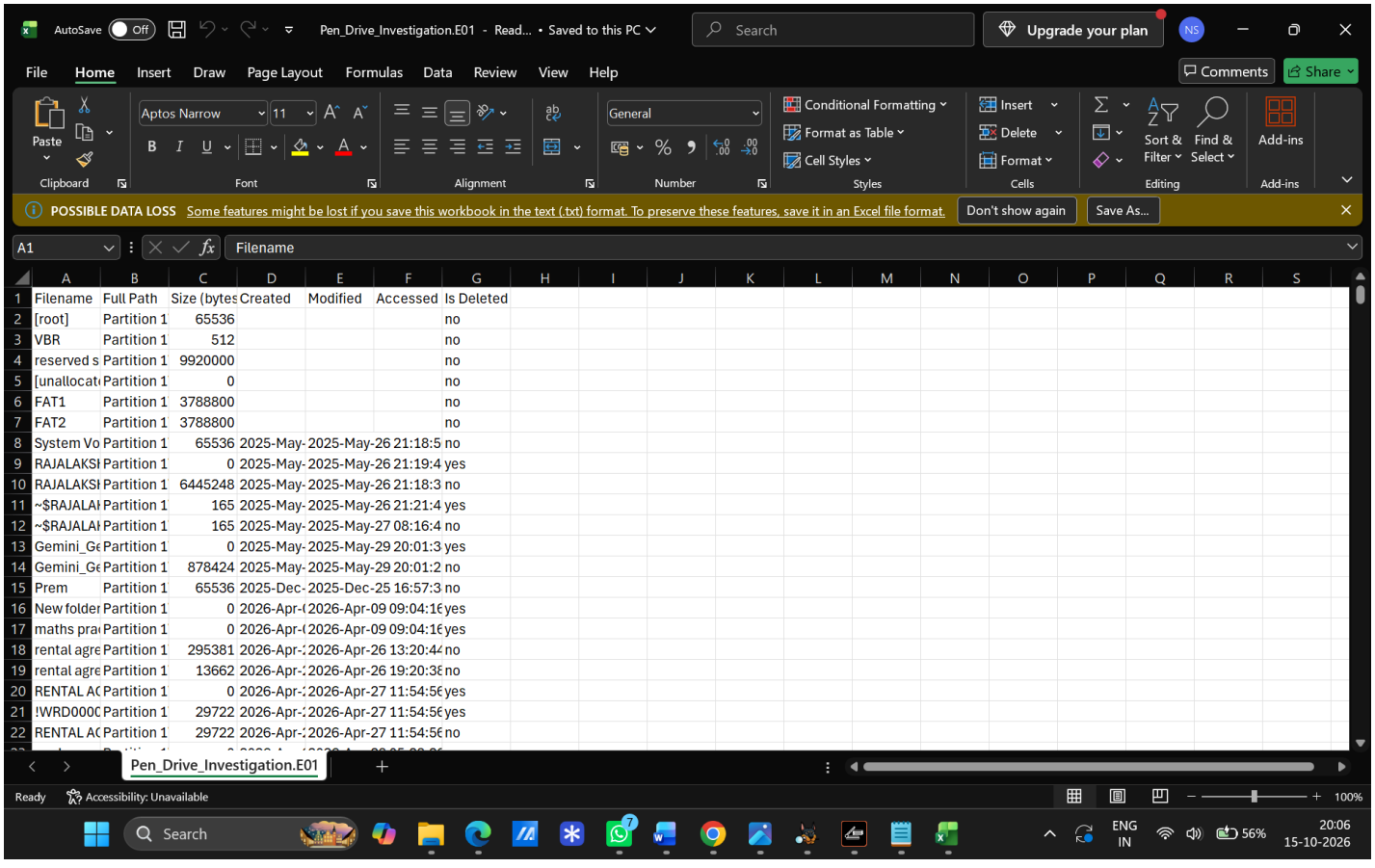

Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
Image Summary
Created By Exterro® FTK® Imager 8.3.0.27
Case Information:
Case Number: 002
Evidence Number: 001
Unique description: Pen_Drive_Investigation
Examiner: Naveen
Notes: investigation
Encrypted Password : N/A
Computer Name: LAPTOP-Q6VPI7T2
Windows Edition: Windows 11 Home Single Language
Windows Version: Microsoft Windows 6.3.26200.9711
OS build: 6.3.26200
Processor: Intel(R) Core(TM) Ultra 5 225H
Installed RAM: 16 GB
Information for C:\Pen_Drive_Investigation:
Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
OK
Custom Content Sources
Evidence:File System|Path|...
New Edit Remove
Properties Hex Value... Custom C...
For User Guide, press F1
Search
ENG IN
44%
19:55
15-10-2026

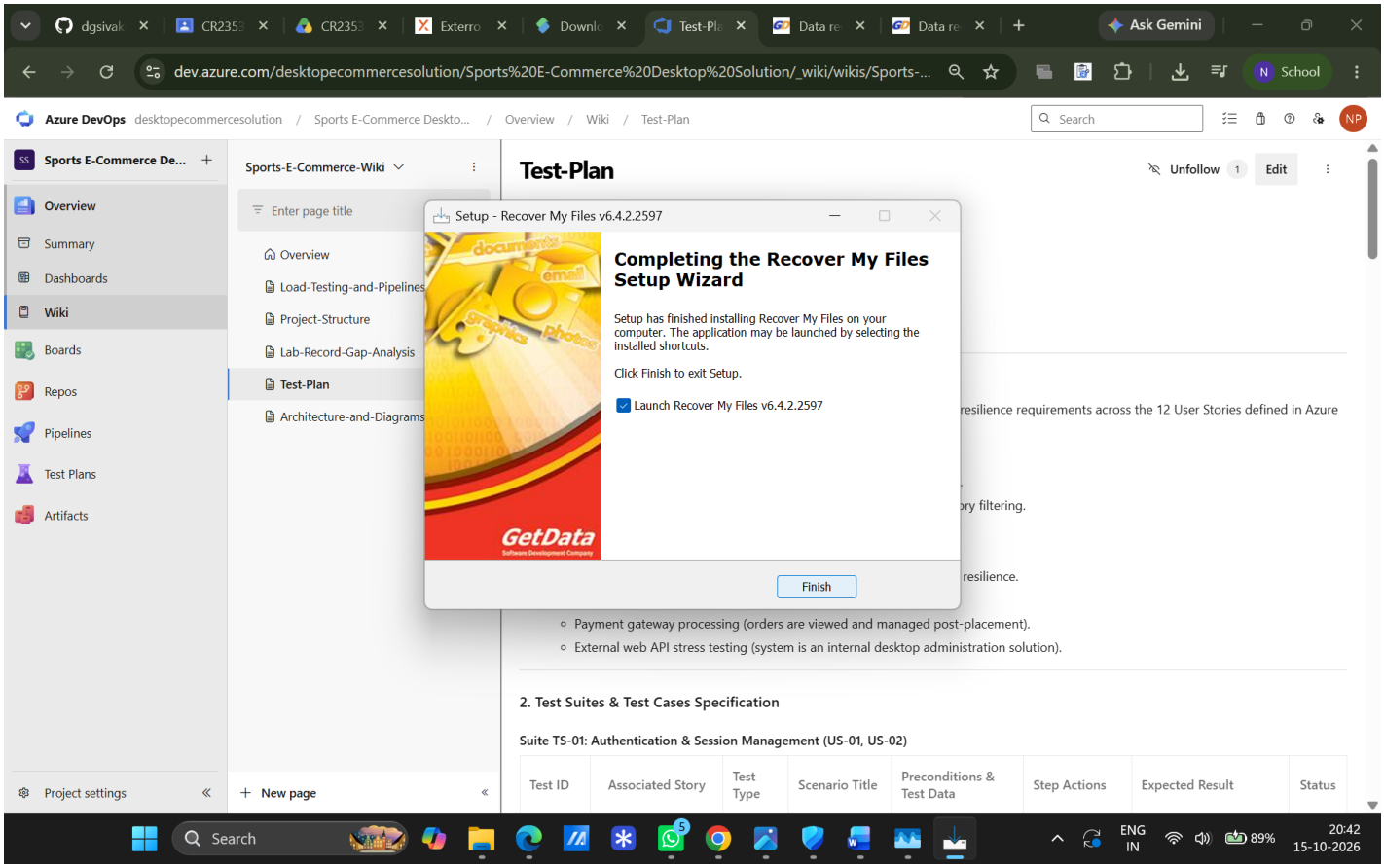

Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
Image Summary
Information for C:\Pen_Drive_Investigation:
Physical Evidentiary Item (Source) Information:
[Device Info]
Source Type: Physical
[Drive Geometry]
Cylinders: 7,540
Tracks per Cylinder: 255
Sectors per Track: 63
Bytes per Sector: 512
Sector Count: 121,145,344
[Physical Drive Information]
Drive Model: HP USB Flash Drive USB Device
Drive Serial Number: 077706D14070
Drive Interface Type: USB
Removable drive: True
Source data size: 59153 MB
Sector count: 121145344
[Computed Hashes]
MD5 Hash: 77456f982156a61ead41794d42c60b82
SHA1 Hash: 32aa4d268697e4014be29bcf18c427cea9eba935
Verification Definitions:
Computed hash: The hash calculated of the source media during the creation or verify process
Stored verification hash: The hash calculated during image acquisition and stored inside the image
OK
Custom Content Sources
Evidence:File System|Path|...
New Edit Remove
Properties Hex Value... Custom C...
For User Guide, press F1
Search
ENG IN
44%
19:55
15-10-2026

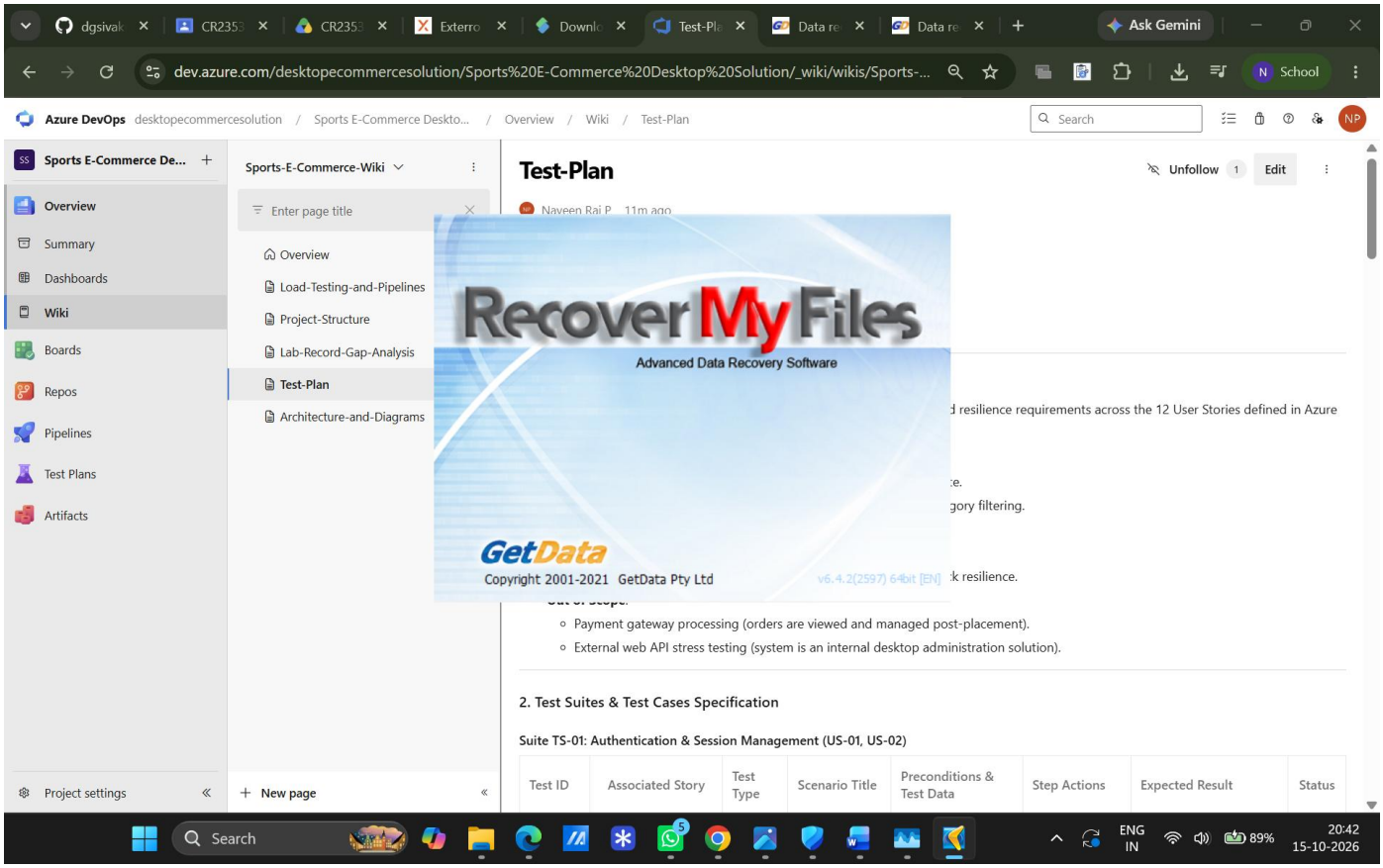

recover-my-files.en.softonic.com/support?dt=internalDownload&installerType=riseInstaller
Exterro FTK Imager 8.3.0.27
File View Mode Help Upgrade to PRO
Evidence Tree
Image Summary
[Computed Hashes]
MD5 Hash: 77456f982156a61ead41794d42c60b82
SHA1 Hash: 32aa4d268697e4014be29bcf18c427cea9eba935
Verification Definitions:
Computed hash: The hash calculated of the source media during the creation or verify process
Stored verification hash: The hash calculated during image acquisition and stored inside the image
Report hash: The hash calculated during image acquisition and stored in the image summary report
Image Information:
Acquisition started: Thu Oct 15 19:19:59 2026 (2026-10-15T13:49:59Z)
Acquisition finished: Thu Oct 15 19:52:10 2026 (2026-10-15T14:22:10Z)
Segment list:
C:\Pen_Drive_Investigation.E01
Image Verification Results:
Verification started: Thu Oct 15 19:52:10 2026 (2026-10-15T14:22:10Z)
Verification finished: Thu Oct 15 19:53:16 2026 (2026-10-15T14:23:16Z)
MD5 - Computed hash: 77456f982156a61ead41794d42c60b82 : verified
MD5 - Stored verification hash: 77456f982156a61ead41794d42c60b82 : verified
MD5 - Report hash: 77456f982156a61ead41794d42c60b82 : verified
SHA1 - Computed hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
SHA1 - Stored verification hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
SHA1 - Report hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
OK
Custom Content Sources
For User Guide, press F1
Search
ENG IN
44%
19:55
15-10-2026

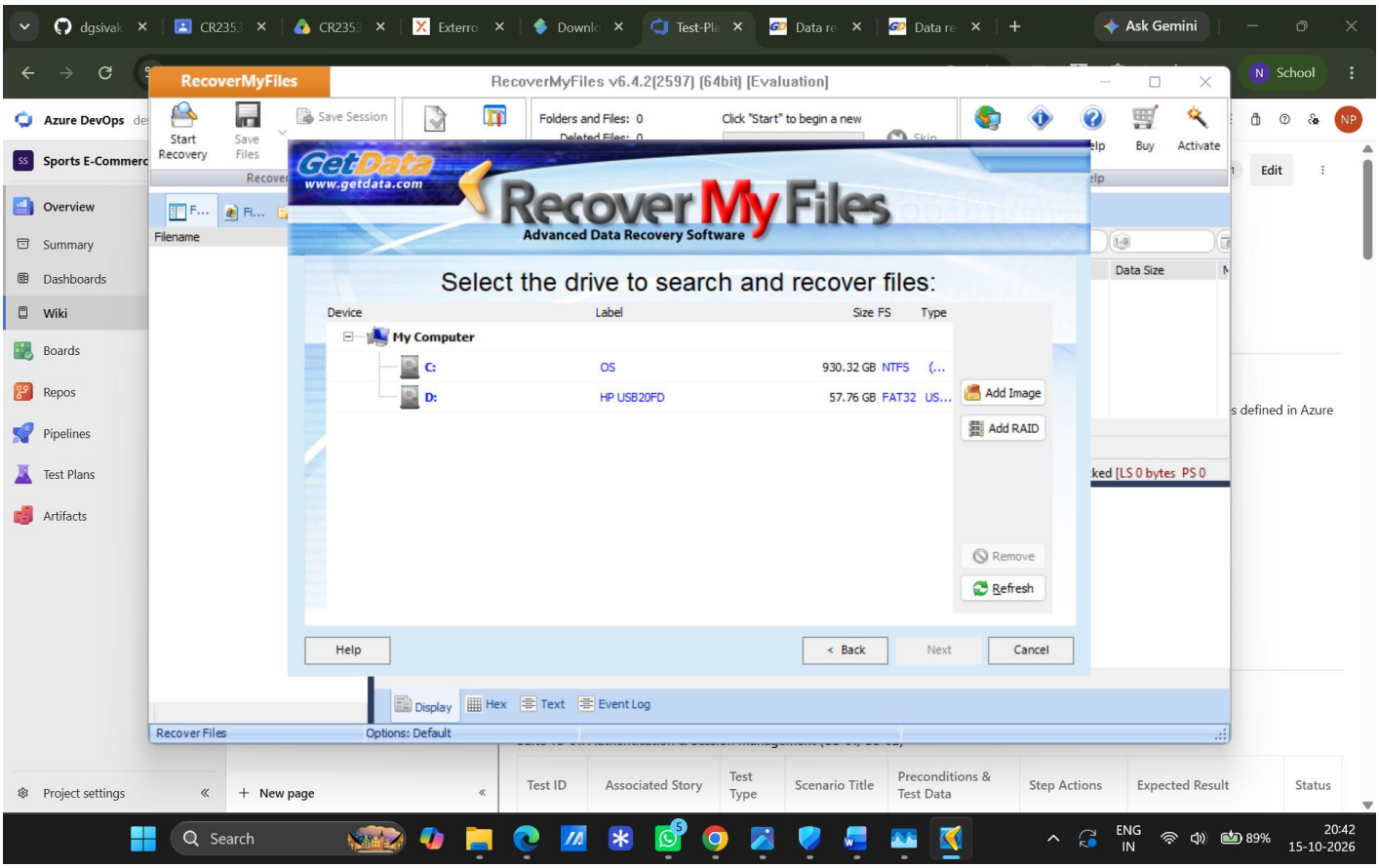

OS (C:)
This PC > OS (C:) >
Search OS (C:)
New
Sort
View
Details
Home
Gallery
Naveen - Person
Desktop
Downloads
Documents
Pictures
Music
Videos
FOAI Project
Screenshots
cdf
OS (C:)
This PC
OS (C:)
HP USB20FD (I
Name
Date modified
Type
Size
Drive_Investigation 15-10-2026 17:38 File folder
eSupport 11-02-2026 01:58 File folder
inetpub 25-07-2025 08:55 File folder
Program Files 15-10-2026 19:06 File folder
Program Files (x86) 15-10-2026 19:06 File folder
Python314 18-08-2026 08:22 File folder
Users 16-08-2026 16:49 File folder
Windows 15-10-2026 19:05 File folder
appverifUI.dll 15-04-2026 11:15 Application extension 110 KB
Pen_Drive_Investigation.E01 15-10-2026 19:52 E01 File 1,46,432 KB
Pen_Drive_Investigation.E01 15-10-2026 19:52 Microsoft Excel Com... 387 KB
Pen_Drive_Investigation.E01 15-10-2026 19:53 Text Document 3 KB
vfcompat.dll 15-04-2026 11:15 Application extension 67 KB
13 items | 1 item selected 2.35 KB
Search
ENG IN
48%
19:59
15-10-2026

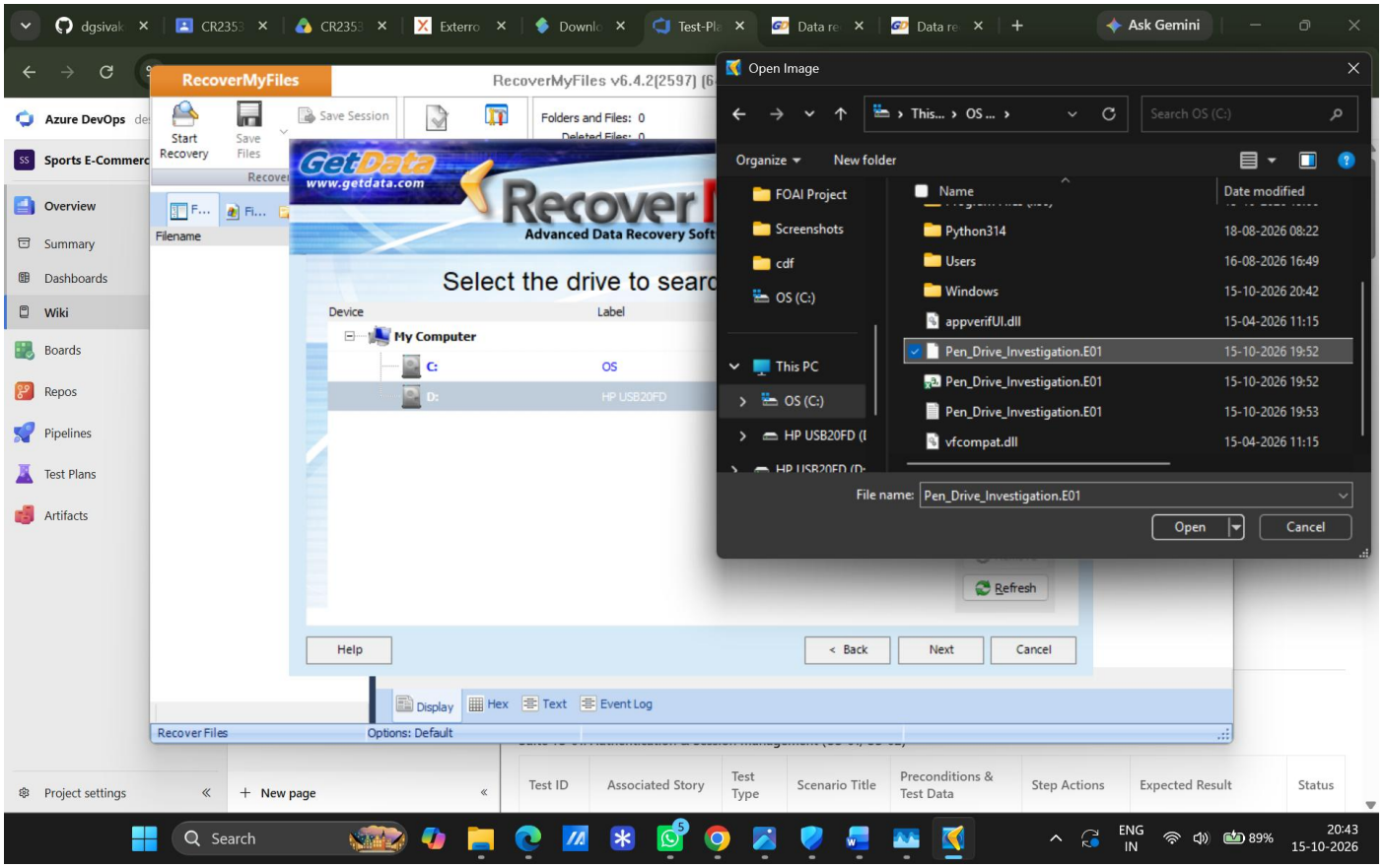

OS (C:)
This PC > OS (C:) >
Search OS (C:)
README.md
Submitted by
A.txt
MALICIOUS.txt
Pen_Drive_Inves
File Edit View
[Computed Hashes]
MD5 Hash: 77456f982156a61ead41794d42c60b82
SHA1 Hash: 32aa4d268697e4014be29bcf18c427cea9eba935
Verification Definitions:
Computed hash: The hash calculated of the source media during the creation or verify process
Stored verification hash: The hash calculated during image acquisition and stored inside the image
Report hash: The hash calculated during image acquisition and stored in the image summary report
Image Information:
Acquisition started: Thu Oct 15 19:19:59 2026 (2026-10-15T13:49:59Z)
Acquisition finished: Thu Oct 15 19:52:10 2026 (2026-10-15T14:22:10Z)
Segment list:
C:\Pen_Drive_Investigation.E01
Image Verification Results:
Verification started: Thu Oct 15 19:52:10 2026 (2026-10-15T14:22:10Z)
Verification finished: Thu Oct 15 19:53:16 2026 (2026-10-15T14:23:16Z)
MD5 - Computed hash: 77456f982156a61ead41794d42c60b82 : verified
MD5 - Stored verification hash: 77456f982156a61ead41794d42c60b82 : verified
MD5 - Report hash: 77456f982156a61ead41794d42c60b82 : verified
SHA1 - Computed hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
SHA1 - Stored verification hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
SHA1 - Report hash: 32aa4d268697e4014be29bcf18c427cea9eba935 : verified
Ln 62, Col 1
2,346 characters
Plain text
100%
Windows (CRLF)
UTF-8 with BOM
13 items | 1 item selected 2.35 KB
20:06
15-10-2026

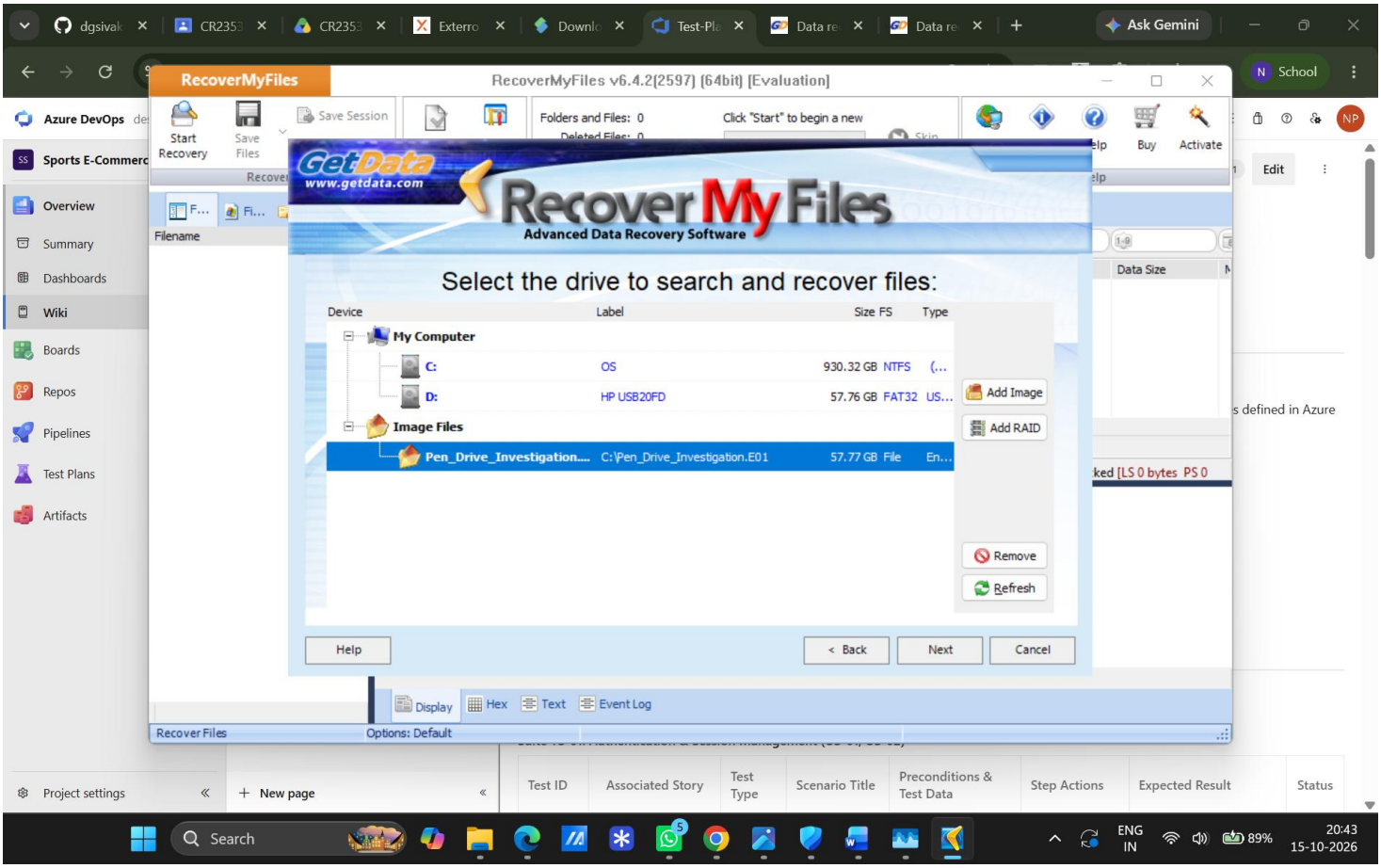

Pen_Drive_Investigation.E01 - Read... • Saved to this PC
Upgrade your plan
POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the text (.txt) format. To preserve these features, save it in an Excel file format.
Don't show again
Save As...
A1
Filename
Filename Full Path Size (bytes Created Modified Accessed Is Deleted
[root] Partition 1 65536 no
VBR Partition 1 512 no
reserved s Partition 1 9920000 no
[unallocat Partition 1 0 no
FAT1 Partition 1 3788800 no
FAT2 Partition 1 3788800 no
System Vo Partition 1 65536 2025-May- 2025-May-26 21:18:5 no
RAJALAKSI Partition 1 0 2025-May- 2025-May-26 21:19:4 yes
RAJALAKSI Partition 1 6445248 2025-May- 2025-May-26 21:18:3 no
~$RAJALAI Partition 1 165 2025-May- 2025-May-26 21:21:4 yes
~$RAJALAI Partition 1 165 2025-May- 2025-May-27 08:16:4 no
Gemini_Ge Partition 1 0 2025-May- 2025-May-29 20:01:3 yes
Gemini_Ge Partition 1 878424 2025-May- 2025-May-29 20:01:2 no
Prem Partition 1 65536 2025-Dec- 2025-Dec-25 16:57:3 no
New folder Partition 1 0 2026-Apr-(2026-Apr-09 09:04:1(yes
maths pra Partition 1 0 2026-Apr-(2026-Apr-09 09:04:1(yes
rental agre Partition 1 295381 2026-Apr-: 2026-Apr-26 13:20:4 no
rental agre Partition 1 13662 2026-Apr-: 2026-Apr-26 19:20:3 no
RENTAL AC Partition 1 0 2026-Apr-: 2026-Apr-27 11:54:5(yes
!WRD000(Partition 1 29722 2026-Apr-: 2026-Apr-27 11:54:5(yes
RENTAL AC Partition 1 29722 2026-Apr-: 2026-Apr-27 11:54:5(no
Pen_Drive_Investigation.E01
Ready
Accessibility: Unavailable
20:06
15-10-2026

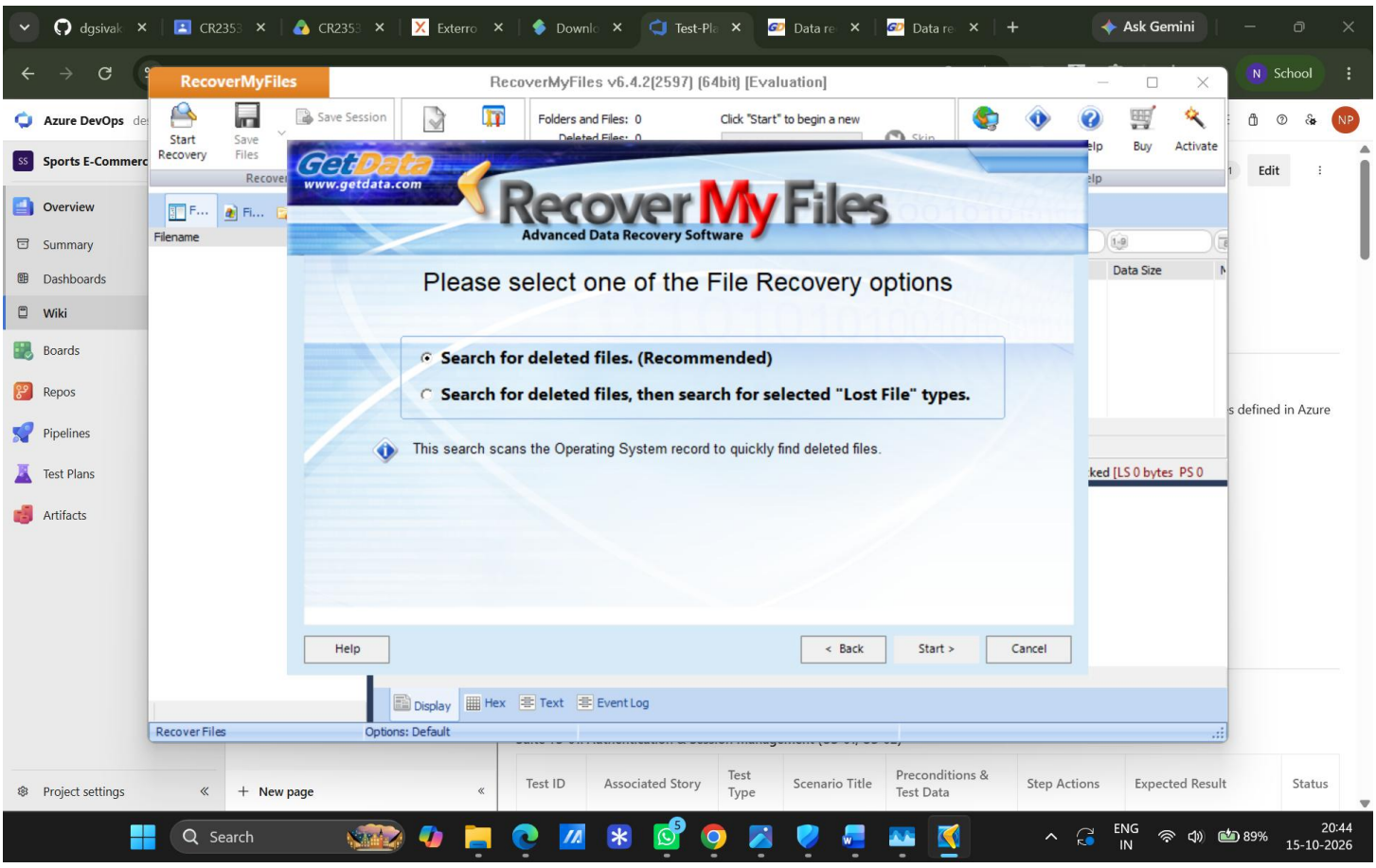

Setup - Recover My Files v6.4.2.2597
Completing the Recover My Files Setup Wizard
Setup has finished installing Recover My Files on your computer. The application may be launched by selecting the installed shortcuts.
Click Finish to exit Setup.
Launch Recover My Files v6.4.2.2597
Finish
GetData
Test-Plan
Unfollow
Edit
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
Sports-E-Commerce-Wiki
Enter page title
Load-Testing-and-Pipelines
Project-Structure
Lab-Record-Gap-Analysis
Test-Plan
Architecture-and-Diagrams
New page
2. Test Suites & Test Cases Specification
Suite TS-01: Authentication & Session Management (US-01, US-02)
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status

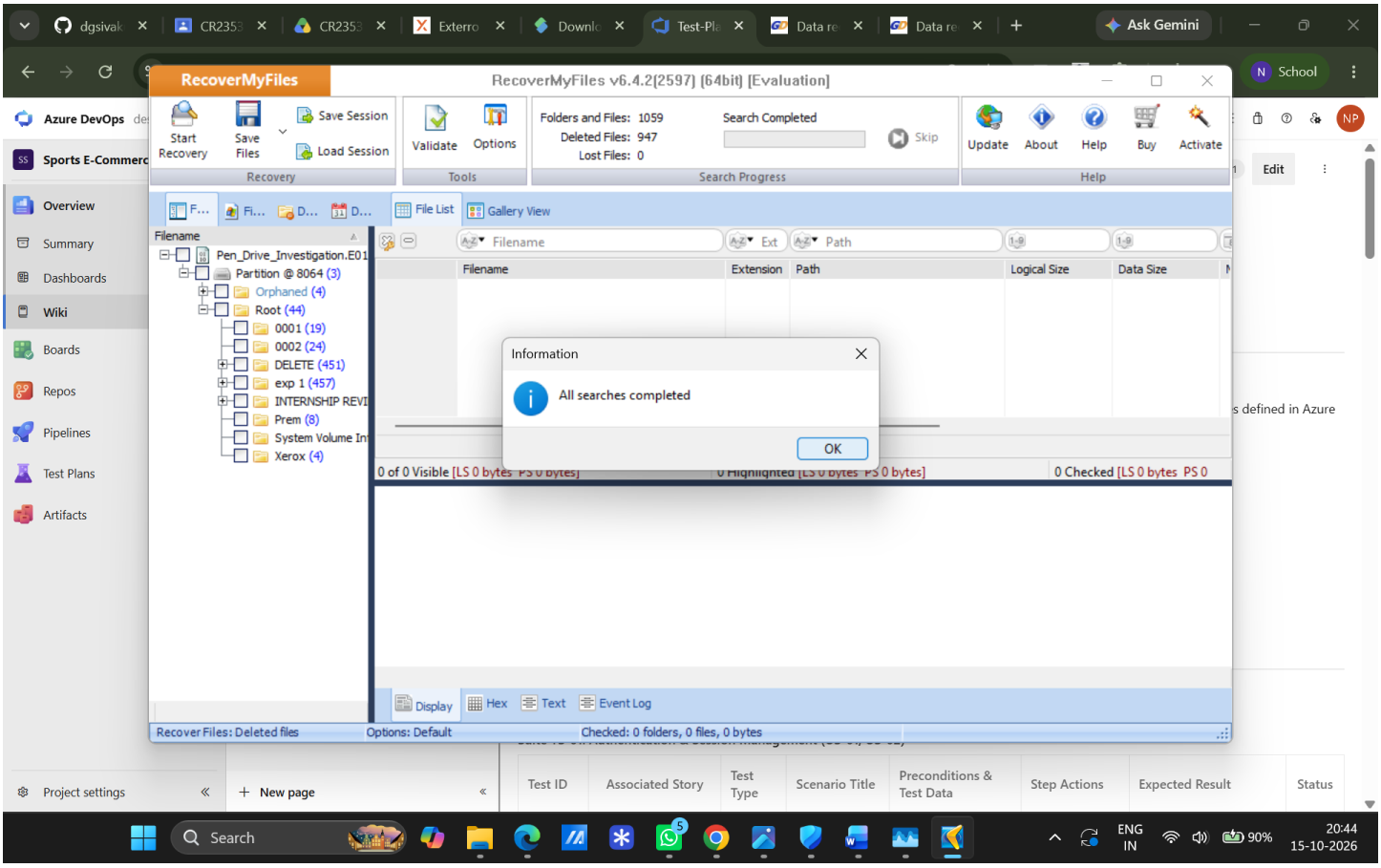

Recover My Files
Advanced Data Recovery Software
GetData
Copyright 2001-2021 GetData Pty Ltd
Test-Plan
Unfollow
Edit
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
Sports-E-Commerce-Wiki
Enter page title
Load-Testing-and-Pipelines
Project-Structure
Lab-Record-Gap-Analysis
Test-Plan
Architecture-and-Diagrams
New page
2. Test Suites & Test Cases Specification
Suite TS-01: Authentication & Session Management (US-01, US-02)
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status

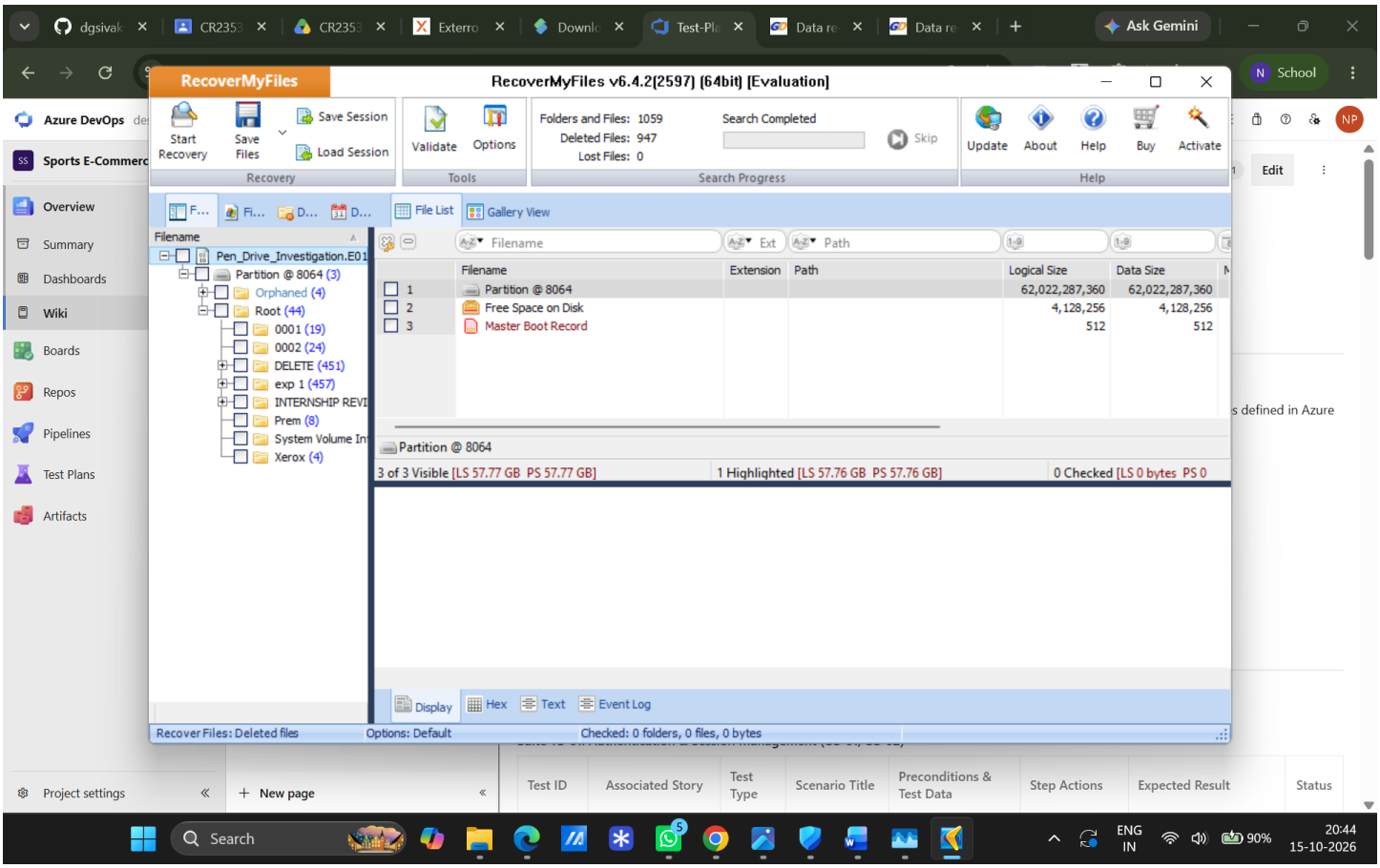

RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Folders and Files: 0
Click "Start" to begin a new
Buy
Activate
GetData
www.getdata.com
Recover My Files
Advanced Data Recovery Software
Select the drive to search and recover files:
Device
Label
Size FS
Type
My Computer
C:
OS
930.32 GB NTFS
D:
HP USB20FD
57.76 GB FAT32
Add Image
Add RAID
Remove
Refresh
Help
< Back
Next
Cancel
Display
Hex
Text
Event Log
Recover Files
Options: Default
Azure DevOps
Sports E-Commerc
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
89%
20:42
15-10-2026

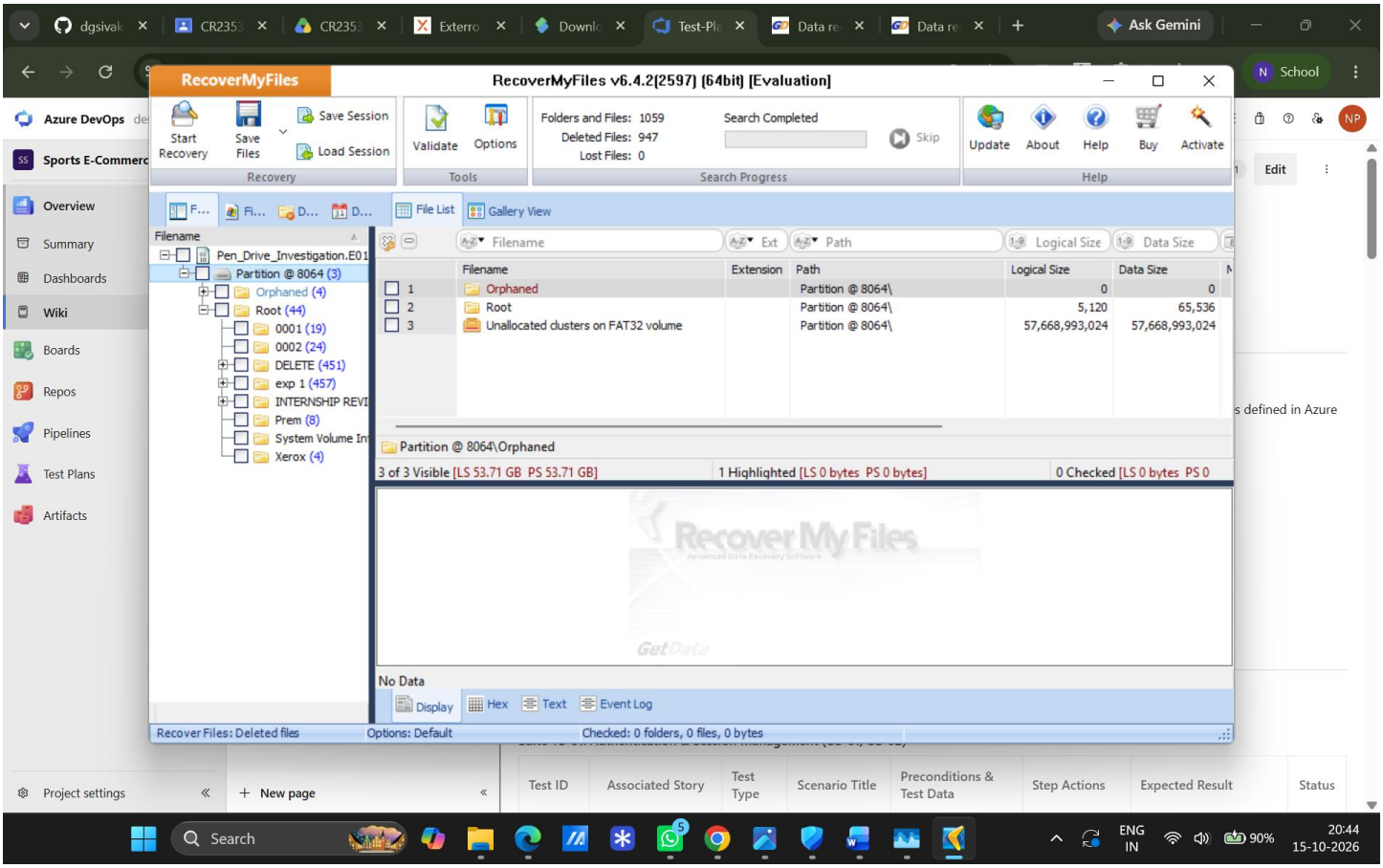

Open Image
Organize
New folder
FOAI Project
Screenshots
cdf
OS (C:)
This PC
HP USB20FD (I
Name
Date modified
Python314
18-08-2026 08:22
Users
16-08-2026 16:49
Windows
15-10-2026 20:42
appverifUI.dll
15-04-2026 11:15
Pen_Drive_Investigation.E01
15-10-2026 19:52
Pen_Drive_Investigation.E01
15-10-2026 19:52
Pen_Drive_Investigation.E01
15-10-2026 19:53
vfcompat.dll
15-04-2026 11:15
File name:
Pen_Drive_Investigation.E01
Open
Cancel
RecoverMyFiles
Start Recovery
Save Files
Save Session
Folders and Files: 0
GetData
www.getdata.com
Advanced Data Recovery Soft
Select the drive to searc
Device
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My Computer
C:
OS
D:
HP USB20FD
Refresh
Help
< Back
Next
Cancel
Display
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Event Log
Recover Files
Options: Default
Azure DevOps
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Summary
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Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
89%
20:43
15-10-2026

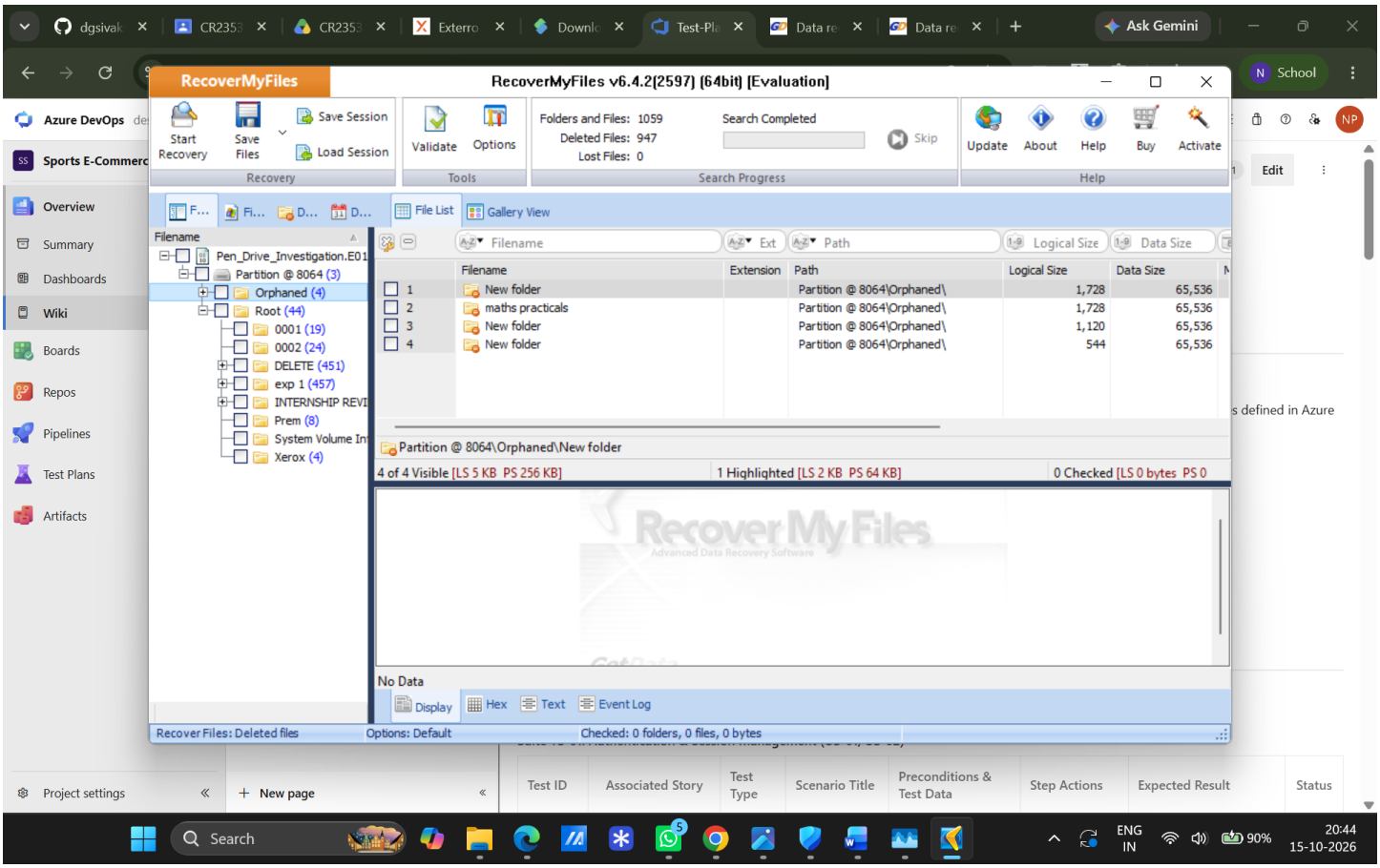

RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Folders and Files: 0
Click "Start" to begin a new
Buy
Activate
GetData
www.getdata.com
Recover My Files
Advanced Data Recovery Software
Select the drive to search and recover files:
Device
Label
Size FS
Type
My Computer
C:
OS
930.32 GB NTFS
D:
HP USB20FD
57.76 GB FAT32
Image Files
Pen_Drive_Investigation....
C:\Pen_Drive_Investigation.E01
57.77 GB File
Add Image
Add RAID
Remove
Refresh
Help
< Back
Next
Cancel
Display
Hex
Text
Event Log
Recover Files
Options: Default
Filename
Data Size
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Project settings
New page

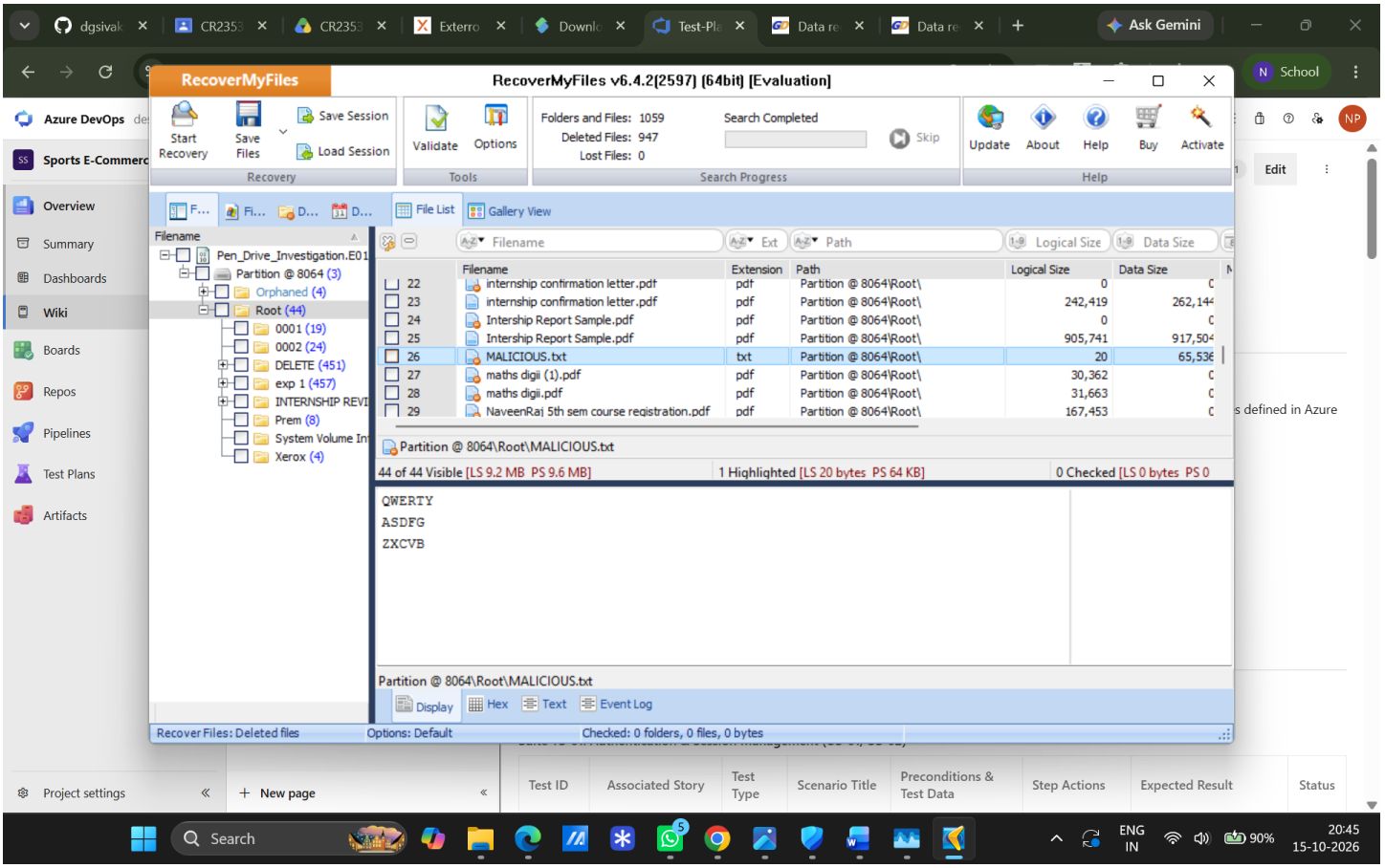

RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Folders and Files: 0
Click "Start" to begin a new
Buy
Activate
GetData
www.getdata.com
Recover My Files
Advanced Data Recovery Software
Please select one of the File Recovery options
Search for deleted files. (Recommended)
Search for deleted files, then search for selected "Lost File" types.
This search scans the Operating System record to quickly find deleted files.
Help
< Back
Start >
Cancel
Display
Hex
Text
Event Log
Recover Files
Options: Default
Filename
Data Size
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Project settings
New page

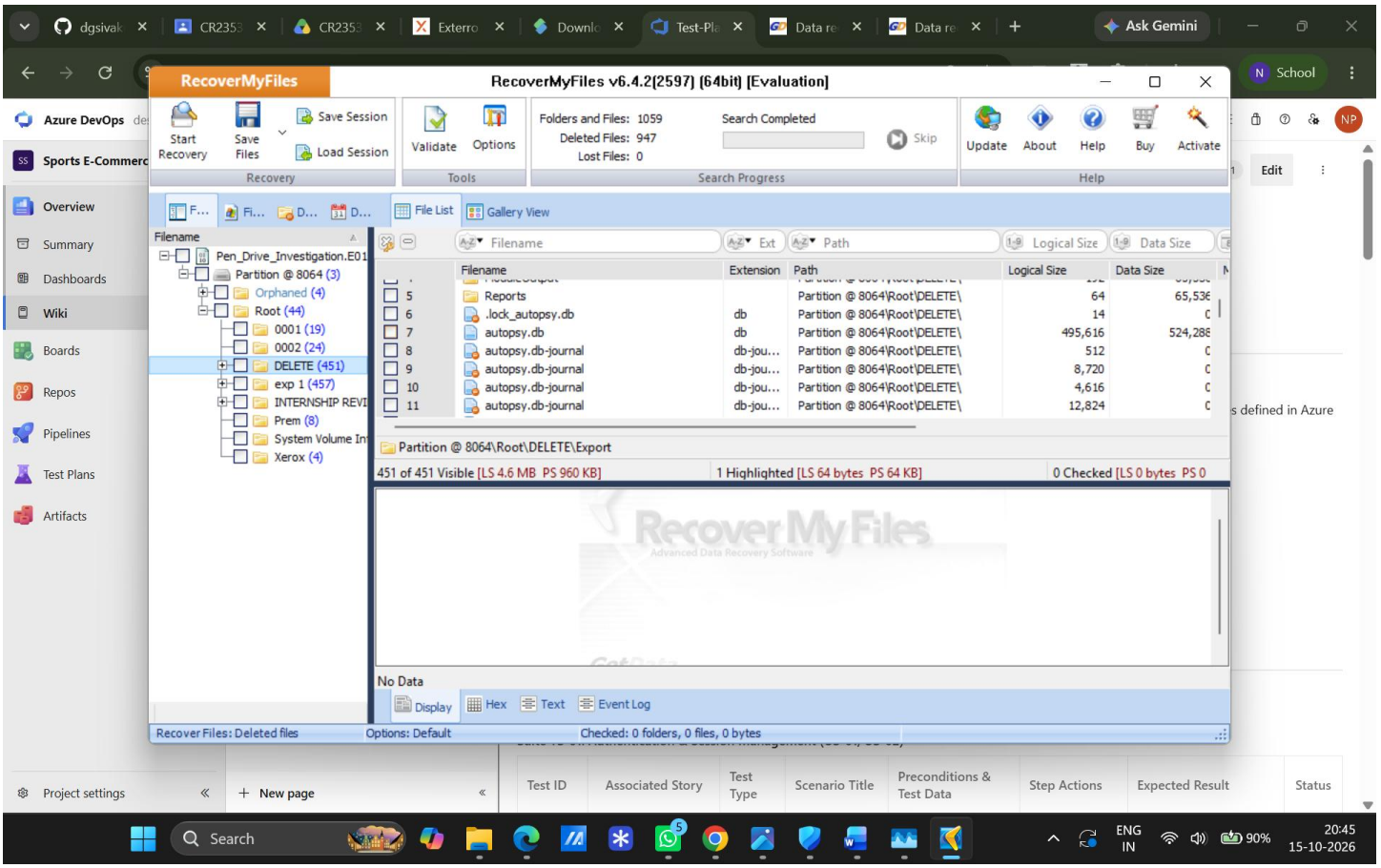

RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
Filename
Extension
Path
Logical Size
Data Size
Information
All searches completed
OK
0 of 0 Visible [LS 0 bytes PS 0 bytes]
0 Checked [LS 0 bytes PS 0
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
Sports E-Commerc
Overview
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Pipelines
Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
90%
20:44
15-10-2026


RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
Filename
Extension
Path
Logical Size
Data Size
1
Partition @ 8064
62,022,287,360
62,022,287,360
2
Free Space on Disk
4,128,256
4,128,256
3
Master Boot Record
512
512
Partition @ 8064
3 of 3 Visible [LS 57.77 GB PS 57.77 GB]
1 Highlighted [LS 57.76 GB PS 57.76 GB]
0 Checked [LS 0 bytes PS 0
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
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Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
90%
20:44
15-10-2026


RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
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About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
Filename
Extension
Path
Logical Size
Data Size
1 Orphaned Partition @ 8064\ 0 0
2 Root Partition @ 8064\ 5,120 65,536
3 Unallocated clusters on FAT32 volume Partition @ 8064\ 57,668,993,024 57,668,993,024
Partition @ 8064\Orphaned
3 of 3 Visible [LS 53.71 GB PS 53.71 GB]
1 Highlighted [LS 0 bytes PS 0 bytes]
0 Checked [LS 0 bytes PS 0
No Data
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
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Test Plans
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Project settings
New page
Edit
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
90%
20:44
15-10-2026


RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
Filename
Extension
Path
Logical Size
Data Size
1 New folder Partition @ 8064\Orphaned\ 1,728 65,536
2 maths practicals Partition @ 8064\Orphaned\ 1,728 65,536
3 New folder Partition @ 8064\Orphaned\ 1,120 65,536
4 New folder Partition @ 8064\Orphaned\ 544 65,536
Partition @ 8064\Orphaned\New folder
4 of 4 Visible [LS 5 KB PS 256 KB]
1 Highlighted [LS 2 KB PS 64 KB]
0 Checked [LS 0 bytes PS 0
No Data
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
Sports E-Commerc
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
New page
Edit
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
90%
20:44
15-10-2026


RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
22 internship confirmation letter.pdf pdf Partition @ 8064\Root\ 0
23 internship confirmation letter.pdf pdf Partition @ 8064\Root\ 242,419 262,144
24 Intership Report Sample.pdf pdf Partition @ 8064\Root\ 0
25 Intership Report Sample.pdf pdf Partition @ 8064\Root\ 905,741 917,504
26 MALICIOUS.txt txt Partition @ 8064\Root\ 20 65,536
27 maths digii (1).pdf pdf Partition @ 8064\Root\ 30,362
28 maths digii.pdf pdf Partition @ 8064\Root\ 31,663
29 NaveenRaj 5th sem course registration.pdf pdf Partition @ 8064\Root\ 167,453
Partition @ 8064\Root\MALICIOUS.txt
44 of 44 Visible [LS 9.2 MB PS 9.6 MB]
1 Highlighted [LS 20 bytes PS 64 KB]
0 Checked [LS 0 bytes PS 0
QWERTY
ASDFG
ZXCVB
Partition @ 8064\Root\MALICIOUS.txt
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
Sports E-Commerc
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
90%
20:45
15-10-2026


RecoverMyFiles
RecoverMyFiles v6.4.2(2597) (64bit) [Evaluation]
Start Recovery
Save Files
Save Session
Load Session
Validate
Options
Folders and Files: 1059
Deleted Files: 947
Lost Files: 0
Search Completed
Skip
Update
About
Help
Buy
Activate
Recovery
Tools
Search Progress
Help
File List
Gallery View
Pen_Drive_Investigation.E01
Partition @ 8064 (3)
Orphaned (4)
Root (44)
0001 (19)
0002 (24)
DELETE (451)
exp 1 (457)
INTERNSHIP REVI
Prem (8)
System Volume In
Xerox (4)
5 Reports Partition @ 8064\Root\DELETE\ 64 65,536
6 .lock_autopsy.db db Partition @ 8064\Root\DELETE\ 14
7 autopsy.db db Partition @ 8064\Root\DELETE\ 495,616 524,288
8 autopsy.db-journal db-jou... Partition @ 8064\Root\DELETE\ 512
9 autopsy.db-journal db-jou... Partition @ 8064\Root\DELETE\ 8,720
10 autopsy.db-journal db-jou... Partition @ 8064\Root\DELETE\ 4,616
11 autopsy.db-journal db-jou... Partition @ 8064\Root\DELETE\ 12,824
Partition @ 8064\Root\DELETE\Export
451 of 451 Visible [LS 4.6 MB PS 960 KB]
1 Highlighted [LS 64 bytes PS 64 KB]
0 Checked [LS 0 bytes PS 0
No Data
Display
Hex
Text
Event Log
Recover Files: Deleted files
Options: Default
Checked: 0 folders, 0 files, 0 bytes
Azure DevOps
Sports E-Commerc
Overview
Summary
Dashboards
Wiki
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Repos
Pipelines
Test Plans
Artifacts
Project settings
New page
Test ID
Associated Story
Test Type
Scenario Title
Preconditions & Test Data
Step Actions
Expected Result
Status
Search
ENG IN
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20:45
15-10-2026

Observations

- **Imaging:** FTK Imager acquired a complete physical image of the drive without altering original evidence.
- **Recovery:** RecoverMyFiles successfully parsed unallocated space in the forensic image and fully restored the deleted text file.

Result

The intentionally deleted digital evidence was accurately captured, analyzed, and recovered.

EXPERIMENT 04

Find Last Connected USB on Your System (USB Forensics)

Aim

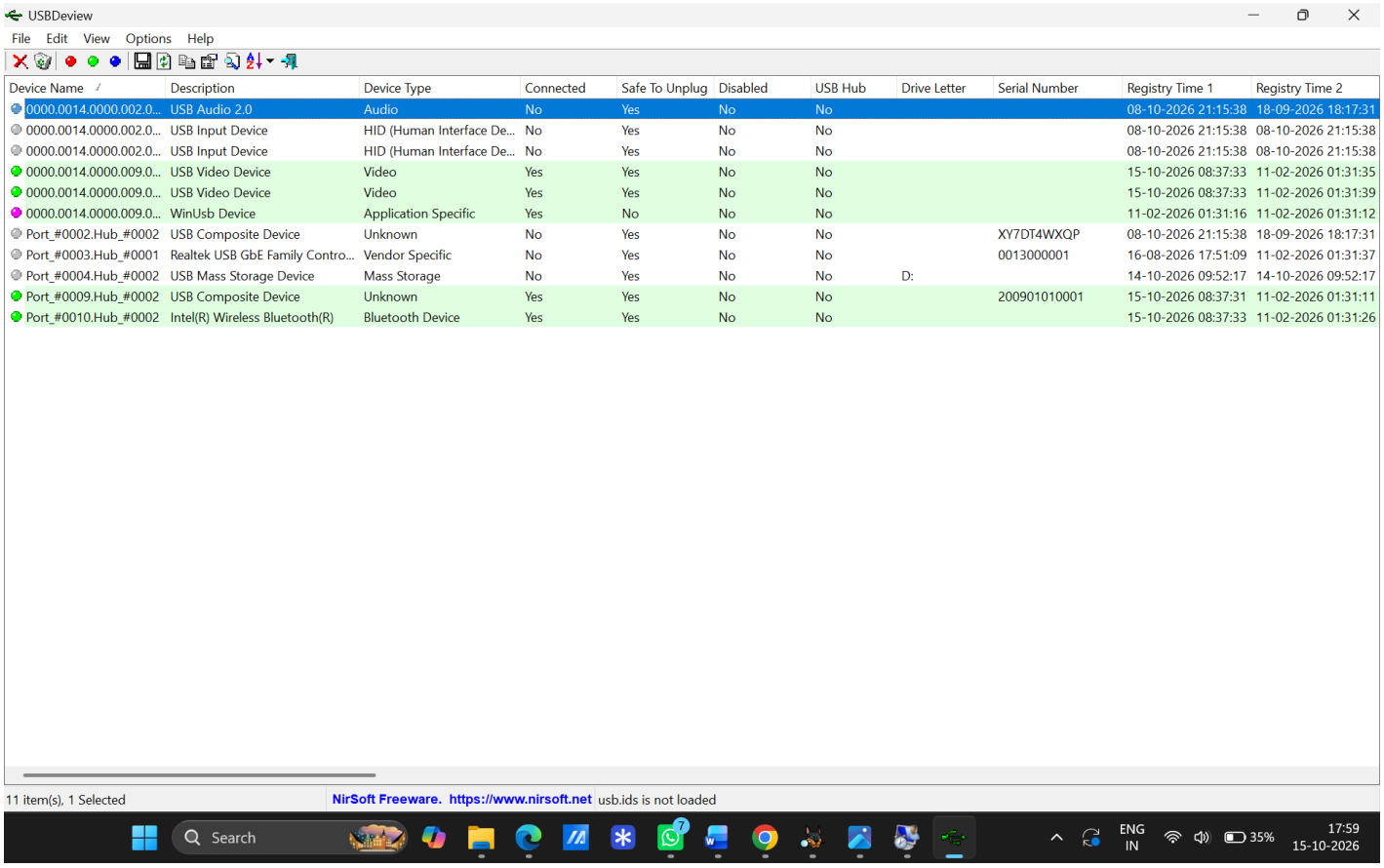
To identify and analyze history, connection logs, and details of USB devices previously or currently connected to the system.

Software / Tools Required

- USBDeview Utility
- Target Windows Host PC

Procedure

1. Executed **USBDeview** on the target Windows system with administrator privileges.
2. Retrived the generated list of all currently connected and previously attached USB devices.
3. Extracted device attributes (Device Name, Serial Number, Vendor ID, Product ID, and Last Plug/Unplug Timestamps).



Observations

Attribute Analyzed	Findings / Recorded Data
Device Description & Type	Mass Storage, Communication, and Vendor-Specific devices.
Identifiers	Vendor ID (VID), Product ID (PID), and Serial Numbers.
Connection Timestamps	Exact date and time when each device was last plugged or unplugged.
Connection Status	Indicates whether the device is currently active/connected.

Result

The USB forensic examination using USBDeview was successfully executed, and historical USB connection artifacts were analyzed.

EXPERIMENT 05

View Last Activity of Your PC

Aim

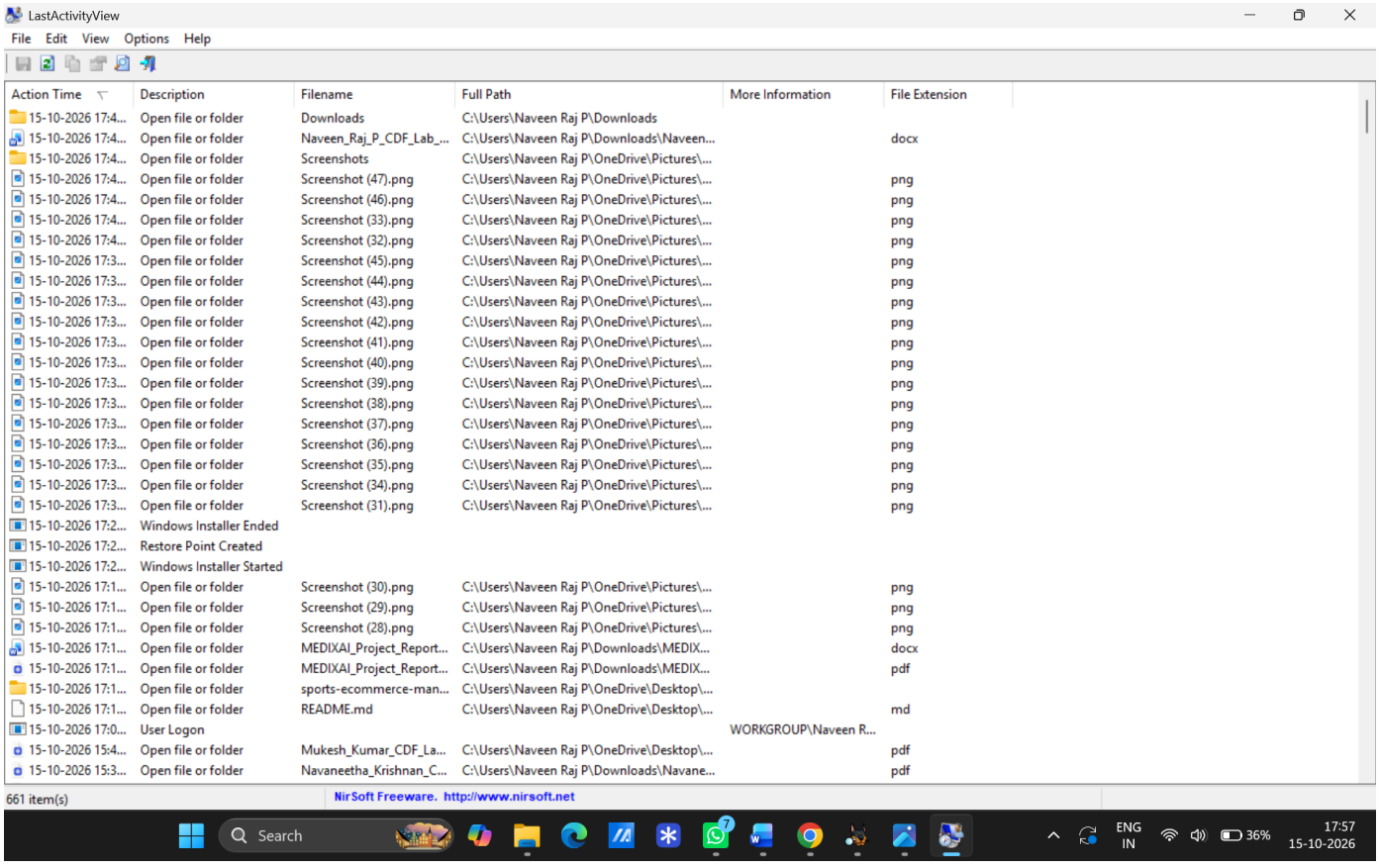
To collect and analyze system logs to view the detailed history of user actions and computer events on a running Windows OS.

Software / Tools Required

- LastActivityView Utility
- Windows Host PC

Procedure

1. Launched **LastActivityView** on the active Windows host.
2. Allowed the utility to gather event records from system sources (Registry, log files, system folders).
3. Filtered and reviewed the chronological log entries showing user activities, app executions, and system events.



Observations

Log Category	Artifacts & Event Data Observed
User Activity	User logins, folder views, and opened files/folders.
Application History	Installed, executed, or crashed software applications.
System Events	System startup, shutdown, and network connection timestamps.

Result

System activity forensic examination using LastActivityView was successfully performed and historical logs were analyzed.

EXPERIMENT 06

Extraction of Browser Artifacts

Aim

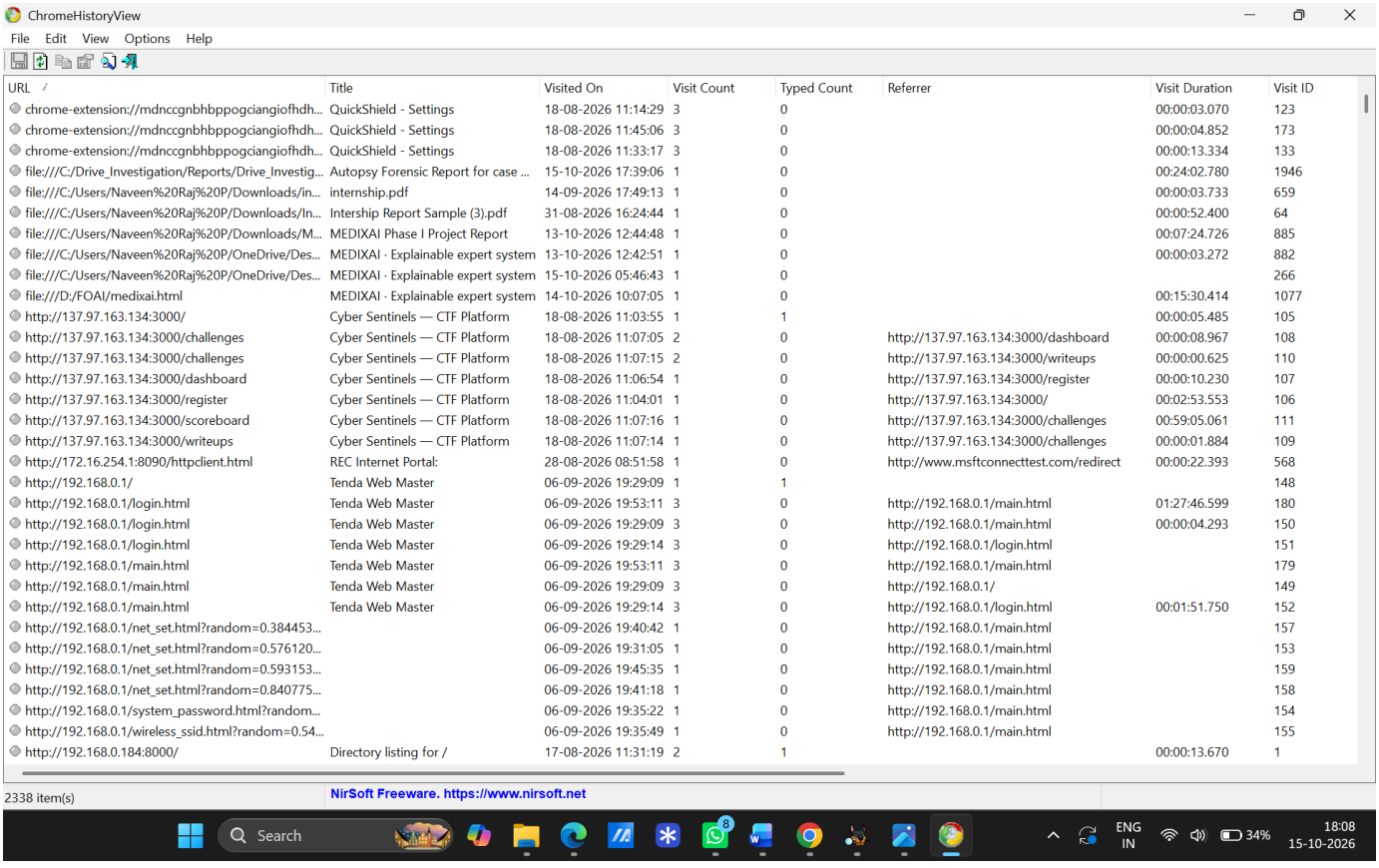
To extract and analyze web browser history, cache, and artifact data using specialized forensic utilities.

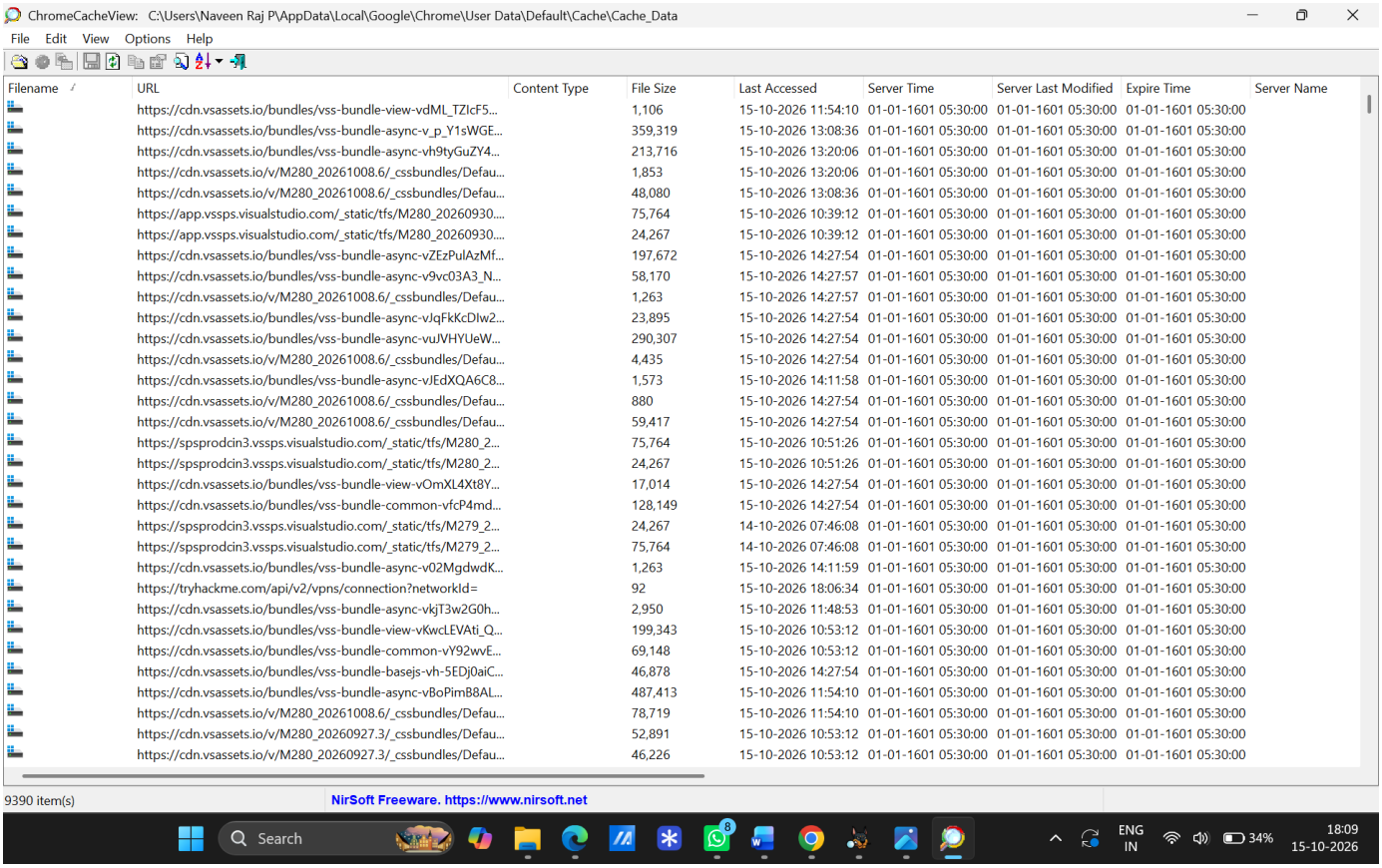
Software / Tools Required

- ChromeHistoryView & ChromeCacheView
- IEHistoryView & IECacheView
- Windows Host PC with Google Chrome and Internet Explorer installed

Procedure

1. **Chrome Extraction:** Launched **ChromeHistoryView** to view browsing history files and **ChromeCacheView** to inspect cache files.
2. **IE Extraction:** Executed **IEHistoryView** to read visited URL records and **IECacheView** to inspect cached file details.
3. **Data Analysis:** Evaluated browsing attributes (URLs, access timestamps, file paths, visit counts) and exported entries to HTML/text reports.





Observations

Tool	Analyzed Artifacts & Displayed Information
ChromeHistoryView	Visited page URLs, page titles, visit dates/times, visit counts, typed counts, and referrers.
ChromeCacheView	Cached file URLs, content types, file sizes, last access times, and server response details.
IEHistoryView	Complete list of visited URLs and historical web page accesses.
IECacheView	Cached file names, full file paths, folder names, content types, file sizes, and hit counts.

Result

Browser forensic tools were executed successfully, and browser artifacts were accurately extracted and analyzed.

EXPERIMENT 07

Comparison of Two Files for Forensics Investigation by Compare IT Software

Aim

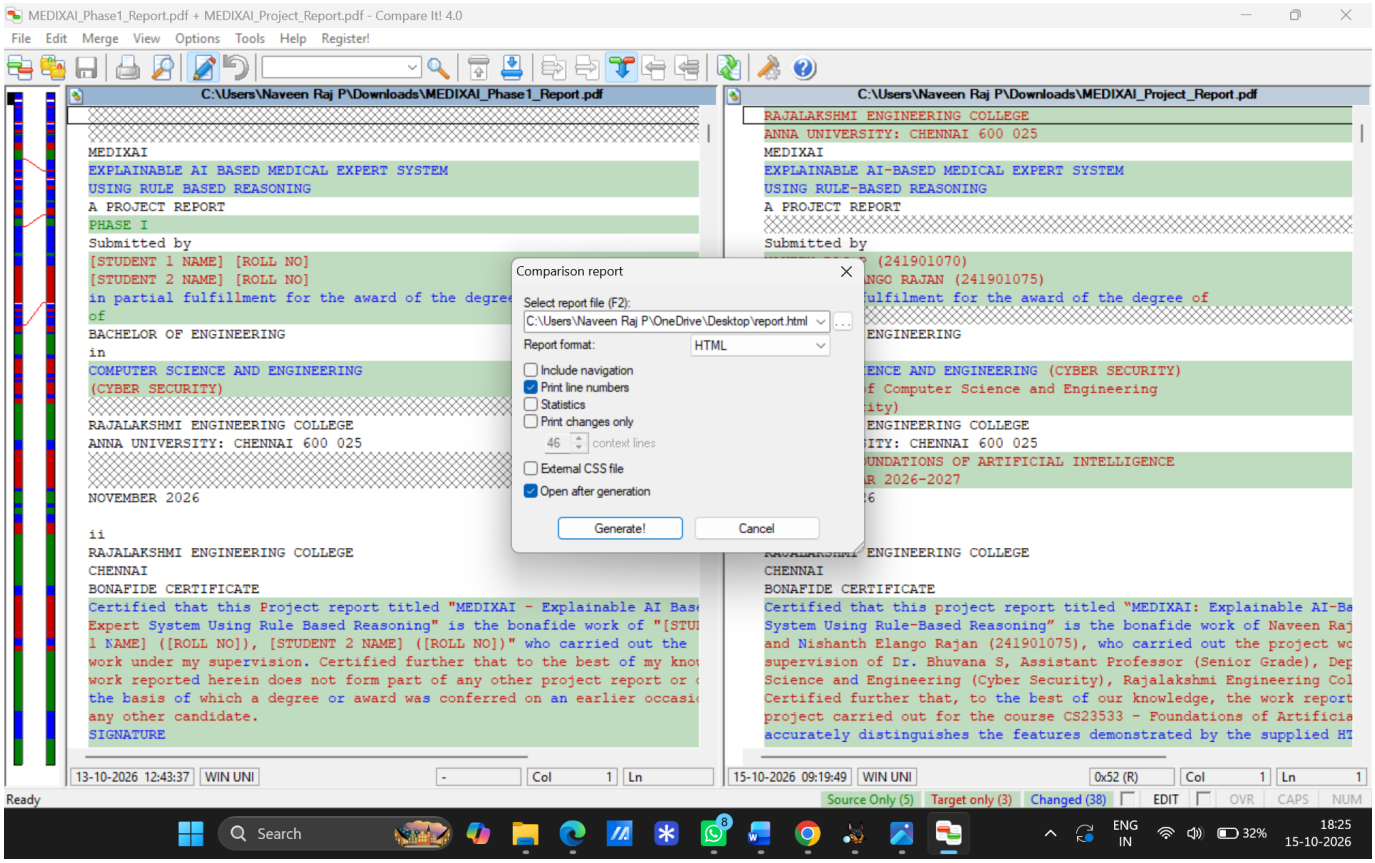
To compare two files side-by-side using **Compare IT** software to detect structural and content differences during a digital forensic investigation.

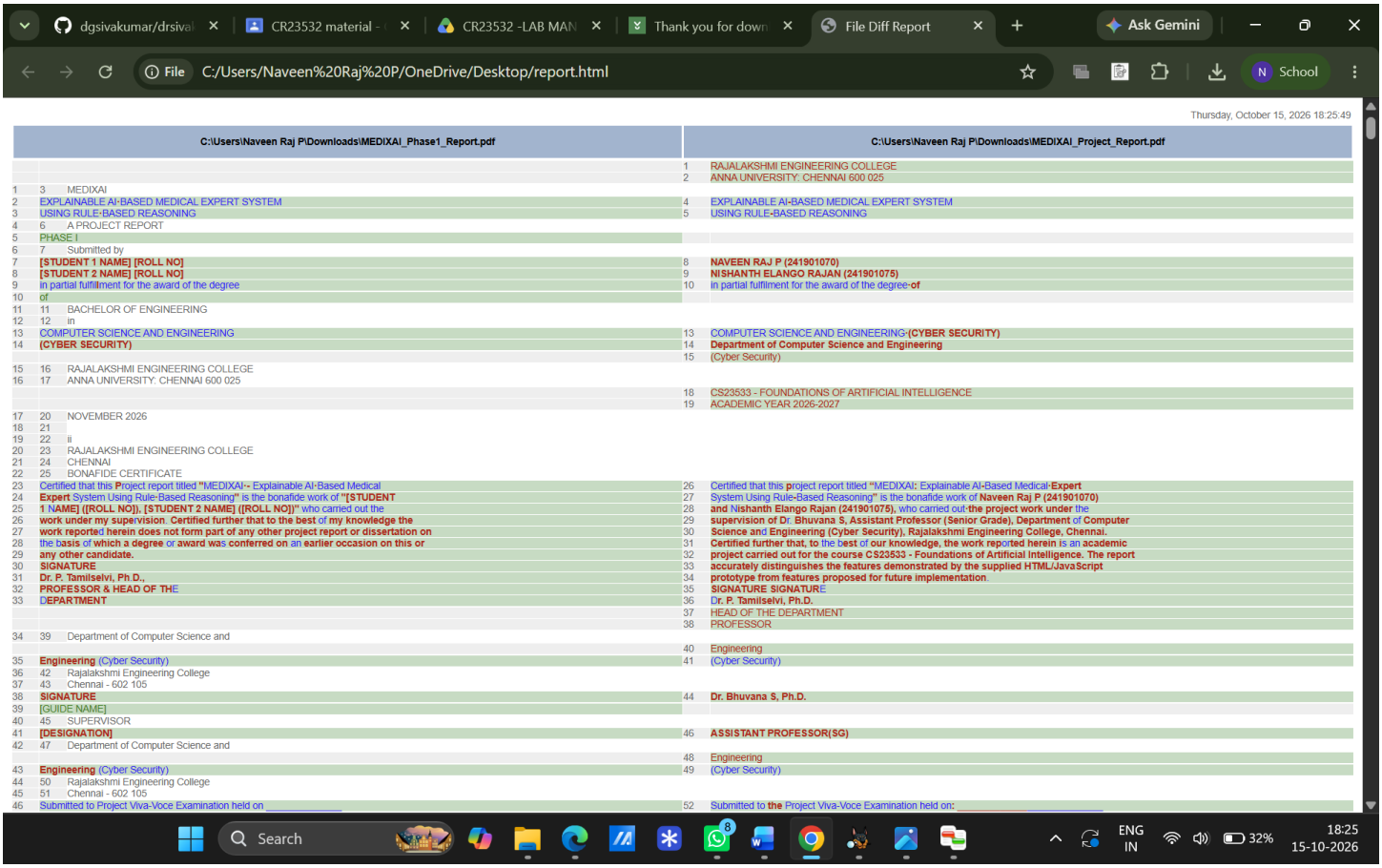
Software / Tools Required

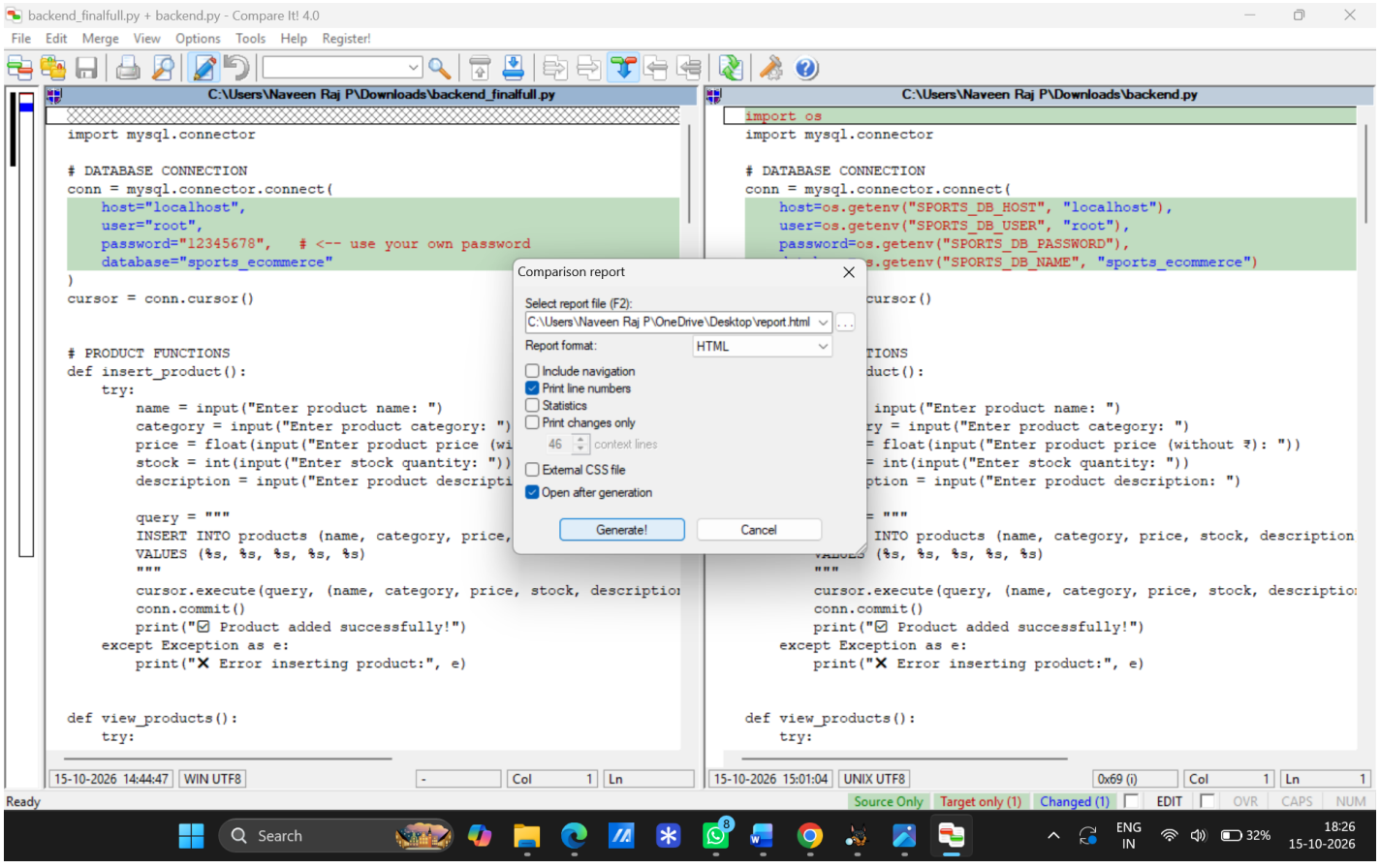
- Compare IT Software
- Target files for comparison

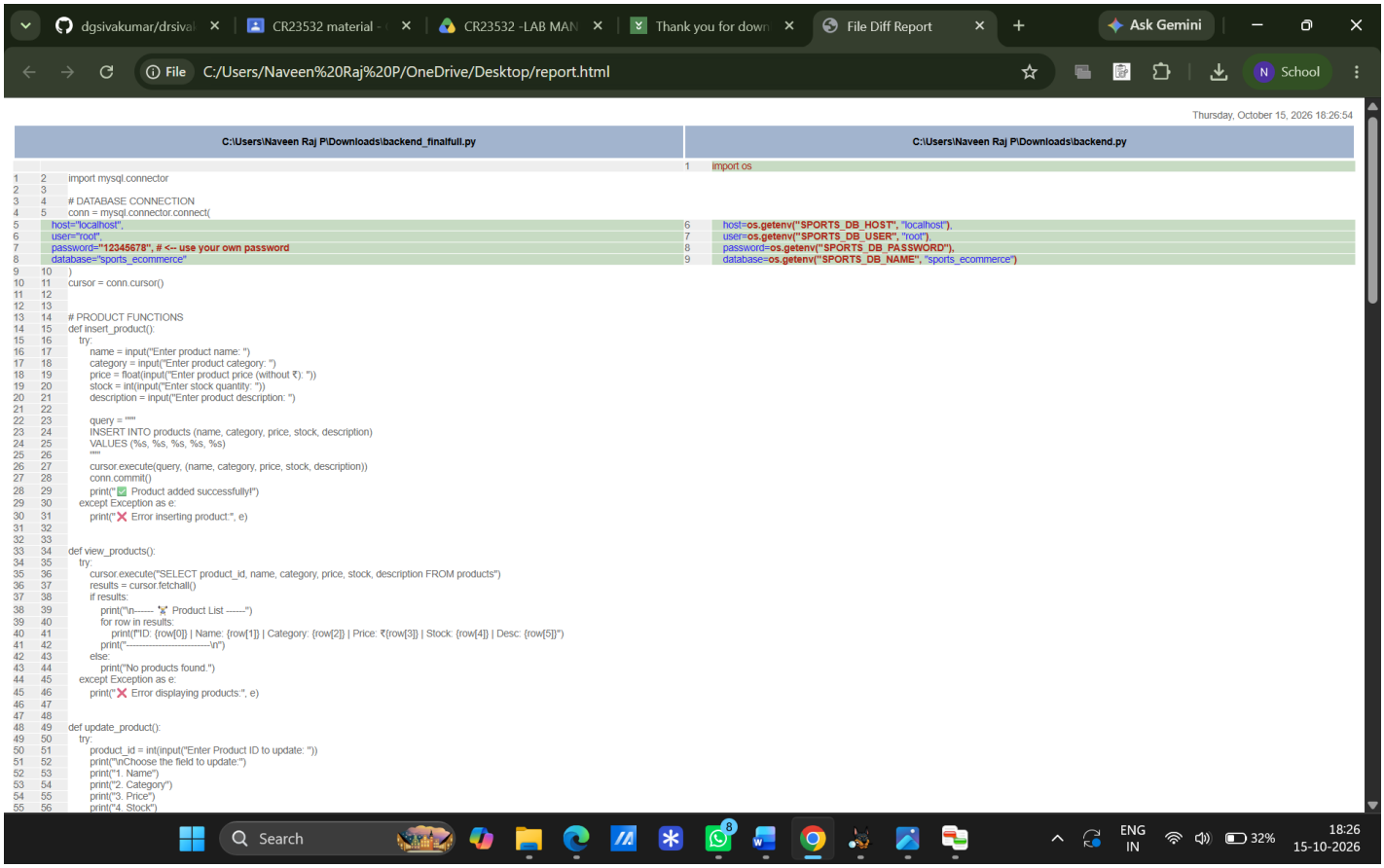
Procedure

1. Installed and opened the **Compare IT** application.
2. Loaded the first file into the left pane and the second file into the right pane.
3. Reviewed the side-by-side display with color-coded highlighted differences.
4. Saved/printed the color-coded difference report summarizing the findings.




dgsivakumar/drsiva
CR23532 material -
CR23532 -LAB MAN
Thank you for down
File Diff Report
Ask Gemini
File C:/Users/Naveen%20Raj%20P/OneDrive/Desktop/report.html
School
Thursday, October 15, 2026 18:25:49
C:\Users\Naveen Raj P\Downloads\MEDIXAI_Phase1_Report.pdf
C:\Users\Naveen Raj P\Downloads\MEDIXAI_Project_Report.pdf
MEDIXAI
EXPLAINABLE AI-BASED MEDICAL EXPERT SYSTEM
USING RULE-BASED REASONING
A PROJECT REPORT
PHASE I
Submitted by
[STUDENT 1 NAME] [ROLL NO]
[STUDENT 2 NAME] [ROLL NO]
in partial fulfilment for the award of the degree
of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)
RAJALAKSHMI ENGINEERING COLLEGE
ANNA UNIVERSITY: CHENNAI 600 025
NOVEMBER 2026
ii
RAJALAKSHMI ENGINEERING COLLEGE
CHENNAI
BONAFIDE CERTIFICATE
Certified that this Project report titled "MEDIXAI – Explainable AI-Based Medical
Expert System Using Rule-Based Reasoning" is the bonafide work of "[STUDENT
1 NAME] ([ROLL NO]), [STUDENT 2 NAME] ([ROLL NO])" who carried out the
work under my supervision. Certified further that to the best of my knowledge the
work reported herein does not form part of any other project report or dissertation on
the basis of which a degree or award was conferred on an earlier occasion on this or
any other candidate.
SIGNATURE
Dr. P. Tamilselvi, Ph.D.,
PROFESSOR & HEAD OF THE
DEPARTMENT
Department of Computer Science and
Engineering (Cyber Security)
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RAJALAKSHMI ENGINEERING COLLEGE
ANNA UNIVERSITY: CHENNAI 600 025
CS23533 – FOUNDATIONS OF ARTIFICIAL INTELLIGENCE
ACADEMIC YEAR 2026-2027
EXPLAINABLE AI-BASED MEDICAL EXPERT SYSTEM
USING RULE-BASED REASONING
NAVEEN RAJ P (241901070)
NISHANTH ELANGO RAJAN (241901075)
in partial fulfilment for the award of the degree of
COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)
Department of Computer Science and Engineering
(Cyber Security)
Certified that this project report titled "MEDIXAI: Explainable AI-Based Medical Expert
System Using Rule-Based Reasoning" is the bonafide work of Naveen Raj P (241901070)
and Nishanth Elango Rajan (241901075), who carried out the project work under the
supervision of Dr. Bhuvana S, Assistant Professor (Senior Grade), Department of Computer
Science and Engineering (Cyber Security), Rajalakshmi Engineering College, Chennai.
Certified further that, to the best of our knowledge, the work reported herein is an academic
project carried out for the course CS23533 - Foundations of Artificial Intelligence. The report
accurately distinguishes the features demonstrated by the supplied HTML/JavaScript
prototype from features proposed for future implementation.
SIGNATURE SIGNATURE
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ASSISTANT PROFESSOR(SG)
Engineering
(Cyber Security)
Submitted to the Project Viva-Voce Examination held on: _____
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15-10-2026


backend_finalfull.py + backend.py - Compare It! 4.0
File Edit Merge View Options Tools Help Register!
C:\Users\Naveen Raj P\Downloads\backend_finalfull.py
C:\Users\Naveen Raj P\Downloads\backend.py
import mysql.connector
DATABASE CONNECTION
conn = mysql.connector.connect(
host="localhost",
user="root",
password="12345678", # <-- use your own password
database="sports_ecommerce"
)
cursor = conn.cursor()
PRODUCT FUNCTIONS
def insert_product():
try:
name = input("Enter product name: ")
category = input("Enter product category: ")
price = float(input("Enter product price (wi
stock = int(input("Enter stock quantity: "))
description = input("Enter product descripti
query = """
INSERT INTO products (name, category, price,
VALUES (%s, %s, %s, %s, %s)
"""
cursor.execute(query, (name, category, price, stock, descriptio
conn.commit()
print("☑ Product added successfully!")
except Exception as e:
print("✕ Error inserting product:", e)
def view_products():
try:
import os
import mysql.connector
DATABASE CONNECTION
conn = mysql.connector.connect(
host=os.getenv("SPORTS_DB_HOST", "localhost"),
user=os.getenv("SPORTS_DB_USER", "root"),
password=os.getenv("SPORTS_DB_PASSWORD"),
s.getenv("SPORTS_DB_NAME", "sports_ecommerce")
cursor()
TIONS
duct():
input("Enter product name: ")
ry = input("Enter product category: ")
= float(input("Enter product price (without ₹): "))
= int(input("Enter stock quantity: "))
ption = input("Enter product description: ")
= """
INTO products (name, category, price, stock, description
(%s, %s, %s, %s, %s)
"""
cursor.execute(query, (name, category, price, stock, descriptio
conn.commit()
print("☑ Product added successfully!")
except Exception as e:
print("✕ Error inserting product:", e)
def view_products():
try:
Comparison report
Select report file (F2):
C:\Users\Naveen Raj P\OneDrive\Desktop\report.html
Report format:
HTML
Include navigation
Print line numbers
Statistics
Print changes only
46 context lines
External CSS file
Open after generation
Generate!
Cancel
15-10-2026 14:44:47
WIN UTF8
Col 1
Ln
15-10-2026 15:01:04
UNIX UTF8
0x69 (i)
Col 1
Ln 1
Ready
Source Only
Target only (1)
Changed (1)
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Observations

Feature / Aspect	Findings & Analysis Observed
Visual Comparison	Files are displayed side-by-side with color-highlighted differences for visual tracking.
Modification Capabilities	Direct editing of files and single-click transfer of changes between files.
Output / Reporting	Generates an accurate, color-coded printout or report of the file differences.

Result

File comparison using Compare IT software was successfully conducted, and differences between target files were documented.

EXPERIMENT 08

Collecting Email Evidence in Victim PC

Aim

To collect and extract email evidence and volatile data from a victim PC by capturing and analyzing physical memory (RAM).

Software / Tools Required

- DumpIt Utility
- Bulk Extractor (or RAM analyzer)
- Target Windows Host PC

Procedure

1. Executed **DumpIt.exe** directly from an external USB drive on the victim machine.
2. Confirmed the execution prompt to generate a full, raw physical memory dump (.raw file).
3. Used **Bulk Extractor** to scan the generated RAM image for email addresses, headers, and message fragments.

Observations

Feature / Step	**Findings & Analysis Observed**
Volatile Data Acquisition	DumpIt successfully captured physical RAM across x86/x64 Windows architecture without installation.
Evidence Extraction	Memory dump contained unencrypted email fragments, URLs, and active session artifacts residing in volatile memory.

Result

Physical RAM from the victim PC was successfully dumped using DumpIt, and email evidence was accurately extracted.

EXPERIMENT 09

Hide and Extract Text File Behind an Image/Audio File Using Command Prompt

Aim

To study the steps for hiding and extracting a text file inside an image file using the Windows Command Prompt (Steganography).

Software / Tools Required

- Command Prompt (cmd.exe)
- Source Image file (B.jpg)
- Text file containing secret data (A.txt)
- WinRAR / Archiving tool

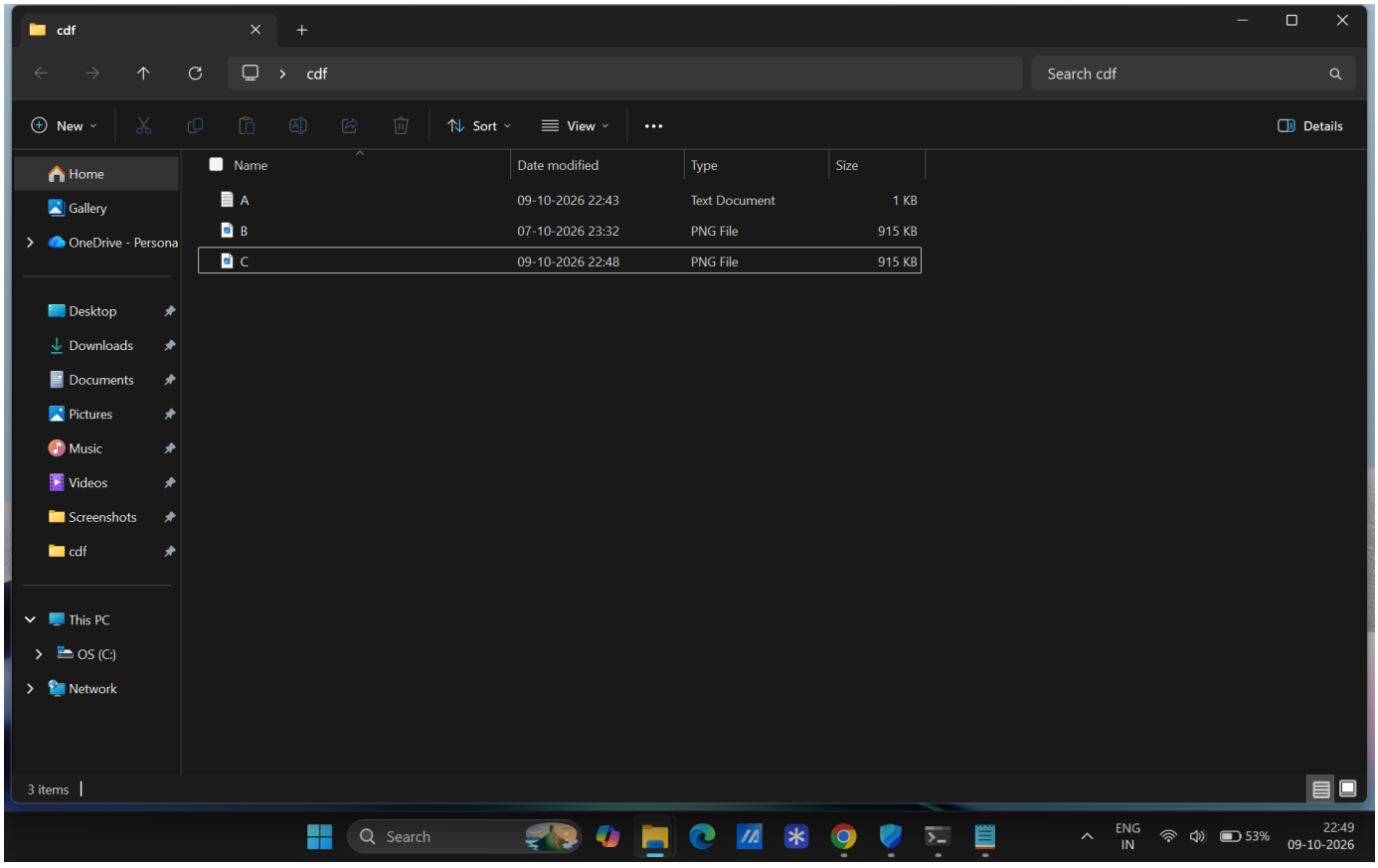
Procedure

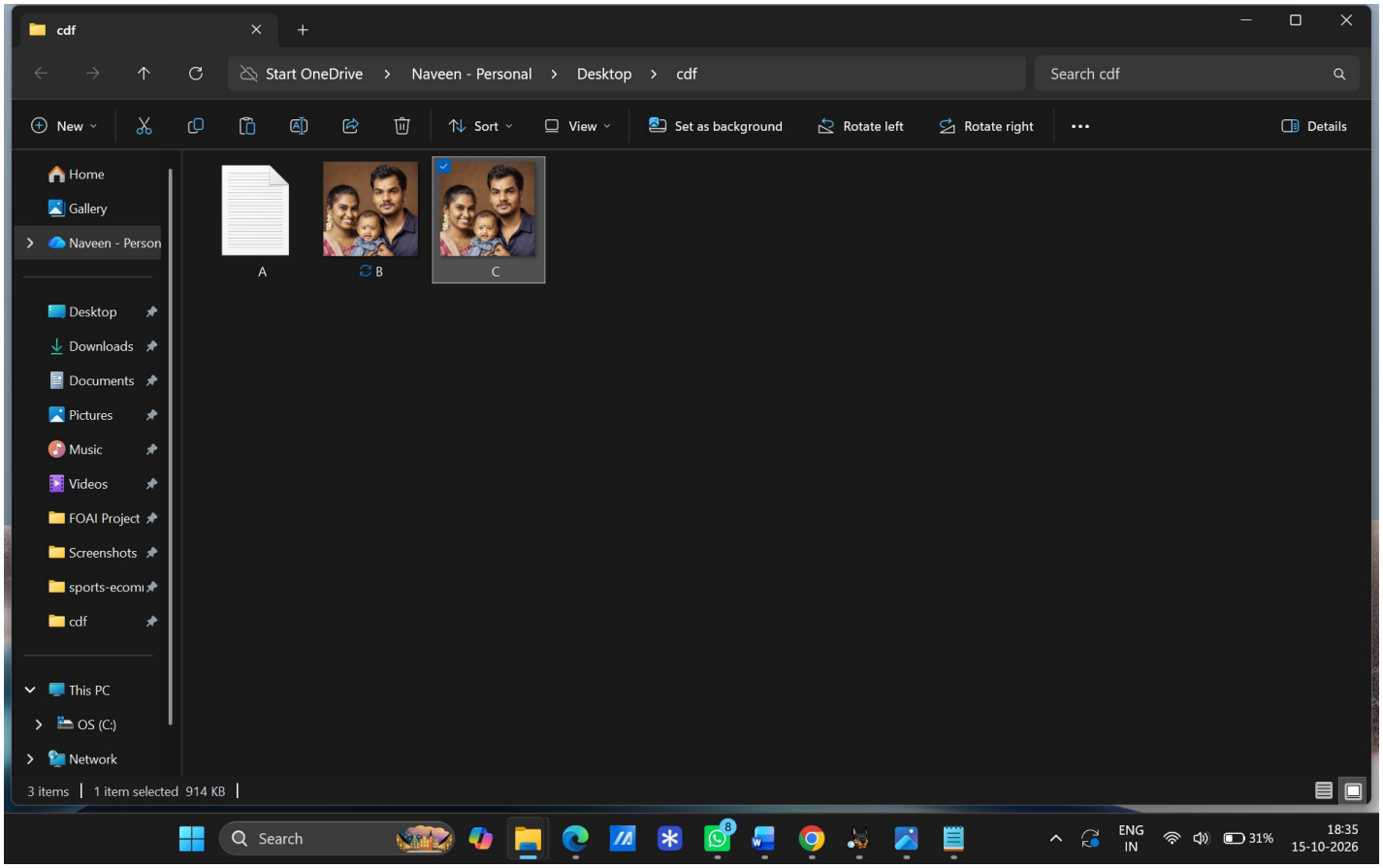
1. Placed cover image (B.jpg) and text file (A.txt) in the same folder directory.
2. Opened Command Prompt, navigated to the folder, and executed the binary copy command:

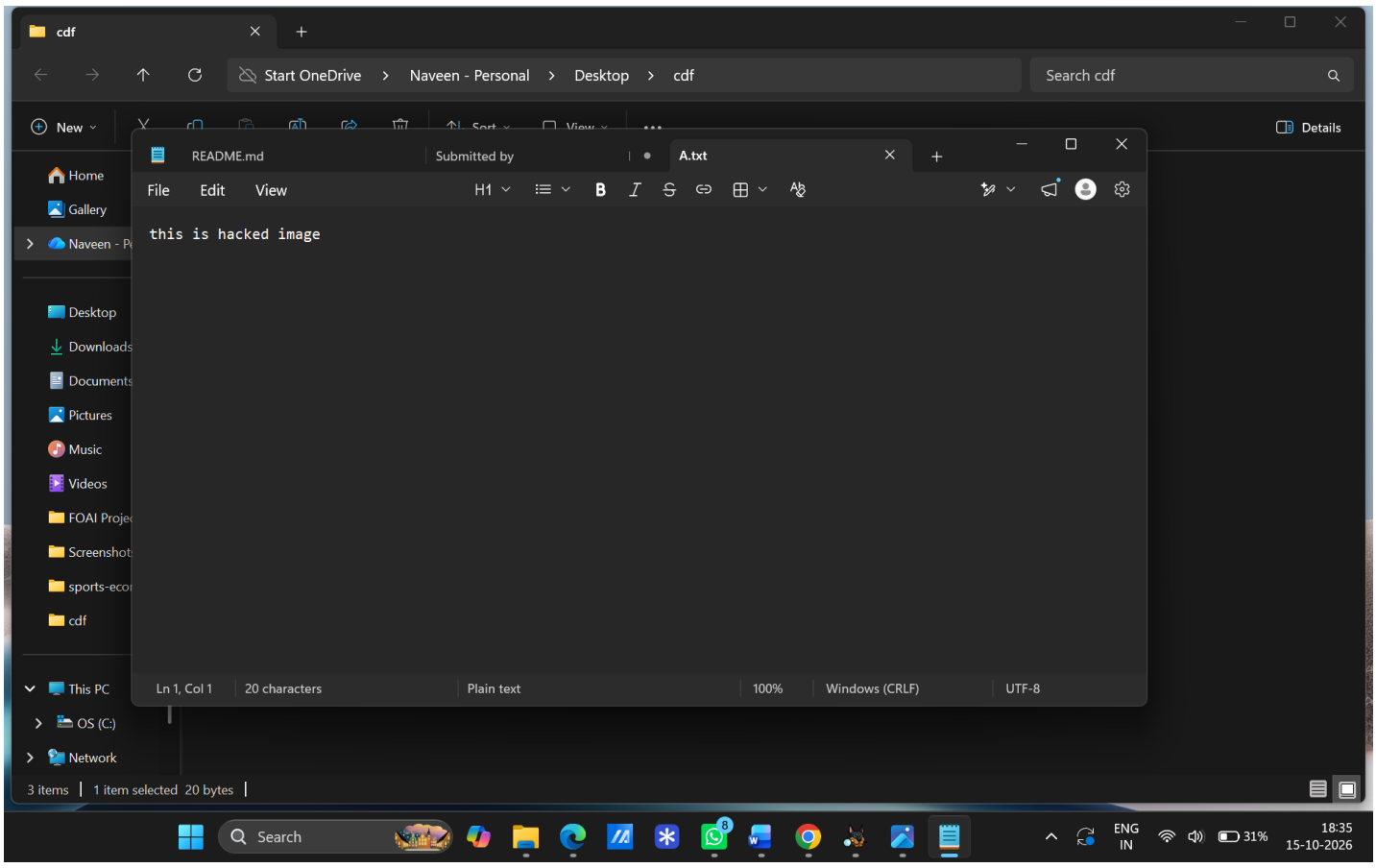
DOS

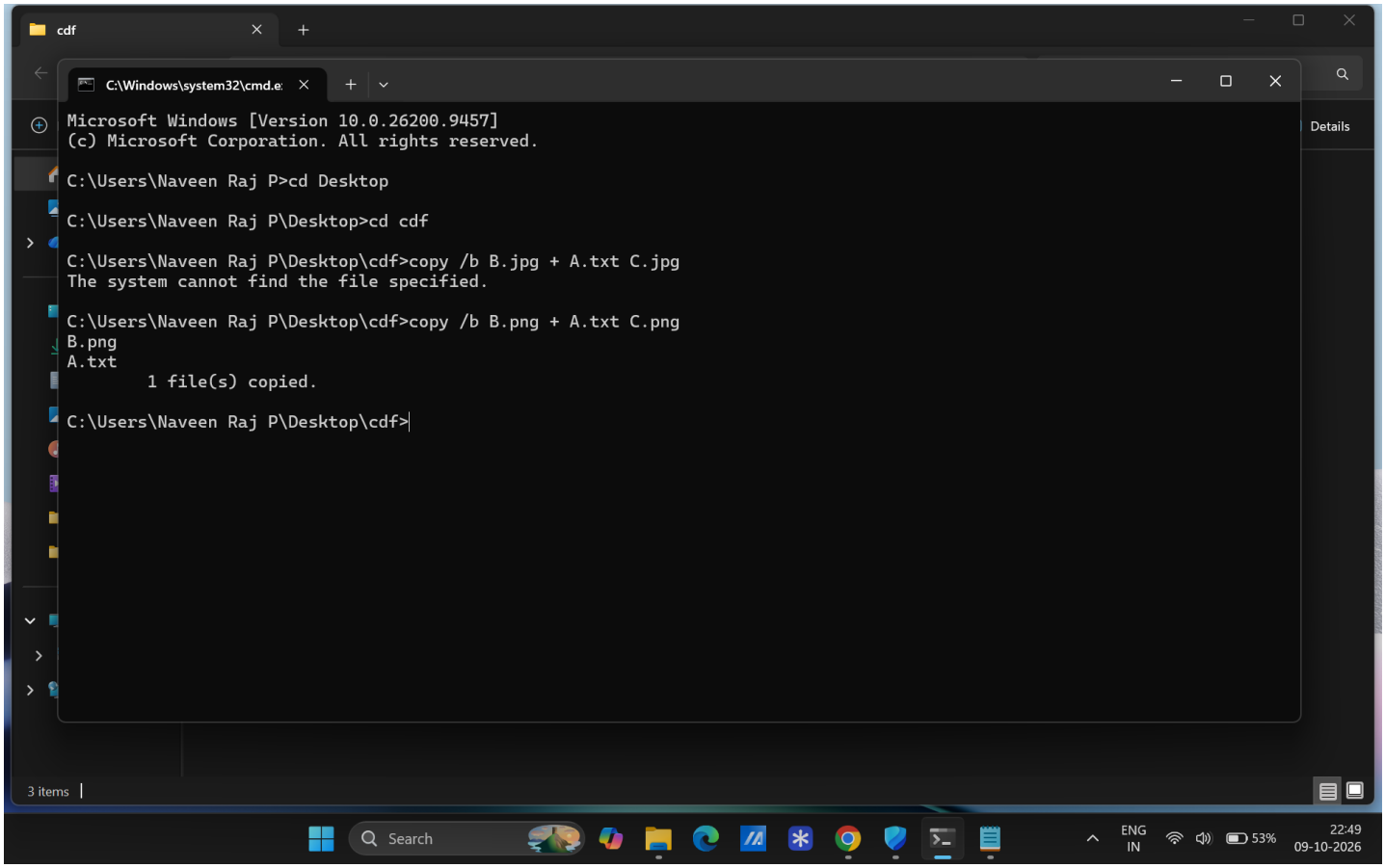
copy /b B.jpg + A.txt C.jpg

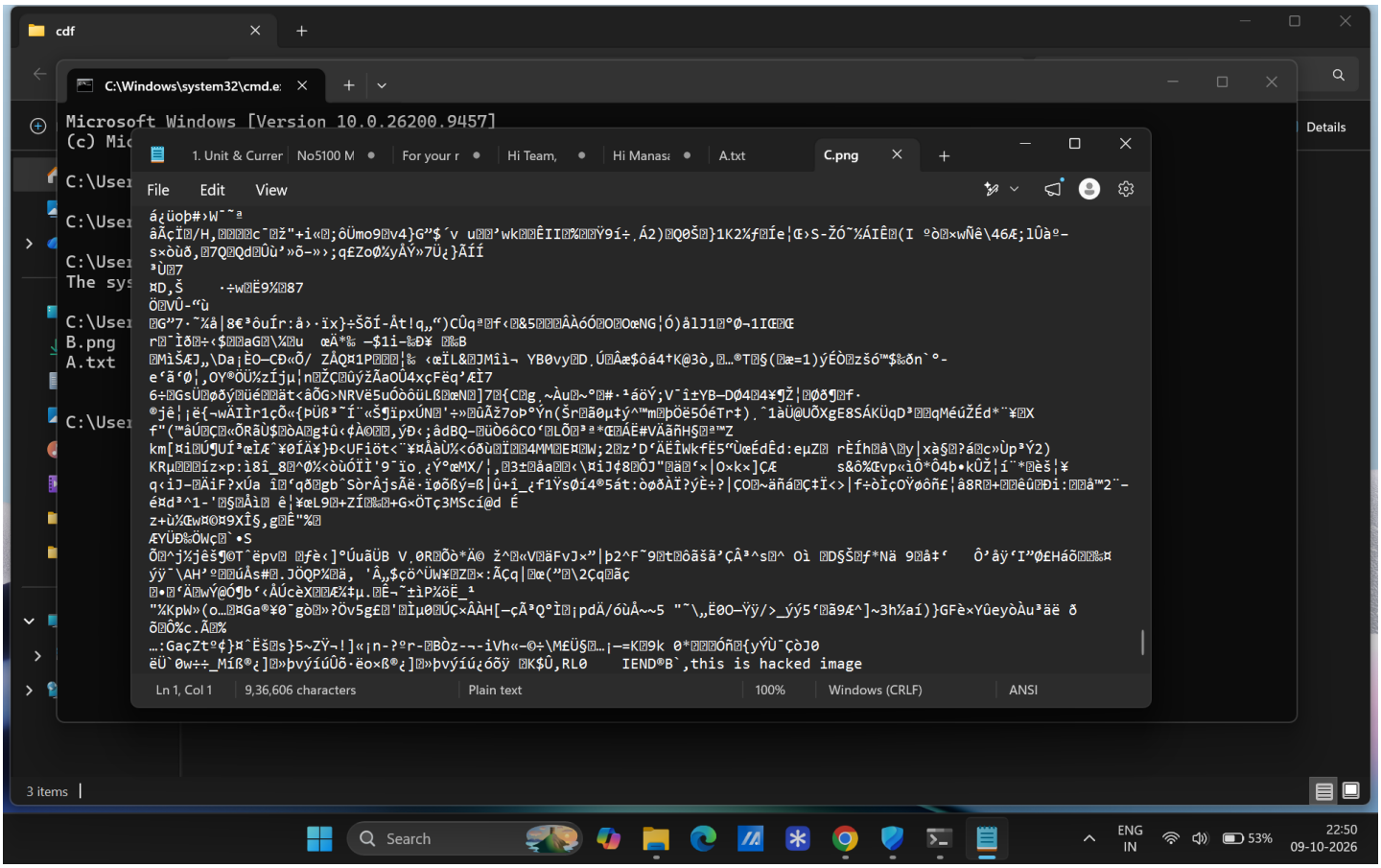
3. Opened the output image (C.jpg) using WinRAR or command-line utilities to view and extract A.txt.




cdf
Start OneDrive
Naveen - Personal
Desktop
cdf
Search cdf
New
Sort
View
Set as background
Rotate left
Rotate right
Details
Home
Gallery
Naveen - Person
Desktop
Downloads
Documents
Pictures
Music
Videos
FOAI Project
Screenshots
sports-ecomi
cdf
This PC
OS (C:)
Network
A
B
C
3 items | 1 item selected 914 KB
Search
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15-10-2026


cdf
Start OneDrive
Naveen - Personal
Desktop
cdf
Search cdf
New
Details
README.md
Submitted by
A.txt
File
Edit
View
H1
this is hacked image
Ln 1, Col 1
20 characters
Plain text
100%
Windows (CRLF)
UTF-8
Home
Gallery
Desktop
Downloads
Documents
Pictures
Music
Videos
FOAI Proje
Screenshot
sports-eco
cdf
This PC
OS (C:)
Network
3 items | 1 item selected 20 bytes
Search
ENG
IN
31%
18:35
15-10-2026


cdf
C:\Windows\system32\cmd.e
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Naveen Raj P>cd Desktop
C:\Users\Naveen Raj P\Desktop>cd cdf
C:\Users\Naveen Raj P\Desktop\cdf>copy /b B.jpg + A.txt C.jpg
The system cannot find the file specified.
C:\Users\Naveen Raj P\Desktop\cdf>copy /b B.png + A.txt C.png
B.png
A.txt
1 file(s) copied.
C:\Users\Naveen Raj P\Desktop\cdf>
Details
3 items
Search
ENG
IN
53%
22:49
09-10-2026


cdf
C:\Windows\system32\cmd.e
Microsoft Windows [Version 10.0.26200.9457]
1. Unit & Currer
No5100 M
For your r
Hi Team,
Hi Manas
A.txt
C.png
File
Edit
View
Ln 1, Col 1
9,36,606 characters
Plain text
100%
Windows (CRLF)
ANSI
Details
3 items
Search
ENG
IN
53%
22:50
09-10-2026

0bservations

Aspect	Findings Observed
Output Image Appearance	Modified image (C.jpg) opens normally in standard image viewers without visual corruption.
File Size Change	The size of C.jpg equals the combined total size of B.jpg and A.txt.
Data Extraction	Opening C.jpg in WinRAR reveals the intact, hidden A.txt file content.

Result

The text file was successfully hidden behind the image file using Command Prompt, and the hidden data was retrieved.

EXPERIMENT 10

Extract Exchangeable Image File Format (EXIF) Data from Image Files using Exifreader Software

Aim

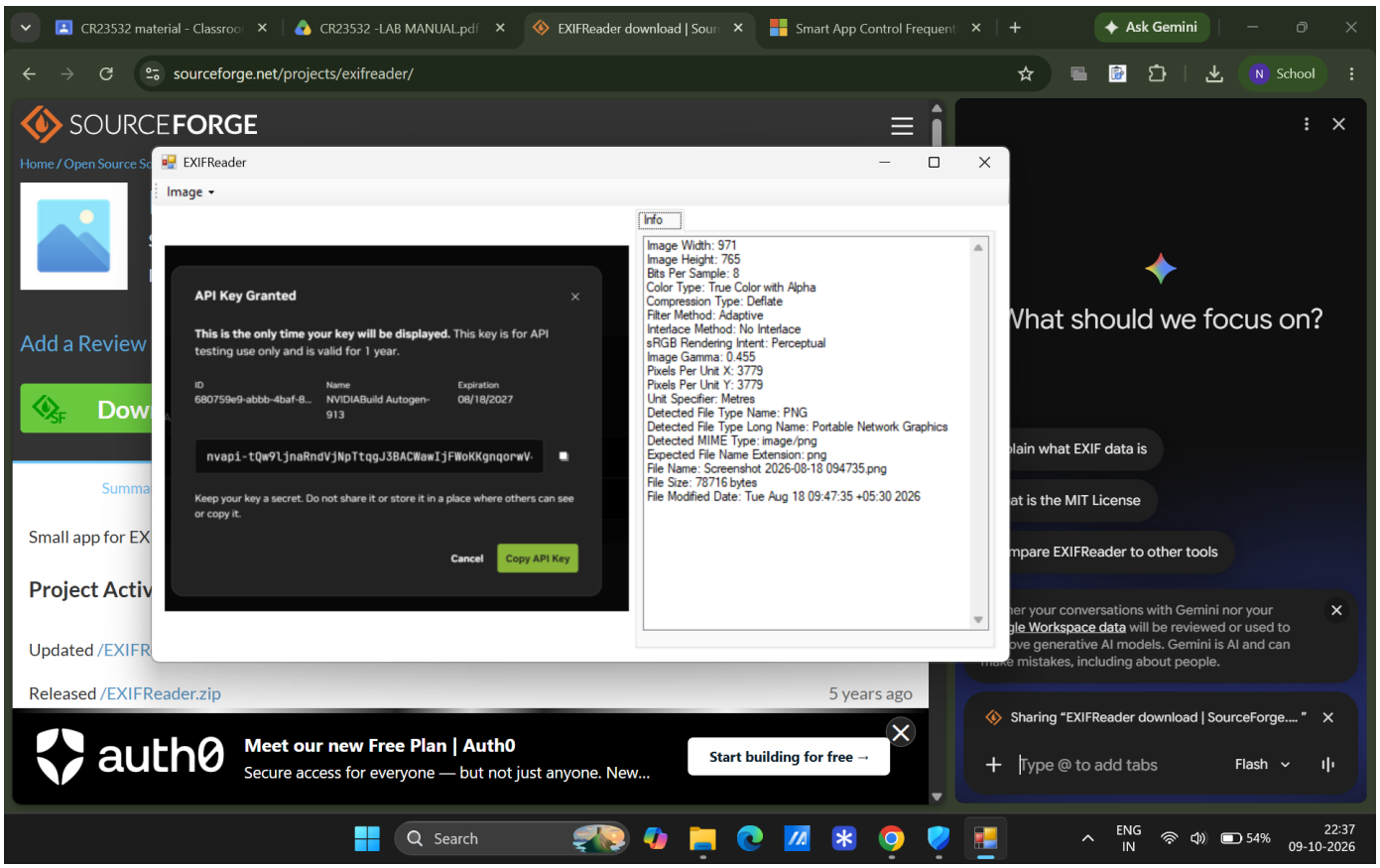
To extract and analyze metadata/EXIF data embedded within digital image files using **Exifreader** software for forensic investigation.

Software / Tools Required

- Exifreader Software
- Target Digital Image File (JPEG/TIFF image from a camera or smartphone)

Procedure

1. Opened the **Exifreader** application on the investigation host.
2. Imported the target digital image file into the software interface.
3. Allowed the tool to automatically parse embedded EXIF header tags.
4. Reviewed camera settings, device details, timestamps, and geolocation information.



EXIFReader

Image



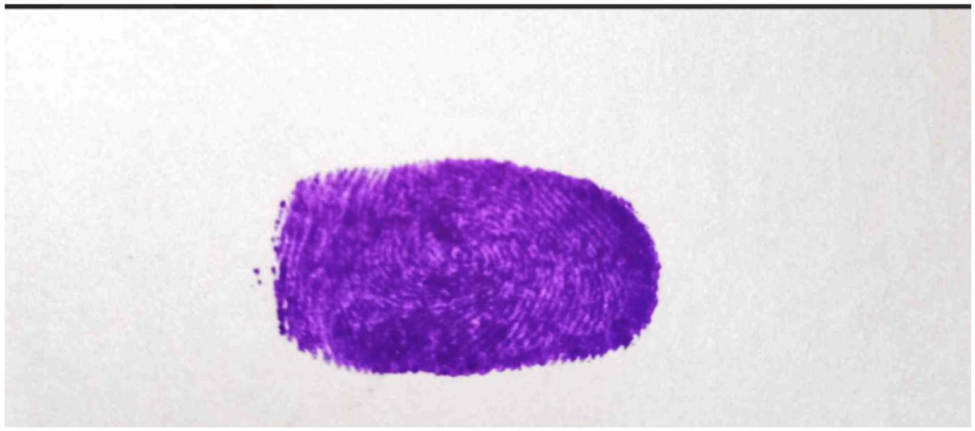
Info

```
Image Width: 971
Image Height: 765
Bits Per Sample: 8
Color Type: True Color with Alpha
Compression Type: Deflate
Filter Method: Adaptive
Interlace Method: No Interlace
sRGB Rendering Intent: Perceptual
Image Gamma: 0.455
Pixels Per Unit X: 3779
Pixels Per Unit Y: 3779
Unit Specifier: Metres
Detected File Type Name: PNG
Detected File Type Long Name: Portable Network Graphics
Detected MIME Type: image/png
Expected File Name Extension: png
File Name: Screenshot 2026-08-18 094735.png
File Size: 78716 bytes
File Modified Date: Tue Aug 18 09:47:35 +05:30 2026
Image Width: 882
Image Height: 883
Bits Per Sample: 8
Color Type: True Color with Alpha
Compression Type: Deflate
Filter Method: Adaptive
Interlace Method: No Interlace
sRGB Rendering Intent: Perceptual
Image Gamma: 0.455
Pixels Per Unit X: 5669
Pixels Per Unit Y: 5669
Unit Specifier: Metres
Detected File Type Name: PNG
Detected File Type Long Name: Portable Network Graphics
Detected MIME Type: image/png
Expected File Name Extension: png
File Name: Screenshot 2026-10-07 233237.png
File Size: 936600 bytes
File Modified Date: Wed Oct 07 23:32:37 +05:30 2026
```

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EXIFReader

Image



Info

```
Image Width: 971
Image Height: 765
Bits Per Sample: 8
Color Type: True Color with Alpha
Compression Type: Deflate
Filter Method: Adaptive
Interlace Method: No Interlace
sRGB Rendering Intent: Perceptual
Image Gamma: 0.455
Pixels Per Unit X: 3779
Pixels Per Unit Y: 3779
Unit Specifier: Metres
Detected File Type Name: PNG
Detected File Type Long Name: Portable Network Graphics
Detected MIME Type: image/png
Expected File Name Extension: png
File Name: Screenshot 2026-08-18 094735.png
File Size: 78716 bytes
File Modified Date: Tue Aug 18 09:47:35 +05:30 2026
Image Width: 882
Image Height: 883
Bits Per Sample: 8
Color Type: True Color with Alpha
Compression Type: Deflate
Filter Method: Adaptive
Interlace Method: No Interlace
sRGB Rendering Intent: Perceptual
Image Gamma: 0.455
Pixels Per Unit X: 5669
Pixels Per Unit Y: 5669
Unit Specifier: Metres
Detected File Type Name: PNG
Detected File Type Long Name: Portable Network Graphics
Detected MIME Type: image/png
Expected File Name Extension: png
File Name: Screenshot 2026-10-07 233237.png
File Size: 936600 bytes
File Modified Date: Wed Oct 07 23:32:37 +05:30 2026
Image Width: 1648
Image Height: 721
Bits Per Sample: 8
Color Type: True Color with Alpha
Compression Type: Deflate
Filter Method: Adaptive
Interlace Method: No Interlace
sRGB Rendering Intent: Perceptual
Image Gamma: 0.455
Pixels Per Unit X: 5669
Pixels Per Unit Y: 5669
Unit Specifier: Metres
```

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Observations

Metadata Category	Extracted EXIF Artifacts
Device & Hardware Information	Camera make/model, smartphone type, and scanner specifications.
Capture Settings	Shutter speed, aperture, ISO, focal length, flash usage, and resolution details.
Time & Location Data	Exact date/time stamp of image capture and GPS coordinates (if enabled).

Result

EXIF data was successfully extracted from the target image file using Exifreader, providing valuable metadata for linking evidence to a device and timeline.